CHEMICAL BONDING

DISCLAIMER

"The content provided herein are created and owned by various authors and licensed to Sorting Hat Technologies Private Limited ("Company"). The Company disclaims all rights and liabilities in relation to the content. The author of the content shall be solely responsible towards, without limitation, any claims, liabilities, damages or suits which may arise with respect to the same."

Chemical Bonding

Force of attraction between two or more than two atoms to hold them together is called chemical bond.

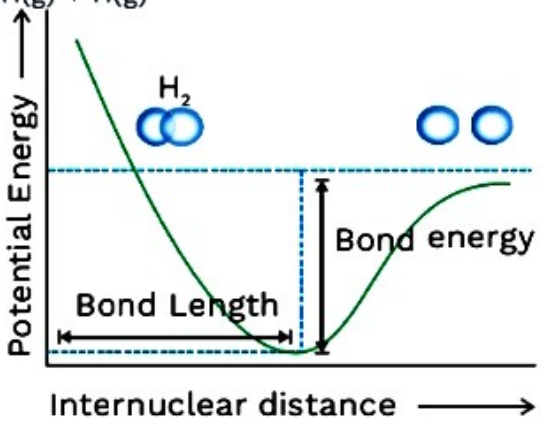
⇒ Atoms combine to gain stability by decrement in energy.

Example :

Formation of H_2 molecule

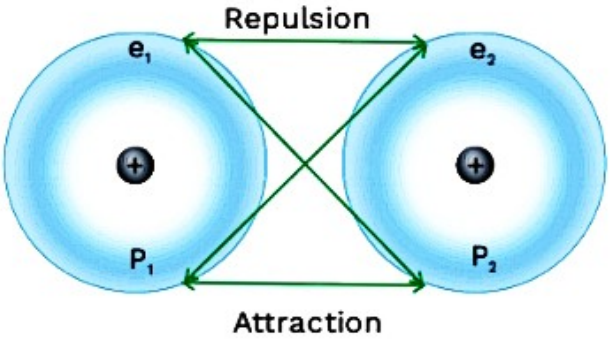
$$H(g) + H(g) \longrightarrow H_2(g) + 435.5\ kJ/mol$$
$$\Delta H = -435.5\ kJ/mol$$

$$H_2 + 435.5 \longrightarrow H(g) + H(g)$$



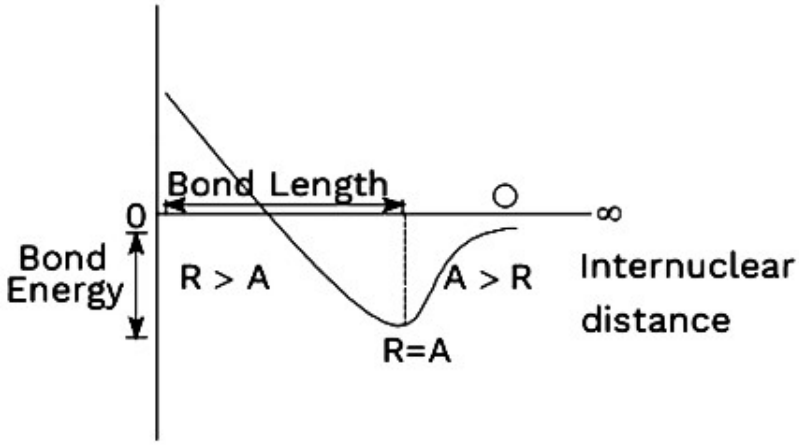
Tendency to acquire minimum energy:

1. When two atoms approach to each other-electrons of one atom attracts the nucleus of another atom.
2. Two electrons and nucleus of both the atoms repel each other's.
3. Total energy of system decreases if net result is attraction. When the net energy of the system is least a chemical bond is formed.
4. Energy released in bond formation is called bond energy.
5. Bond formation is an exothermic process.
6. Bond dissociation is an endothermic process.



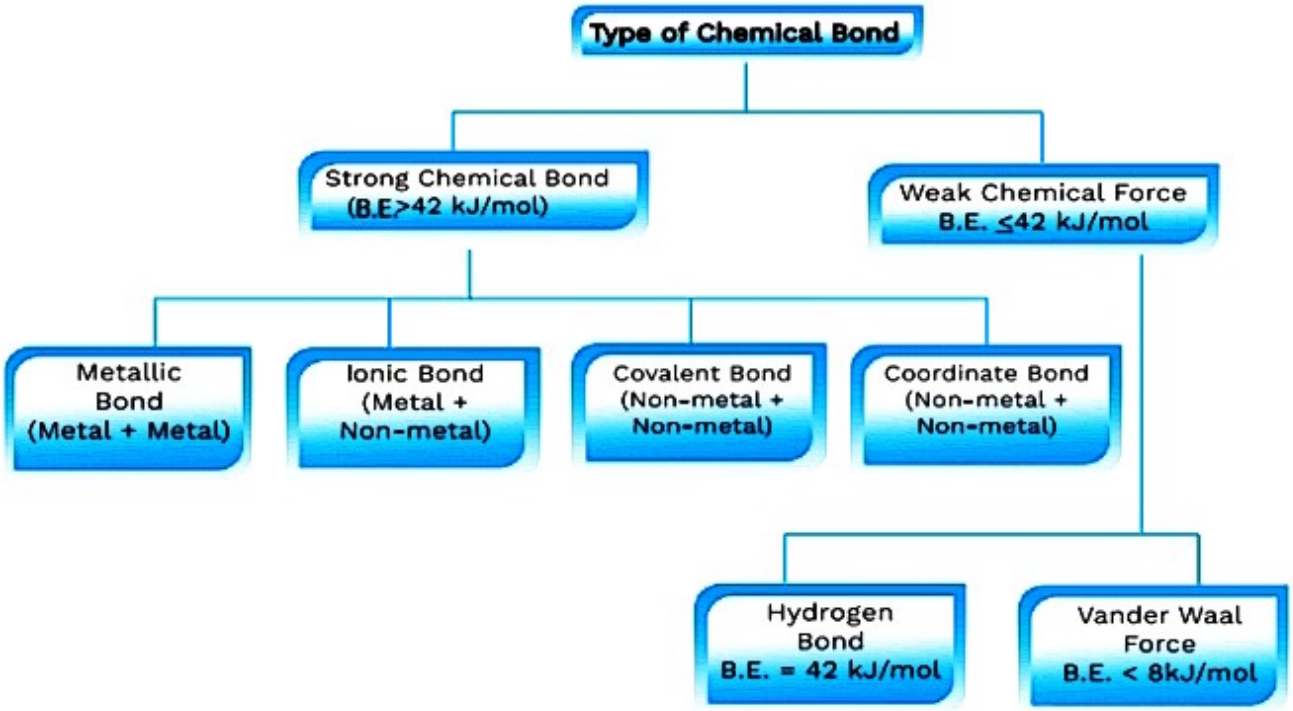
Force of Attraction	Force of Repulsion
Energy ↓	Energy ↑
Stability ↑	Stability ↓
Exothermic	Endothermic
P_1-e_2, P_2-e_1	e_1-e_2, P_1-P_2

Internuclear distance at the time of bond formation is called bond length



O ← r	r	r = ∞
R>A	R = A	A > R
Energy ↑	Min.Energy	Energy ↓
Stability ↓	Max.Stability	Stability ↑

Bond length $H_2 \simeq D_2$
Bond Energy $H_2 < D_2$



- Energy released in bond formation is called bond energy which is calculated in terms of bond dissociation energy because molecule are found naturally and conversion of molecule into atom is easily possible.
- Atoms combine and form stable species molecule and lattice.
- Combination of two or more than two atoms of same or different type having independent existence is called molecule.

Q.1 **Statement-I : Atoms of all element undergo bond formation phenomenon.**
Statement-II : Bond formation phenomenon is always accomplished by decrease in energy

Sol. Statement-I is false whereas statement-II is true because noble gases like (He, Ne, Ar, ...) do not participate in bond formation phenomenon.

Weak Chemical Forces

1. **Hydrogen bond** (8-42 kJ/mol) (For neutral molecule)

When hydrogen atom involved in the two bonds, one is a normal covalent bond and the other is a hydrogen bond. A hydrogen bond is always formed between two small size, high electronegative atoms such as fluorine, oxygen and nitrogen.

$$\overset{+\delta}{A}\text{———}\overset{-\delta}{B}\text{.....}\overset{+\delta}{A}\text{———}\overset{-\delta}{B}$$

Example : $\overset{+\delta}{H}\text{———}\overset{-\delta}{F}\text{.....}\overset{+\delta}{H}\text{———}\overset{-\delta}{F}$

dipole-dipole attraction

- It is also known as polar-polar attraction

$X\text{———}H\text{.....}X\text{———}X$ Where X could be F, O, N

Short trick to learn hydrogen bonding :

[F, O, N, sp-hybridcarbon
or
E.N of C = 3.0
^{es}–CCl_3 – CF_3 species]

Conditions for stronger H–bonding :

- Electronegativity of X should be high
- Atomic size of electronegative element should be small.

$\overset{+\delta}{H}\text{—}\overset{-\delta}{Cl}\cdots\overset{+\delta}{H}\text{—}\overset{-\delta}{Cl}\cdots\overset{+\delta}{H}\text{—}\overset{-\delta}{Cl}$ (Large size of atom)

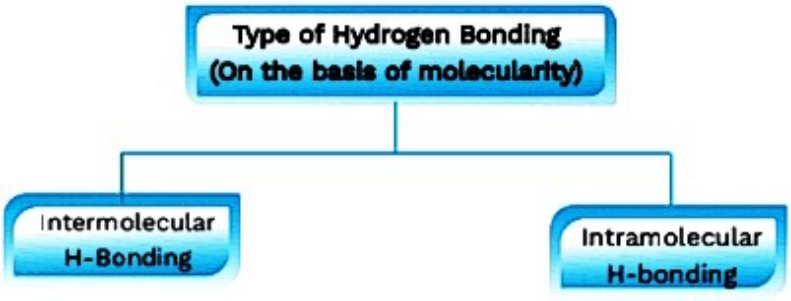
- $\overset{-\delta}{X}\text{—}\overset{+\delta}{H}\cdots y(F, O, N)$

$X\text{—}H\cdots N\text{—}H > X\text{—}H\cdots O\text{—}H > X\text{—}H\cdots F\text{—}H$

Strength↑

donating tendency : N > O > F

Strength ∝ More electronegativity of donor atom



Intramolecular Hydrogen Bonding :

H-Bond formation between two or more molecules of either the same or different compounds known as intermolecular H-bonding.

$\overset{+\delta}{H}\text{—}\overset{-\delta}{X}\cdots\overset{+\delta}{H}\text{—}\overset{-\delta}{X}\cdots\overset{+\delta}{H}\text{—}\overset{-\delta}{X}\cdots$

These are two types:

(a) Homo-intermolecular :- Hydrogen bond between molecules of same compounds.

Examples:

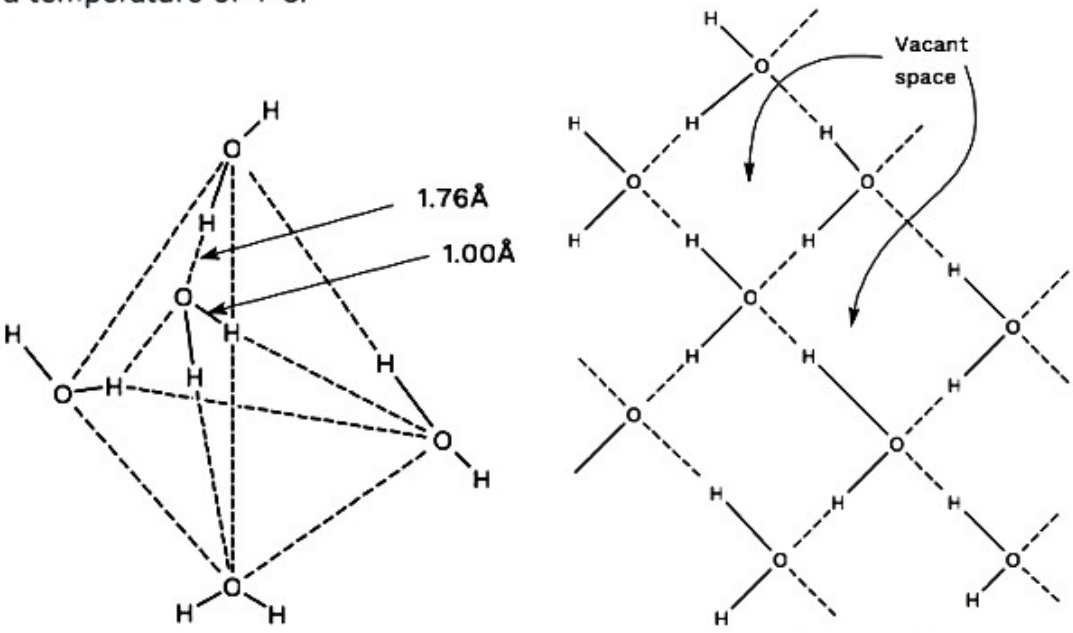
(i) Hydrogen fluoride:

HF is a polar molecule, with the fluorine atom acquiring a slight negative charge and the hydrogen atom acquiring an equal positive charge. The electrostatic attraction between the oppositely charged ends results in hydrogen bonding as shown below.

H–F ..... H–F ..... H–F .....

(ii) Water : The high boiling point of water compared to that of hydrogen sulphide is due to molecular association through hydrogen bonding.

The crystal structure of ice shows a tetrahedral arrangement of water molecules. Each oxygen atom is surrounded tetrahedrally by 4 others. Hydrogen bond link pairs of oxygen atoms together as shown in Fig. The arrangement of water molecules in ice is a very open structure and this explains the low density of ice. When ice melts, the structure breaks down and the molecules pack more closely together so that water has a higher density; this packing goes to a maximum up to a temperature of 4°C.

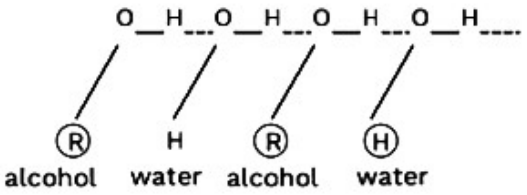


(b) Hetero intermolecular: Hydrogen bond between molecules of different compounds.

Example:

(i) Alcohols and water :

Lower alcohols and phenols are associated due to intermolecular hydrogen bonding. Methanol, ethanol and phenol have relatively much higher boiling points than methane or chloromethane, ethane or chloroethane, benzene or chlorobenzene respectively.



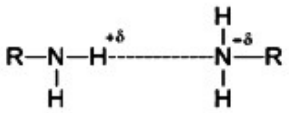
Order of strength of H-bond :

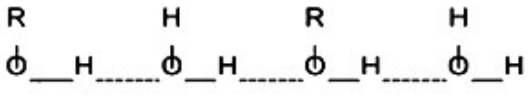
- $H-F > H_2O > NH_3$
- $O—H\cdots\cdots O > N—H\cdots\cdots N > N-H\cdots\cdots O$
- $T_2O > D_2O > H_2O$

For Better H-bonding

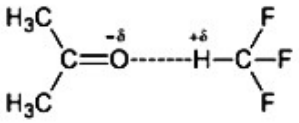
Extent of H – bonding $\propto$ Number of hydrogenbond $\propto$ bond strength

Some Examples of intermolecular hydrogen bonding

(i)  (Homo intermolecular)

(ii)  (Hetro intermolecular)

(iii) 

(iv) 

(A) Application of intermolecular H-bonding

1. Boiling point

(a) $HCl < HBr < HI < \textcircled{HF}$ Halogen family

(b) $H_2S < H_2Se < H_2Te < \textcircled{H_2O}$ Oxygen family

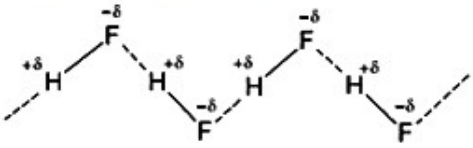
(c) $PH_3 < AsH_3 < NH_3 < SbH_3 < BiH_3$ Nitrogen Family

2. **Order of melting point :**

(a) $HCl < HBr < HF < HI$

(b) $H_2S < H_2Se < H_2Te < H_2O$

Important point :

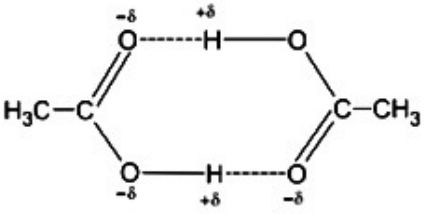


Angular shape/zig-zig → in solid and liquid phase
linear → In gas phase

(C) $PH_3 < AsH_3 < SbH_3 < NH_3$ (no flipping in solid state)

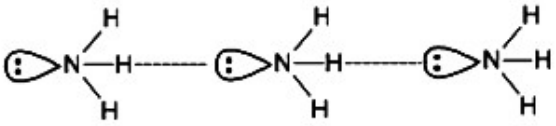
(d) Melting point $H_2O > NH_3 > HF$
Boiling point $H_2O > HF > NH_3$

3. **Dimerization:** CH_3COOH exist as a dimer in vapour phase



Observed molecular mass ⇒ 2 × 60 = 120, due to intra molecular H-bonding dimerization is observed.

4. NH_3-linear H-bonding



5. • $\boxed{HCl_2^-, HBr_2^-, HI_2^-}$ not exist

In case of HCl_2^- :

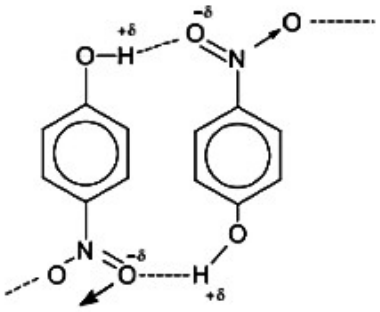
$HCl_2^- \rightarrow Cl^{+\delta}—H\cdots Cl^{-\delta}$ → large size

• In case of K^+F^-, due to similar size of F^- :

$K^+F^- + H - F \rightarrow [F \cdots H - F]^- K^+$ or $K^+[HF_2]^-$

Symmetrical H – bonding

6. p-nitrophenol

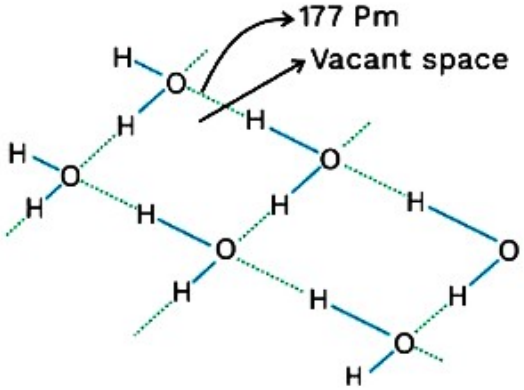


B.P ↑
V.P ↓

O – nitrophenol B.P ↑ V.P ↓

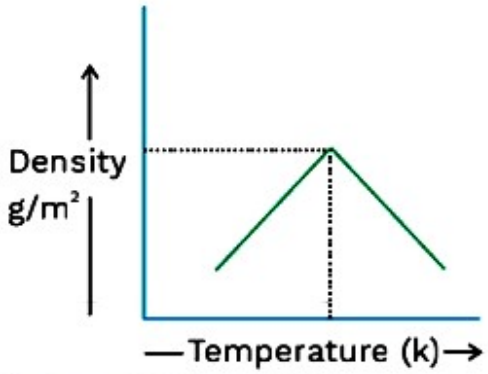
Q.2 Why ice floats on water?

Sol.



Density of ice is less than water. Due to cage like structure of Ice, it has vacant space inside making it less denser.

- Maximum density of H_2O at 4°C (277 K)



7. Order of M.P and B.P in amines

RNH_2	>	R_2NH	>	R_3N
1°		2°		3°

8. Viscosity $\propto$ Number of Hydrogen bonding

9. Solubility $\propto$ Number of Hydrogen bonding

Q.3 **Correct order of solubility in water**

(i) 1° amine > Alcohol

(ii) Glycerol > ethyl glycol > ethanol

(iii) $CH_3CH_2CH_3 > CH_3CH_2OH > CH_3-O-CH_3$

Sol. (ii)

Glycerol > ethyl glycol > ethanol

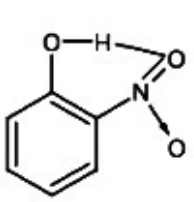
(B) Application of interamolecular H-bonding

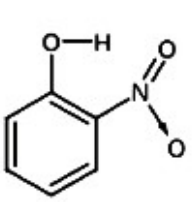
1. Chelation or intramolecular hydrogen bonding (with in a molecule)

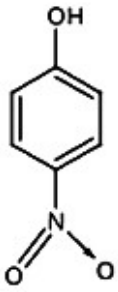
Sometimes hydrogen bonding may take place within a molecule; this is known as intramolecular (or internal) hydrogen bonding. It may lead to the linkage of two groups to from a ring; such an effect is known as chelation,

Examples:

(i) Nitrophenols







o-nitrophenol
less soluble in water

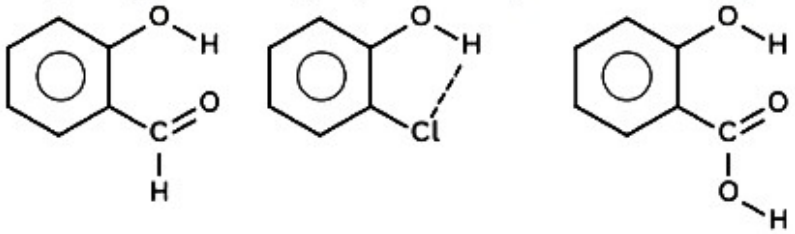
m-nitrophenol

p-nitrophenol

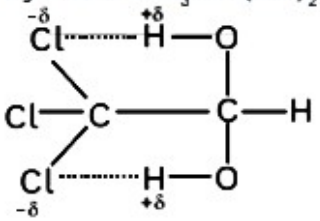
Comparatively more soluble

Because of the proximity of –OH and $-NO_2$ groups in o-nitrophenol there is intramolecular hydrogen bonding which prevents intermolecular hydrogen bonding between two or more molecules. Since molecular association cannot take place, the boiling point of o-nitrophenol is lower than that of the other two. Because of the distance between –OH and $-NO_2$ groups in m- and p-nitrophenols there is no possibility of intramolecular hydrogen bonding. Intermolecular hydrogen bonding may take place to a certain extent which cause some degree of molecular association; this is responssible for the higher boiling points of the two nitrophenols.
Further the formation of intramolecular hydrogen bonding in o-nitrophenol prevents it from entering into intermolecular hydrogen bonding with water and this explains its reduced solubility.

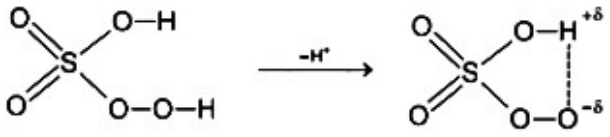
(ii) o-hydroxybenzaldehyde, o-chlorophenol and o-hydroxybenzoic acid.



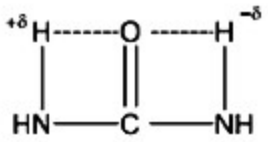
(iii) Chloralhydrate $CCl_3.CH(OH)_2$



(iv) Anions of Caro's acid (H_2SO_5)



(v) Urea



Ring formed is 4 – membered

It is having angle strain
Weaker H-bond cannot tole rate that C–C covalent bonds can do, so cyclo propane and cyclobutene exist.

Q.4 **Which of the given statements is/are true?**
(a) H_2S is gas while H_2O is liquid.
(b) Ethanol has higher boiling point than diethyl ether.
(c) Both (a) and (b).
(d) None of these.

Sol. (c)

Q.5 ***o*-Nitrophenol is**
(a) less volatile than *p*-nitrophenol (b) equally volatile as *p*-nitrophenol
(c) more volatile than *p*-nitrophenol (d) none of these.

Sol. (c)

o-Nitrophenol is more volatile than *p*-nitrophenol as it has lower boiling point than *p*-nitrophenol due to intramolecular hydrogen bonding while

p-nitrophenol has stronger intermolecular hydrogen bonding.

Q.6 **Among the following pairs, the pairs which are having dipole-induced dipole interaction are**
(a) HCl — He, H_2O — alcohols, Cl_2 — CCl_4
(b) HCl — He, H_2O — Xe, H_2O — CO_2
(c) H_2O — alcohols, SiF_4 — He, H_2O — Xe
(d) SiF_4 — He, Cl_2 — CCl_4, H_2O — CO_2

Sol. (b)

HCl — He, H_2O — Xe, H_2O — CO_2

Q.7 **Arrange in increasing order of boiling**

(i) HF, HCl, HBr, HI **(ii) H_2O, HF, NH_3**

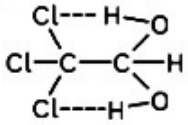
(iii) PH_3, H_2S, HC1 **(iv) H_2O, CH_3OH, CH_3OCH_3**

Sol. (i) $HCl < HBr < HI < HF$ (ii) $NH_3 < HF < H_2O$

(iii) $PH_3 < H_2S < HCl$ (iv) $CH_3OCH_3 < CH_3OH < H_2O$

Q.8 **Why chloral hydrate is a stable molecule ?**

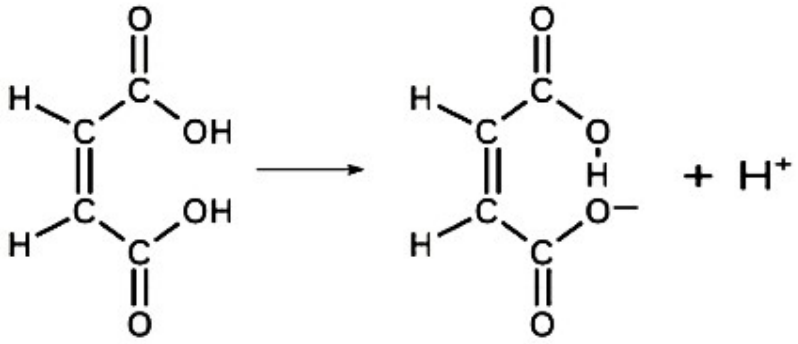
Sol. Chloral hydrate a is stable due to H-bonding (intramolecular)



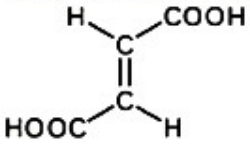
Chloral hydrate

Q.9 **Explain, why cis-maleic acid is stronger acid than trans-fumaric acid?**

Sol. cis -Maleic acid is stronger acid than trans -fumaric acid because conjugate base of maleic acid is stable due to intramolecular H-bonding.

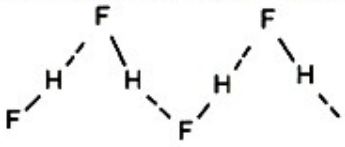


Stable conjugate base of maleic acid

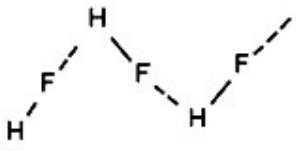


Fumaric acid (No-intramolecular H-bonding)

Q.10 **Which structure of HF is correct ?**



(i)



(ii)

(iii) H – F – –H – F – –H – F – –H – F – –

(iv) None

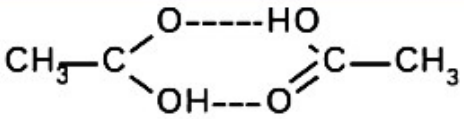
Sol. F has 3 lone pair hence linear structure around F is not possible. Hence correct option is (i).

Q.11 **Heat of vapourisation of water is higher than HF. However strength of H-Bond in HF is higher than water.**

Sol. H_2O is surrounded by 4 other H_2O molecules whereas HF is surrounded by only 2 molecules. Due to this, extent of hydrogen bonding in H_2O is more, hence Heat of vapourisation of water is higher.

Q.12 **Experimental molecular weight of acetic acid is just double than theoritical molecular weight. Why ?**

Sol.



Observed molecular weight = 120
Due to intermolecular, H–Bonding

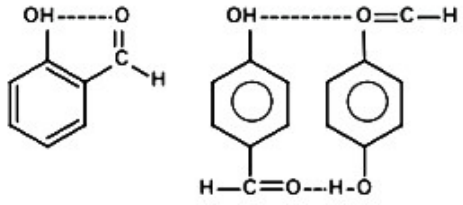
Q.13 **KHF_2 exists but $KHBr_2$ & KHI_2 does not. Why ?**

Sol. $KHF_2 \rightarrow K^+ + HF_2^-$ [F^-..... H–F] H-bonding

$KHBr_2 \rightarrow K^+ + HBr_2^-$ [not stable]
$KHI_2 \rightarrow K^+ + HI_2^-$ [not stable]
} No hydrogen bonding

Q.14 **o-hydroxybenzaldehyde is liquid while p-hydroxybenzaldehyde is solid at room temperature? Why?**

Sol.



Intramolecular H-bonding

Intermolecular H-bonding

Due to intramolecular hydrogen bonding in p-hydroxybenzaldehyde.

2. Molecular forces :

Strength : ion-ion attraction forces > ion-dipole attraction forces > dipole-dipole attraction forces > ion-induced dipole attraction forces > dipole induced dipole attraction forces > Instantaneous dipole induced dipole attraction forces

Weak intermolecular attraction forces :

(1) Ion-dipole attraction

Polar molecules are attracted by ions. The -ve pole is attracted by cation and +ve pole attracted by the anion.



Polar

Strength depend on

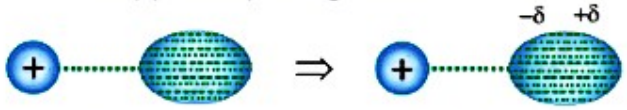
(i) Charge density of ion

(ii) Polarity of polar molecule

$(Li^+ + H_2O)$, $(Cl^- + HCl)$, $(NO_3^- + CH_3OH)$, $(KCl + H_2O)$

(2) Ion-induced dipole attraction

When non polar molecules come in contact with ions, its electron cloud gets polarized and the oppositely charged end of it is attracted by the ion



$Cl^- + Cl_2$ $\qquad$ $K^+ + \phi$

$Br^- + CCl_4$

$NO_3^- + I_2$

3. Van der Waal's forces :

These are the weakest type of forces that happen amongst the chemical species which bring a considerable change in physical properties.

⇒ These are non-valence cohesive forces, non-directional. These attractive forces being played among the 2 molecules, are independent of the presence of other molecules.

⇒ Solid, liquid or gaseous states of many molecules are described on the basis of inter molecular forces other than metallic, covalent, ionic bonds. Although inert gases do not form any type of bond but may exist in liquid and solid states. This shows that the atoms of inert gases are attracted by each other through some type of intermolecular forces. These intermolecular forces are called Vander Waals forces.

1. Dipole-dipole attraction forces (Keesom forces)

The oppositely charged poles of two polar molecules (for example: H_2S, HCl, PH_3 etc.) is known as dipole-dipole attraction.



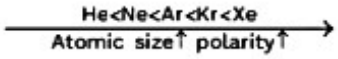
2. Dipole induced-dipole attraction forces

This is force occurs in a mixture of polar and non-polar molecules. For example, force of attraction between Cl_2 and H_2O.

- Size of non polar molecules increases then interaction between molecules increases.



Debye force

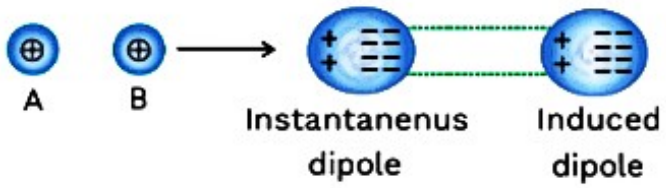


- Solubility of noble gases in water and formation of noble gas hydrates like $Ar.6H_2O$, $Xe.6H_2O$
- He and Ne do not form clathrate compound.

3. Instantaneous dipole induced dipole attraction forces. / London dispersion forces

The weakest intermolecular forces operating in similar non polar gaseous molecules are known as London forces. These forces are very weak in nature and exists only at low temperature. For example, weak intermolecular forces in F_2, Cl_2, N_2, molecules and in nobal gases.

Example : Non-polar atoms, molecule and compound like, He, Ne, Ar, Kr, Xe, F_2, Cl_2, Br_2, I_2, and Benzene, CCl_4, CO_2, CS_2 etc.



Q.15 **Dispersion force is present between**

(A) polar + polar **(B) non polar + non polar**

(C) polar + non polar **(D) all**

Sol. (D)

Q.16 **Which of the following order of density is incorrect ?**

(A) $H_2O(s) < H_2O(l)$ **(B) $Ga(s) < Ga(l)$**

(C) $Bi(s) < Bi(l)$ **(D) $D_2O(s) < D_2O(l)$**

(E) $D_2O(s) < H_2O(l)$

Sol. (E) Mass dominant on volume

Strong Camical Bond

1. Metallic Bond

t is given by drude and explained by Loraze, Bond theory : Based on MOT

$\oplus e^-$	$\oplus e^-$	$\oplus e^-$
$\oplus e^-$	$\oplus e^-$	$\oplus e^-$

Metallic bond strength :

In a group :

$$\text{Metallic bond strength} \propto \frac{1}{\text{Size}}$$

Li > Na > K > Rb > Cs

In a period :

$$\text{Metallic bond strength} \propto \text{Number of electron participating in metallic bond}$$

For d-block

In group $\text{Metallic bond strength} \propto Z_{eff}$

3d < 4d < 5d (Heat of atomisation)

$$\text{Softness} \propto \frac{1}{\text{metallic bond strength}}$$

Softest s-block metal → Cs

Hardest s-block metal → Be

→ Metals are malleable and ductile

→ Metallic lustre is due to oscillation of surface mobile electron by absorption of light energy.

→ Metals are good conductors of heat and electricity
Conductivity order ⇒ Ag > Cu > Au > Al

Q.17 **The conductivity of metal is determined by**
(a) the electronic concentration and the mobility of the free electrons
(b) the number of valence electrons per atom
(c) both (a) and (b) none of these.

Sol. : (c)

The conductivity of metal is determined by the electronic concentration and the mobility of free electrons and the number of valence electrons per atom.

Q.18 **Metallic solids are characterised by .....**
(b) low conductivity of heat and electricity **(c) low melting point**
(d) paired electrons **(e) bright lustre.**

Sol. : (d)

Metallic solids are characterised by bright lustre, hardness, high melting point, ability to resonate sound and excellent conductivity of heat and electricity.

Q.19 **.Metals are malleable because when struck, one plane of metal atoms**
(a) can slide past another plane without breaking bonds
(b) cannot easily move out of the way
(c) moves in a way that maximizes the repulsive forces within the metal
(d) bonds to the plane directly beneath it.

Sol. : (a)

Metals are malleable because when struck, one plane of metal atoms can slide past another plane without breaking bonds.

Q.20 **In metallic crystal, the**
(a) **kernels as well as the electrons move rapidly**
(b) **electrons float freely around metal atom**

(c) **electrons are immobile**
(d) **valence electrons are not available.**

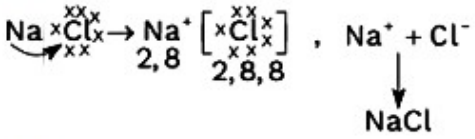
Sol. : (b)

Valence electrons float freely around the metal atom in metallic solid.

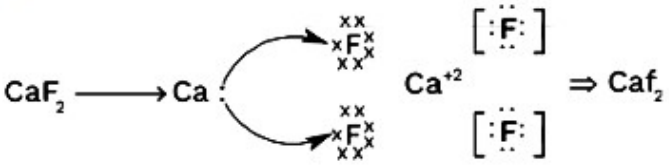
2. Ionic bond/electrovalent bond

Ionic bond formed between metal and non metal.
Ionic bond is a electrostatic force of attraction between cation and anion which have been created by complete removal and acceptance of electron respectively during its formation.

Example :



Example :



Example :

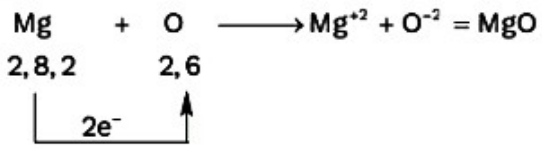
Ionic compound of Li, Be, Mg, Al,
LiF, Li_2O, Li carbides, LiH
MgF_2 MgO Mg_2C_3
AlF_3 Al_2O_3 Al_4C_3
Be_2C

- Metals salt having poly atomic anion are ionic in nature.
  Electrovalency : Number of electron donate and gain to achieve inert gas configuration.

$Na \rightarrow 1$
$Mg \rightarrow 2$
$Al \rightarrow 3$

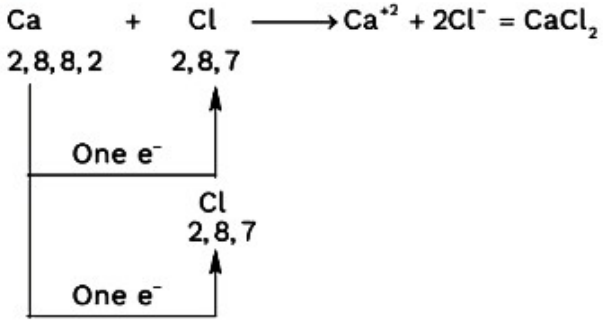
$\begin{bmatrix} F & 2,7 \rightarrow 1 \\ O & 2,6 \rightarrow 2 \\ N & 2,5 \rightarrow 3 \end{bmatrix}$ For non metal

Example :



electrovalency of Mg = 2
electrovalency of O = 2

Example :



electrovalency of Ca = 2
electrovalency of Cl = 1

Q.21 **The compounds containing electrovalent bonds are called**
(a) covalent compounds **(b) ionic compounds**
(c) coordinate compounds **(d) complex compounds.**

Sol. : (b)
The compounds containing electrovalent bonds are called electrovalent compounds or ionic compounds.

Q.22 **In electrovalent compounds, the electrovalency is equal to the**
(a) number of unit charges on the ion **(b) number of p-orbital electrons.**
(c) period number **(d) none of these.**

Sol. : (a)
Electrovalency is equal to the unit charges on the ion.

Q.23 **In the formation of ionic bond in KNO_3, how many electrons gets transferred by potassium atom?**

(a) 2 **(b) 1** **(c) 3** **(d) 0**

Sol. : (b)

In formation of ionic bond in KNO_3, K loses only one electron to get inert gas stability.

Q.24 **In the formation of NaCl, the type of force generated in this ionic compound is**

(a) van der Waals' forces **(b) hydrogen bonding**

(c) electrostatic attraction **(d) none of these.**

Sol. : (c)

In formation of NaCl, Na atom gains one positive charge and chlorine gains one negative charge. The force which attract the opposite ions together is electrostatic force.

Q.25 **The easec of atom to form ionic bond depends on**

(a) high ionisation enthalpy **(b) low ionisation enthalpy**

(c) low electron gain enthalpy **(d) both (a) and (c).**

Sol. : (b)

Lesser the ionisation enthalpy of an atom, greater is the ease of losing the valence electron.

Q.26 **All alkaline earth metals have _____ valence electrons and form ionic bond by gaining _____ charge.**

(a) 2, +2 **(b) 1, +1** **(c) 2, −2** **(d) 1, −1**

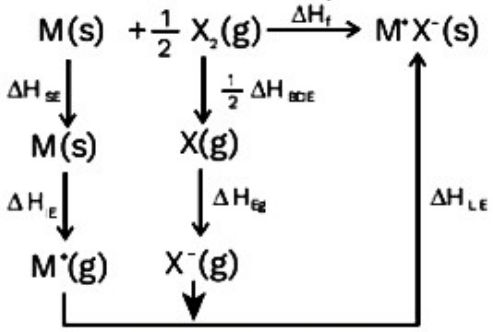
Sol. : (a)

All the alkaline earth metals have two electrons in their valence shell, so they lose two electrons to form cations with + 2 charge.

Energy steps involved in ionic bond formation :

Born Haber's Cycle

A whole series of energy changes are involved when one starts from the elements and finishes with an ionic crystal. These changes are shown in the Born-Haber cycle.



here, SE = heat of sublimation of Na(s)

IE = ionisation enthalpy of Na(g)

BDE = bond dissociation energy of $Cl_2(g)$

EG = electron gain enthalpy of Cl(g)

LE= lattice energy

ΔH_f = heat of formation of NaCl(s)

Hess Law :

$$\Delta HF = \Delta H_{SE} + \Delta H_{IE} + \frac{1}{2}\Delta H_{BDE} + \Delta H_{Eg} + \Delta H_{LE}$$

ΔH_f = -ve $\quad\quad$ $\Delta H_f < 0$

$$\Delta H_f = \underbrace{\Delta H_{SE} + \Delta H_{IE} + \frac{1}{2}\Delta H_{BDE}}_{\text{Required energy}} + \underbrace{(\Delta H_{eg} + \Delta H_{LE})}_{\text{Released energy}}$$

Released energy > Required energy

- It is a necessary condition for formation of ionic bond

Favorable condition for ionic bond formation :

(i) Low S.E. of metal

(ii) Atom forming cation should have low ionisation energy

(iii) Atom forming anion should have high electron gain enthalpy

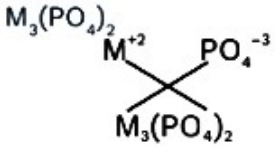
(iv) LE(released amount of LE) should be high

Q.27 **Select the correct formula of magnesium oxide :**
(A) MgO **(B) Mg_2O_2** **(C) MgO_2** **(D) Mg_2O**

sol. (A)

Q.28 **If the formula of a metal sulphuric is MSO_4 then what could be the formula of metal phosphate.**

sol.



Q.29 **Lattice energy is inversely proportional to sum of radii of**
(a) anion **(b) cation** **(c) ions** **(d) both (a) and (b).**

Sol. : (d)
Lattice energy is inversely proportional to interatomic distance of anion and cation.

Q.30 **Which of the following statements is incorrect regarding lattice enthalpy?**
(a) Lattice enthalpy becomes more negative as interionic distance decreases.
(b) Lattice enthalpy is directly proportional to the product of ionic charges.
(c) Crystal with high lattice enthalpy melts at low temperature usually.
(d) None of these

Sol. : (c)
Usually crystal with high lattice enthalpy melts at high temperature.

Q.31 **Born–Haber cycle is an energy cycle for**

(a) aqueous solution **(b) solid compounds**

(c) ionic compounds **(d) gaseous compounds.**

Sol.: (c)

Born–Haber cycle is used to calculate the lattice energy of the ionic compound.

Q.32 **Select the incorrect statement from the following options.**

(a) Every pure solid crystalline substance has a characteristic and unique melting point.

(b) Impure sample of substance has different melting point.

(c) Two different pure substances have same melting points.

(d) Melting point serves as the criteria of purity of a solid substance.

Sol.: (c)

Two different pure substances have different melting points. All the other options are correct. Every pure solid crystalline substance has a characteristic and unique melting point and impure sample of substance has different melting point. Melting point serves as the criteria of purity of a solid substance.

Q.33 **The correct order regarding the solubility of alkaline earth metal halides in water is**

(a) $BeCl_2 < MgCl_2 < CaCl_2 < SrCl_2 < BaCl_2$

(b) $MgCl_2 > BeCl_2 > BaCl_2 > SrCl_2 > CaCl_2$

(c) $BaCl_2 > MgCl_2 > CaCl_2 > BeCl_2 > SrCl_2$

(d) $BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2 > BaCl_2$

Sol. : (d)

Solubility decreases down the group of alkaline earth metal halides in water. So, correct order here is $BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2 > BaCl_2$.

Q.34 **The ionic radius of cation will be smaller and more efficient in polarizing**

(a) carbonate ion **(b) carbonate molecule**

(c) hydrogen ion **(d) water.**

Sol. : (a)

The ionic radius of cation will be smaller and more efficient in polarizing carbonate ion.

Q.35 **So, $y - x = 6 - 2 = 4$ The following compounds have been arranged in order of their increasing thermal stabilities.**
Identify the correct order K_2CO_3 (I) $MgCO_3$ (II) $CaCO_3$ (III) $BeCO_3$ (IV)
(a) I < II < III < IV **(b) IV < II < III < I**
(c) IV < II < I < III **(d) II < IV < III < I**

Sol. : (b)

On increasing the size of alkaline earth metal ions, stability of their carbonate salts increases due to increasing polarity. Alkali metal carbonates are more stable than carbonates of alkaline earth metals due to their higher polar characters, Thus, correct order of stability $BeCO_{3(s)} < MgCO_{3(s)} < CaCO_{3(s)} < K_2CO_{3(s)}$ *i.e.*, IV < II < III < I.

Q.36 **Which of the following is arranged in order of increasing thermal stability?**
(a) $LiClO_3 > NaClO_3 > KClO_3 > RbClO_3 > CsClO_3$
(b) $LiClO_3 < NaClO_3 < KClO_3 < RbClO_3 < CsClO_3$
(c) $RbClO_3 > CsClO_3 < LiClO_3 < NaClO_3 < kClO_3$
(d) $LiClO_3 < NaClO_3 < CsClO_3 < RbClO_3 < kClO_3$

Sol. : (b)

Large cation stabilised the large anion. Thus, thermal stability increases down the group.

- Stability of ionic compound depend on lattic energy, ΔH_f for alkyl halide

LiF > NaF > KF > RbF > CsF

ΔH_{LE} = –ve

MF > MCl > MBr > MI

ΔH_{LE} = –ve

LiX < NaX < KX < RbX < CsX

[X = Cl | Br | I]

Properties of Ionic compound :

1. Ionic compound are non-directional in nature.
2. Due to non-directional nature of bond ionic compound do not show stereoisom erism
3. Due to strong electrostatic attraction ionic compound exist in the form of solid So, these have high melting point.
4. Ionic compound show conductivity in molten and aqueous medium due to presence of free ions but not in solid state.
5. Ionic compound are more soluble in polar solvent.
6. The reaction of ionic compound is fast.
7. Ionic compound show isomorphism.
8. Ionic compound have crystalline structure.
9. These are hard and brittle in nature.
10. In ionic compound, only ions are present not atoms.

Properties :

Isomorphism : If two ionic compounds have same crystalline structure then they are isomorphous.

Condition :

- Both ionic compounds have same number of ions.
- Both cations should be isoelectronic or have same shape (Hybridisation)

OR

If ionic compounds have crystalline water then they should have same general formula.

Q.37 **Pair of compounds which are not isomorphous**

(A) NaF and MgO **(B) $NaNO_3$ and $MgCO_3$**

(C) $KMnO_4$ and $BaSO_4$ **(D) KI_3 and TlI_3**

(E) Na_2SO_3 and $MgCO_3$

Sol. (E)

Number of ions different.

Q.38 **Which of the following is not isomorphous with others?**

(A) White vitriol → $ZnSO_4.7H_2O$ **(B) Epsum salt → $MgSO_4.7H_2O$**

(C) Green salt → $FeSO_4.7H_2O$ **(D) Washing salt → $Na_2CO_3.10\ H_2O$**

Sol. (D)

Lattice energy :

Amount of energy required to break one mole ionic lattice into its gaseous cation and anion is called lattice energy and vice-versa also defined as lattice energy.

$$NaCl(s) \xrightarrow{\text{Endothermic}} Na^+(g) + Cl^-(g)$$

$$\Delta H_{LE} = +ve$$

$$Na^+(g) + Cl^-(g) \xrightarrow{\text{Exothermic}} Na^+Cl^- + \text{Energy}$$

(solid)

$$\Delta H_{LE} = -ve$$

$$\text{Lattice energy} \propto \frac{q_{e^+} \cdot q_{e^-}}{(r_{c^+} + r_{a^-})}$$

Where K is pocking constant

Lattice energy depends upon following three factors :

1. Change on ions
2. Size of ions
3. Packing of cations and anion (Data based)

- Charge factor more dominant factor

Example : $NaF < MgF_2 < AlF_3$

$LiF < Li_2O < Li_3N$

$LiF > NaF > KF > RbF > CsF$

$KF > KCl > KBr > KI$

$NaF < MgO$

Hydration Energy :

Amount of energy released when one mole of gaseous cation and anion are completely dissolved in water is called hydration energy.

(a) Always carries negative sign

$$Na^+ + xH_2O \longrightarrow \dot{N}a\ (H_2O)_x \text{ or } Na^+(aq) + \text{energy}$$

$$\Delta H_{HE} = -ve$$

$$Cl^-(g) + yH_2O \longrightarrow Cl^-(H_2O)y \text{ or } Cl^- + \text{energy}$$

$$\Delta H_{HE}(A^-) = -ve$$

$$\text{Net hydration energy} = -(\Delta H_{HE}{}_{C^+} + \Delta H_{HE}{}_{A^-})$$

$$\boxed{\Delta H_{HE}{}_{C^+} > \Delta H_{HE}{}_{A^-}}$$

Hydration energy of ion $\propto$ charge density of ions

$$\text{Charge density} = \frac{\text{Charge on ion}}{\text{Size of ions}}$$

$$HE = \frac{q_{c^+}}{r_{c^+}} + \frac{q_{A^\ominus}}{r_{A^\ominus}}$$

In hydration energy, cation contribute more than anion due to small size.
Hydrated size of $Li^+(aq)$ > Hydrated size of $Na^+(aq)$

$$\text{Ionic mobility of ions} \propto \frac{1}{\text{Hydrated size of ions}}$$

- **Order of size**
  - **(i)** $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$
  - **(ii)** $Li^+(aq) > Na^+(aq) > K^+(aq) > Rb^+(aq) > Cs^+(aq)$
  - **(iii)** $F^-(aq) > Cl^-(aq) > Br^-(aq) > I^-(aq)$
  - **(iv)** $F^- < Cl^- < Br^- < I^-$

- **Ionic molality of ions in aqueous solution**
  - **(i)** $Li^+(aq) < Na^+(aq) < K^+(aq) > Rb^+(aq) > Cs^+(aq)$
  - **(ii)** $F^-(aq) < Cl^-(aq) < Br^-(aq) < I^-(aq)$
  - **(iii)** $Al^{+3}(aq) < Mg^{+2}(aq) < Na^+(aq)$

Electrical conductivity of ions :
In aqueous solution, It depends upon following two factors
(i) Charge carrying capacity of ions
(ii) Ionic mobility of ions
$Li^+(aq) < Na^+(aq) < K^+(aq) < Rb^+(aq)$
$F^-(aq) < Cl^-(aq) < Br^-(aq) < I^-(aq)$
$Na^+(aq) < Mg^{+2}(aq) < Al^+(aq)$

- Charge carrying capacity is given by preference over ionic mobility.

Alum : $(M_2SO_4.M_2'(SO_4)_3.24H_2O$
Where,
M is monovalent cation
M' is trivalent cation (Cr^{+3}, Al^{+3}, Fe^{+3})
$K_2SO_4.Al_2(SO_4)_3.24H_2O$ (Potash alum)
$K_2SO_4.Cr_2(SO_4)_3.24H_2O$ (Crome alum)
All alums are isomorphous

- Aqueous solution of alum is acidic in nature

$$\underset{}{K_2SO_4} \xrightarrow{H_2O} \underset{\text{S.B.}}{2KOH} + \underset{\text{S.A}}{H_2SO_4} \qquad \text{Neutral}$$

$Al_2(SO_4)_3 \xrightarrow{H_2O} Al(OH)_3 + H_2SO_4$ Acidic

S.B. S.A

Species present in aq. solution/solid state of alum are

$2[M(H_2O)_6]^+ + 2(M(H_2O)_6]^{+3} + HSO_4^{2-}$

Ion-dipole attraction ⇒ distance↓ ⇒ M has vacant orbital ⇒ coordinate bond ⇒ 10 layer of incoming H_2O ⇒ Hydrated ion

- Li does not form alum as its covalency = 4.
- On heating alum converts into amorphous solid (dead alum) due to removal of water molecule.

- Pseudo alum : $MSO_4.M_2'(SO_4)_3.24H_2O$ bivalent

Q.39 **Which of the following show conductivity**

(1) AlF_3(molten) **(2) $AlCl_3$** **(3) HCl(l)** **(4) Graphite**
(5) Na(s) **(6) H_2SO_4(aq)** **(7) $NaNO_3$** **(8) KNO_3(aq)**
(9) $MgCl_2$(aq) **(10) $CuSO_4.5H_2O$**

Sol. **(7)** AlF_3, Graphite, Na(s), H_2SO_4, KNO_3, $MgCl_2$, $CuSO_4.5H_2O$

Q.40 **Solubility of KI is maximum in**

(A) CH_3OH (Er = 34)
(B) CH_3-O-CH_3 (Er = 4.1)
(C) C_2H_5-OH (Er = 27)
(D) $H_3C-\overset{\overset{O}{\|}}{C}-CH_3$ **(Er = 21)**

Sol. **(A) CH_3OH**

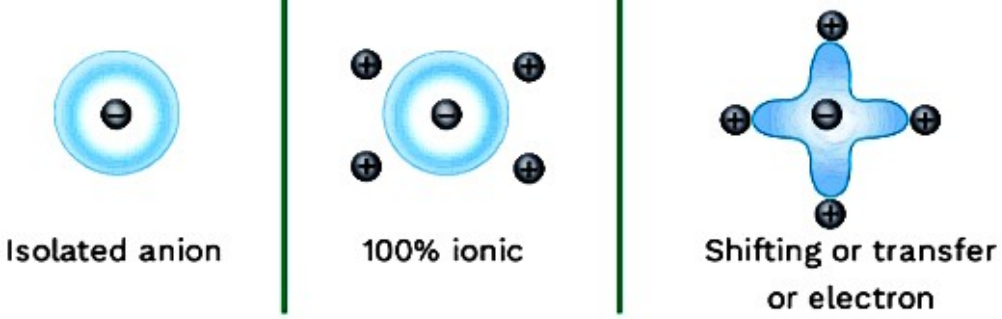
Solvent	HF	H_2SO_4	H_2O_2	H_2O	D_2O
Er	120	102	98	81	79
	Acidic	Oxidising		Universal	

Polarisation :

- No compound exists as 100% ionic
- No bond is 100% ionic. It has some covalent character.
- Calculation of covalent character in ionic bond is practically impossible. So, they can be compared by phenomena of polarisation.

Polarisation (Covalent character in ionic bond)

When two opposite charge attract to each other than ionic bond is formed and cation attract electron cloud of anion. So, symmetrical anion become unsymmetrical distortion of anion by cation is called polarisation due to polarisation electron are shifted toward cation. So, ionic character decreases, covalent character increases.



Sharing of electron ↑, ionic character ↓, Covalent character ↑

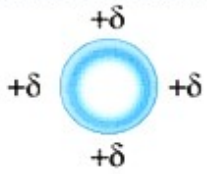
Fajan's Rules

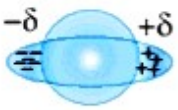
All the covalent bonds have some partial ionic character, the ionic bonds also have partial covalent character. The partial covalent character of ionic bonds was discussed by Fajan's rules:

- The smaller the size of the cation and the larger the size of the anion, the greater the covalent character of an ionic bond.
- The greater the charge on the cation, the greater the covalent character of the ionic bond.
- For cations of the same size and charge, the one, with electronic configuration $(n-1)d^x ns^0$, typical of transition metals, is more polarising than the one with a noble gas configuration, $ns^2 np^6$, typical of alkali and alkaline earth metal cations.
- The cation polarises the anion, pulling the electronic charge toward itself and thereby increasing the electronic charge between the two. This is precisely what happens in a covalent bond, i.e., buildup of electron charge density between the nuclei. The polarising power of the cation, the polarisability of the anion and the extent of distortion (polarisation) of anion are the factors, which determine the per cent covalent character of the ionic bond.
- Polarisation power of a cation is usually called ionic potential or charge density.

$$\text{Ionic potential } \phi \text{ (phi)} = \frac{\text{Charge on cation}}{\text{Size of cation}}$$

1. Size of cation :





Polarisation of electron cloud of anion by cation

Cation Anion

Example : CsCl < RbCl < KCl < NaCl < LiCl

$$\text{Size of cation} \propto \frac{1}{\text{Polarisation}}$$

$$\text{Polarisability} \propto \text{Size of anion}$$

2. Size of anion :

Ability of anion to polarise towards cation is called polarisability.

$$\text{Polarisability} \propto \text{Size of anion}$$

Polarization $\propto$ Charge on anion

Example :

NaF < NaCl < NaBr < NaI

$CaF_2 < CaCl_2 < CaBr_2 < CaI_2$

$PbF_2 < PbCl_2 < PbBr_2 < PbI_2$

AgF < AgCl < AgBr < AbI

$$\xrightarrow[\text{Size of anion}\uparrow,\text{C.C.}\uparrow]{}$$

$F^- < Cl^- < Br^- < I^-$

$$\xrightarrow[\text{Order of polarisability}]{}$$

3. Charge density of cation

$$\text{Ionic potential}(\phi) = \frac{\text{Chare}}{\text{Size}}$$

Polarising power $\propto \phi \propto$ covalent character

Period $\rightarrow \phi \uparrow$ covalent character$\uparrow$

Group $\rightarrow \phi \downarrow$ covalent character $\downarrow$

In period
Example :

(i) $\overset{+}{Li}Cl < \overset{+2}{Be}Cl_2 < \overset{+3}{Be}Cl_3 < \overset{+4}{C}Cl_4$ → ($\phi\uparrow$ covalent character$\uparrow$)

(ii) $\overset{+1}{Na}Cl < \overset{+2}{Mg}Cl_2 < \overset{+3}{Al}Cl_3 < \overset{+4}{Si}Cl_4$ → ($\phi\uparrow$ covalent character$\uparrow$)

In group :
Example

(i) $LiCl < NaCl < KCl < RbCl < CsCl$ → ($\phi\downarrow$, covalent character$\downarrow$ Ionic coharacter$\uparrow$)

(ii) $BeCl_2 < MgCl_2 < CaCl_2 < BaCl_2$ → ($\phi\downarrow$, covalent character$\downarrow$ Ionic coharacter$\uparrow$)

$$\text{Ionic character} \propto \frac{1}{\text{Covalent character}}$$

4. Pseudo Inert gas configuration :
$ns^2np^6nd^{10}$

Example :
Au^+, Bi^{+5}, Pb^{+4}, Sn^{+4}, Ag^+, Cu^+, Zn^{+2}, Cd^{+2}, Hg^{+2}

AgCl > KCl
AuCl > RbCl
] order of covalent character

Cu^+ $\phi = 0.96$ CuCl > Na^+ $\phi = 0.95$ NaCl
2, 8, 182, 8

$Cu^+ = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$
$(ns^2\ np^6\ nd^{10})$

5. Non inert gas configuration :
$ns^2\ np^6\ nd^{10}\ (n+1)s^2$

Example : Tl^+, Pb^{+2}, Bi^{+3}
$PbCl_2 > MgCl_2$
$BiCl_3 > CaCl_2$

← Covalent Character$\uparrow$

For d-block : $\xrightarrow[Z_{eff}\uparrow\ P.P\uparrow]{d^5<d^6<d^7<d^8<d^9}$

$\xrightarrow[Z_{eff}\uparrow\ P.P\uparrow]{3d<4d<5d}$

$\xrightarrow[P.P.\uparrow]{4^{th}\text{ period} < 5^{th}\text{ period} < 6^{th}\text{ period}}$

Pseudo > Noninert > inertgas → Covalent

Example :

(i) $Zn^{+2} < Cd^{+2} < Pb^{+2}$

(ii) $Hg^{+2} > Pb^{2}$

(iii) $Zn^{+2} < Cd^{+2} < Hg^{+2}$

(iv) $Cu^{+} < Ag^{+} < Au^{+}$

(v) $Zn^{+2} < Cd^{+2} < Hg^{+2}$

(vi) $Tl^{+} < Pb^{+2} < Bi^{+3}$

Some order of polarising power :

(i) Be^{+2} Al^{+3}

C.C.↑ P.P↑

Charge <

Size <

P.P. <

(ii) Li^{+} Mg^{+2}

Size >

Charge <

P.P. <

Polarisability ∝ Charge on anion ∝ size of anion

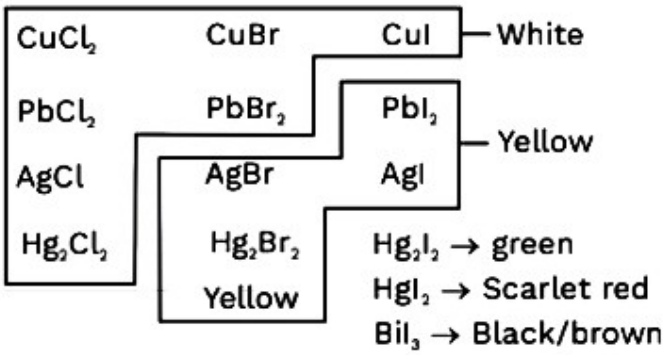
$F^{-} < Cl^{-} < Br^{-} < I^{-}$ $P^{-3} > S^{-2} > Cl^{-}$

$F^{-} < O^{-2} < N^{-3}$

- Important order

$P^{-3} > S^{-2} > O^{-2} \simeq I^{-} > Br^{-} > Cl^{-} > F^{-}$

As polarisation increases transition of electron from anion to cation↑ intensity↑ darkness ↑

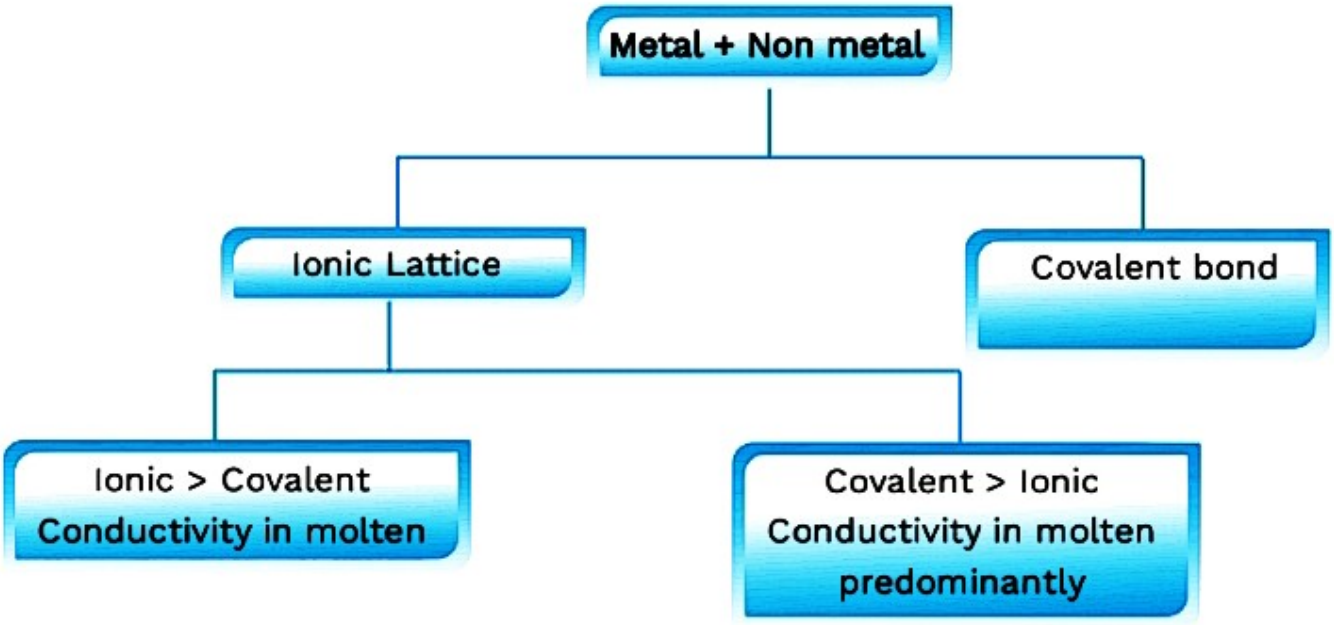


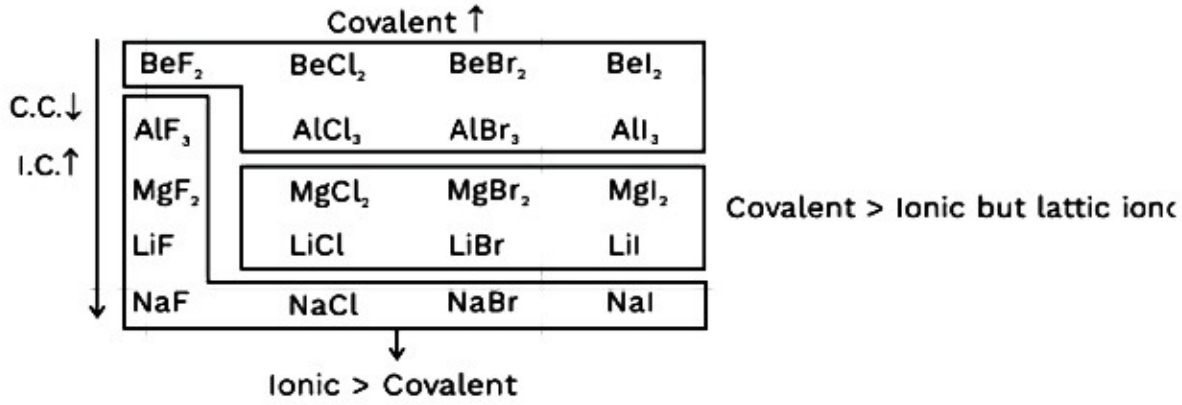
- All phosphide are in black colour
- Mostly sulphide are black coloured
  Cds → yellow ZnS → White
  MnS → Pink

As^{+3}	Sb^{+3}	Sn^{+2}	Sn^{+4}
Yellow	Orange	Brown	Yellow

Application of polarisation :

1. **Covalent character/ionic character :**
   As polarisation ↑, Sharing tendency ↑ , Covalent character ↑ , Ionic character ↑





Q.41 **The degree of polarization depends on**

(a) charge density of cations only

(b) charge density of anions only

(c) charge density of cations and polarizability of the anion

(d) all of these.

Sol. : (c)

Polarizing power depends on the tendency of cation to polarize and the polarizability of the anion.

Q.42 **Many ionic compounds have some covalent characters due to**

(a) ion polarization **(b) charge polarization**

(c) proton polarization **(d) electron polarization.**

Sol. : (a)

As the charge on the ion increases, the electrostatic attraction of the cation for the outer electrons of the anion also increases, with the result its ability for forming the covalent bond increases.

Q.43 **Even though the crystal radii of both K^+ and Ag^+ are almost same, the melting point of AgCl is only 455°C, while that of KCl is 776°C. The best explanation is**

(a) AgCl has more ionic character than KCl due to greater polarizing power of potassium ion

(b) KCl has more ionic character than AgCl due to greater polarizing power of potassium ion

(c) AgCl has more covalent character than KCl due to greater polarizing power of silver ion

(d) KCl has more covalent character than AgCl due to greater polarizing power of silver ion.

Sol. : (c)

The K^+ ion has octet configuration in the outer shell ($3s^2 3p^6$). Therefore, it has less polarizing power and hence KCl is more ionic. Whereas, in Ag^+ ion, there is pseudo octet configuration ($4s^2 4p^6 4d^{10}$) in its outer shell. Therefore, it has greater polarizing power due to ineffective shielding of nuclear charge by *d*-electrons.

Q.44 **The order of covalent nature for aluminium halides is $AlF_3 < AlCl_3 < AlBr_3 < AlI_3$. The best explanation according to Fajans' rule is**

(a) Al^{3+} is larger in size and hence it has greater polarizing power

(b) the polarizability of halide ions increases with increase in their size and hence the covalent nature also increases

(c) the polarizability of halide ions decreases with increase in their size and hence the covalent nature decreases

(d) the given order of covalent nature is due to greater polarizing power of Al^{3+} ion.

Sol. : (b)

The polarizability of halide ions increases with increase in their size and hence, the covalent nature also increases.

Q.45 **The correct order of covalent nature of alkali metal chlorides is**

(a) LiCl < NaCl < KCl < RbCl < CsCl **(b) CsCl < NaCl < KCl < RbCl < LiCl**

(c) LiCl > NaCl > KCl > RbCl > CsCl **(d) LiCl = NaCl = KCl = RbCl < CsCl**

Sol. : (c)

The covalent nature decreases with increase in the size of cation since the polarizing power decreases with increase in the size.

2. Thermal stability :

Stability towards heat/temperature compound having high thermal stability means high temperature required to decompose.

(a) T.S of compound having only covalent bonds

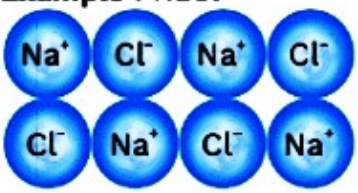
Thermal stability ∝ BE × number of covalent bonds

	NH_3		PH_3	
BE		>		
T.S		>		
	PH_3		H_2S	HCl
BE		<		<
T.S		>		>

(b) T.S of ionic compound having mono atomic anion and cation

CO^{-2}, S^{-2}, F^-, Cl^-, Br^-, I^-, H^-, N^{-3}, P^{-3})

Example : NaCl



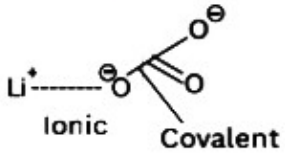
Only ionic bond present

Thermal stability ∝ Lattice Energy

$$\underset{\text{T.S}\uparrow\ \text{L.S}\uparrow}{\xleftarrow{\quad}\ LiH > NaH > KH > RbH > CsH}$$

(C) T.S of ionic compound having polyatomic anion (OH^-, O_2^{2-}, O_2^-, N_3^-, CO_3^{2-}, Oxyanion)

Li_2CO_3



$$Li_2CO_3 \xrightarrow{\Delta} Li^+ + CO_3^{2-}$$

$$\xrightarrow{\Delta} Li_2O + CO_2$$

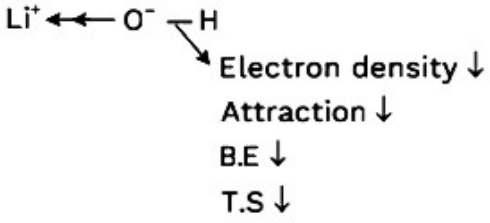
LiOH

$$< K^+ \quad < \quad Cl^-$$

$$\downarrow \qquad\qquad \downarrow$$

isoelectronic Larger

Thermal stability ∝ $\frac{1}{\text{Polarising power}}$

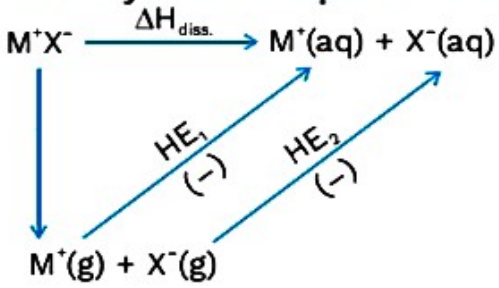


ΔEN ↑ Charge ↑ B.S ↑ T.S ↑
0.9 1.4 1.9

- $LiHCO_3$ and II A bicarbonates donot exist in solid state (exist and solution)
- $BeCO_3$ is kept in CO_2 atmosphere due to less T.S.

3. Solubility :

Solubility of ionic compound in water :



$\Delta H_{diss.} = \underbrace{\Delta H_{LE}}_{+} + \underbrace{\Delta H_{HE}}_{-}$

Required Released

($\Delta H_{diss.}$ = –ve)

$\Delta G = \Delta H - T\Delta S$

ΔG = –ve, ΔH = –ve, ΔS = +ve

ΔG = +ve, ΔH = +ve, L.E >> HE

$\boxed{\text{Solubility} \propto \frac{HE}{LE}}$ Mainly s – block compounds

Note (1)

If common ion is smaller like F^-, O^{-2}, OH^-, Li^+, Na^+, Ba^{+2}
II A cation → L.E. → dominant

$\boxed{LE \downarrow \ S \uparrow}$

Note (2)

If common ion is larger like I^-, Br^-, Cs^+, Rb^+, Polyatomic anions
HE → dominant

Note 3

$$\text{Solubility} \propto \frac{1}{\text{Polarisation / covalent character}}$$

Applicable for Be-halide, p and d-block compounds

Note 4

Solubility decide by ionic character
IA → Carbonate, sulphate bicarbonate

Note 5

Solubility ion organic μ polarisation or C.C

Examples :

(i) $LiF < NaF < KF < RbF < CsF$ (LE)
(ii) $BeCO_3 > MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$
(iii) $BaF_2 < BaCl_2 < BaBr_2 < BaI_2$ (LE)
(iv) $BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2 > BaCl_2$
(v) $BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$
(vi) $Li_2O < Na_2O < K_2O < Rb_2O < Cs_2O$
(vii) $LiClO_4 > NaClO_4 > KClO_4 > RbClO_4 > CsClO_4$
(viii) $Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2$
(ix) $BeF_2 > BeCl_2 > BeBr_2 > BeI_2$
(xi) $AgF > AgCl > AgBr > AgI$
(x) $PbF_2 > PbCl_2 > PbBr_2 > PbI_2$
(xi) $HgF_2 > HgCl_2 > HgBr_2 > HgI_2$

(xii) $Na_2S > CuS < ZnS$

s-block P.P ↓ Cation ↑
Solubility ↑ pol.↓
Sol. ↑

$$\boxed{Na_2S > ZnS > CuS}$$

LiCl > NaCl in acetone/pyridine
covalent↑ Ionic↑
$Ag_2O > Ag_2S$
$NaF < SnCl_4$
Na^+ small ⇒ LE → ON
size↑ LE↓

$SaCl_2 > SnCl_4$
$Fe(OH)_2 > Fe(OH)_3$
charge ↑ pol. ↑ solution ↓

(xiii) $MgF_2 < MgCl_2 < MgBr_2 < MgI_2$

(xiv) $LiCl > NaCl > KCl < RbCl < CsCl$

Cl^- larger Cl^- Smaller

Important points regarding solubility :

(i) All alkali metal salts are water soluble except LiF, Li_2CO_3, $Li_2C_2O_4$, Li_3PO_4

(ii) All metal nitrates are water soluble.

(iii) All metal sulphates are water soluble except $SrSO_4$, $BaSO_4$, $HgSO_4$, Ag_2SO_4, $PbSO_4$

(iv) s-block metal sulphides are water soluble, rest are insoluble.

(v) II A carbonate are insoluble in water

(vi) IA hydroxide and $Ba(OH)_2$ are water soluble. Rest hydroxide are insoluble or sparingly soluble.

(vii) Generally, metal halides are water soluble except Cl^-, Br^-, I^- of Ag, Hg, Pb

Q.46 **Which of the following is least soluble in water.**

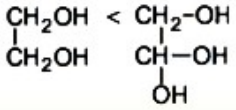
(A) Na_2CO_3 **(B)** KNO_3 **(C)** $SrSO_4$ **(D)** $BaCl_2$

Sol. (C)

Q.47 **Incorrect solubility order :**

(A) $CH_3OH > C_2H_5OH > C_3H_7OH$

(B) $CH_3OH > CH_3-O-CH_3 > C_2H_5-O-C_2H_5$

(C) $C_2H_5OH <$  **(D) O-nitrophenol > p-nitorphenol**

Sol. (D) Solubility ⇒ Inter molecular H-bonding > Intramolecular H-bonding

Melting point and Boiling point :

M.P. → Diamond(s) > NaCl(s) > $H_2O(s)$ > $CO_2(s)$
4000°C ionic H-bond V.W.

M.P. of Metalic + Non-metalic compound :

(i) $\boxed{MP \propto LE}$ applicable for
Na^+, K^+, Pb^+, Cs^+, Al^{+3}, or F^-, O^{-2}, H^-, N^{-3}

(ii) $\boxed{MP \propto \frac{1}{\text{Polarisation}}}$ for others

(iii) $\boxed{M.P \rightarrow \text{Ionic} > \text{Covalent}}$
Except giant molecule like diamond, silica borazone $(BN)_x$, Carborundem (SiC) Norbide (B_4C)

⇒ Hardness → Diamond > B_4C > SiC > Al_2O_3
natural Artificial Corundum

Example :

(i) $BeCl_2 < MgCl_2 < CaCl_2 < SrCl_2 < BaCl_2$ (P.P)
(ii) $CaF_2 > CaCl_2 > CaBr_2 > CaI_2$ (LE)
(iii) $NaF < MgF_2 < AlF_3$ (LE)
(iv) $NaCl > MgCl_2 > AlCl_3$ (P.P)
(v) $Na_2O < MgO < Al_2O_3$ (LE)
(vi) $\overset{+1}{Na}Cl > \overset{+1}{Cu}Cl > \overset{+2}{Cu}Cl_2$
(vii) NaCl > KCl > RbCl > CsCl > LiCl
Covalent
(viiii) $MgF_2 > CaF_2 > SrF_2 > BaF_2 > BeF_2$
126.3°C 1418°C Covalent
(ix) $CaF_2 > MgF_2 > SrF_2 > BaF_2 > BeF_2$
Metal fluoride < Metal oxide
(x) $SnCl_2 > SnCl_4$
(xi) $AlF_3 > AlCl_3$
(xii) NaF < MgO < SCN < TiC
Charge↑ Le↑

Melting point of non metal compound

Intermolecular force $\propto$ M.P $\propto$ B.P $\propto$ T.S $\propto$ Viscosity $\propto \frac{1}{\text{Volatility}} \propto \frac{1}{\text{V.P}}$

CH_3-O-CH_3 <	C_2H_5OH
(VWA)	H-bond
CH_3-N_2 <	HN_3
CVWF	H-bond
$H_2 < CO_2 < H_2O$	B.P
$CH_3OH < H_2O$	T.S
$H_2O < D_2O$	B.P
$HCl < HNO_3 < H_3PO_4$	M.P
VWF	Extent of H-bonding

- $C_2H_5OH < CH_2OH-CH_2OH < CH_2(OH)-CH(OH)-CH_2OH$

$O_2 < S_8 < Se_8 < Te_8$

$H_2 > He$ B.P

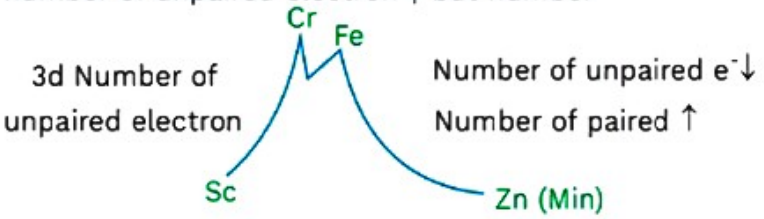
Melting point of metals

M.P $\propto$ Metal bond strength

(i) Li > Na > K > Rb > Cs Size↑ M.B.S↓

(ii) 3d < 4d < 5d Z_{eff}↑ M.B.S↓

- Zn > Cd > Hg (volatile metal)
  Number of paired electron ↑
  Inter electron repulsion between pairs ↑,
  MBS↓
  Interatomic distance↑
  Size ↑
  No unpaired electron here So, this can't be related with others as every where number of unpaired electron ↑ but number



Be > Ca > Sr > Ba > Mg

Innerd is vacant

- High participation of resonation structure (delocalizing) there, which increase

stability (Pauling theory)

- Mg doesnot have vacant d. So, it is less stable ® less M.P.
  Be has Size ↓ → MBS ↓
- Minimum M.P in all elements → He
- Minimum M.P in all elements → C (Diamond)
- Minimum M.P in all metal → W
- Minimum M.P in all metal → Hg
- Minimum M.P in Boron family → Ga
- Minimum M.P in all Boron family → B(Covalent)

Applications of Fajan Rule :

Heating effect

1. In polyatomic compound like CO_3^{2-}, SO_4^{2-}, NO_3^-, HCO_3^-, $C_2O_4^{2-}$, CH_3COO^-, OH^-, O_2^{2-}, O_2^-, ClO_4^-, O_3^- with increase in % ionic character, thermal stability increase.

(i) **CO_3^{2-}(carbonate ion) :**

Thermal decomposition of carbonate

$$\text{Metal carbonate} \xrightarrow{\Delta} \text{Metaloxide} + CO_2$$

Exception :

Na_2CO_3, K_2CO_3, Rb_2CO_3, Cs_2CO_3 possess high thermal stability and they do not decompose even when heated up to red hot rather than melt.

Na_2CO_3 → Soda ash

K_2CO_3 → Pearlash

$$MgCO_3 \xrightarrow{\Delta} MgO + CO_2\uparrow$$

Magnestic

$$ZnCO_3 \xrightarrow{\Delta} ZnO + CO_2\uparrow$$

Calamine

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2\uparrow$$

Calcite

$$PbCO_3 \xrightarrow{\Delta} PbO + CO_2\uparrow$$

Corrusite

$$FeCO_3 \xrightarrow{\Delta} FeO + CO_2\uparrow$$

Siderite

$$CuCO_3.Cu(OH)_2 \xrightarrow{\Delta} 2CuO + CO_2 + H_2O$$

Molacite (gree) black

(basic copper carbonate)

$$2PbCO_3.Pb(OH)_2 \xrightarrow{\Delta} 3PbO + 2CO_2 + H_2O$$

White lead

basic lead carbonate
$PbO \rightarrow$ letharge
$Pb_3O_4 \rightarrow$ Red lead or sindoor
Cu, Hg, Ag, Au, Pt $\rightarrow$ Very less reaction metal

$$Ag_2CO_3 \xrightarrow{\Delta} Ag + CO_2 + \frac{1}{2}O_2$$

$$HgCO_3 \xrightarrow{\Delta} Hg + CO_2 + \frac{1}{2}O_2 \uparrow$$

Diamagnetic Paramagnetic

Colourless

$$\underset{\text{Washing soda}}{Na_2CO_3.10H_2O} \xrightarrow[\Delta]{ROOH} \underset{\text{NATRON}}{Na_2CO_3.H_2O} \xrightarrow{\Delta} \underset{\text{Soda ash}}{Na_2CO_3}$$

No decomposition

(ii) Sulphate (SO_4^{2-})

$$\text{Metal sulphate} \xrightarrow{\leq 800^\circ C} \text{Metal oxide} + SO_3\uparrow$$

$$\text{Metal sulphate} \xrightarrow{>800^\circ C} \text{Metal oxide} + SO_2 + \frac{1}{2}O_2$$

Except $Na_2SO_4, K_2SO_4, Rb_2SO_4, Cs_2SO_4$

$$MgSO_4 \xrightarrow{>800} MgO + SO_2 + \frac{1}{2}O_2 \uparrow$$

$$PbSO_4 \xrightarrow{>800} PbO + SO_2 + \frac{1}{2}O_2$$

$$ZnSO_4 \xrightarrow{>800} ZnO + SO_2 + \frac{1}{2}O_2$$

$$Li_2SO_4 \xrightarrow{>800} Li_2O + SO_2 \uparrow + \frac{1}{2}O_2 \uparrow$$

Some important reaction of sulphate

1. $$FeSO_4 7H_2O \xrightarrow[-7H_2O]{\leq 300^\circ C} FeSO_4 \xrightarrow{>300^\circ C} \underset{\text{Brown solid}}{Fe_2^{+3}O_3} + SO_2 + \frac{1}{2}O_2$$

Example of intramolecular redox reaction

2. $Fe_2(SO_4)_3 \xrightarrow{\Delta} Fe_2O_3 + SO_3$ Non Redox reaction

Ferric sulphate → brown solid

3. $SnSO_4 \xrightarrow{\Delta} \overset{+4}{SnO_2} + \overset{+4}{SO_2}$

Stannous sulphate

Intramolecular Redox reaction

$\overset{+2}{SnO} + \overset{+6}{SO_3}$

Good R.A. Good O.A

4. $CaSO_4 2H_2O \xrightarrow[\Delta]{120°C} CaSO_4 \frac{1}{2}H_2O + \frac{3}{2}H_2O \xrightarrow[\Delta]{T>120°C}$ $CaSO_4$ dead burnt plaster

$CaO + SO_2 + \frac{1}{2}O_2 \xleftarrow{>300°C}$

5. $CuSO_4 5H_2O \xrightarrow{\text{Room temperature}} CuSO_4 8H_2O \xrightarrow[-2H_2O]{100°C} CuSO_4 H_2O$ bluish white

$-2H_2O$ | 100°C

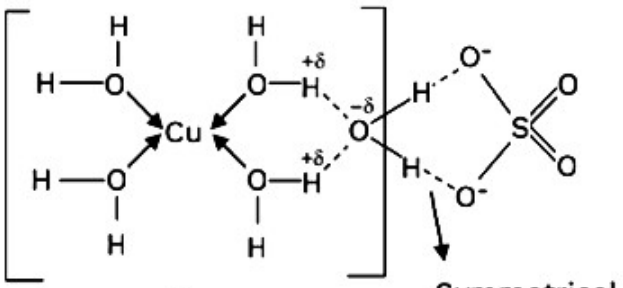
$CuSO_4 H_2O$ bluish white

H_2O | 220°C

$CuO + SO_3$ Black

$Cu_2O + SO_2 + \frac{1}{2}O_2 \xleftarrow{>800°C} CuSO_4$ (white)

Red



$[Cu(H_2O)_4]SO_4H_2O$

(iii) **Nitrate** (NO_3^-):- Water of Crystallization Thermal decomposition reaction of nitrates

$$\text{Metalnitrate} \xrightarrow{\Delta} \text{Metal oxide} + \underset{\text{(Brown colour porcongnetic gas)}}{NO_2} + \underset{\text{colourless pass gas}}{O_2}$$

Exception

$MNO_3 \xrightarrow{\Delta} MNO_2 + O_2$

$(M = Na, k, Rb, G)$

Q↓ Stable oxide donot form

$NaNO_3 \xrightarrow{\Delta} NaNO_2 + O_2$

$kNO_3 \xrightarrow{\Delta} kNO_2 + O_2$

$RbNO_3 \xrightarrow{\Delta} RbNO_2 + O_2$

$$2MnO_3 \xrightarrow{\geq 800^\circ C} M_2O + \underset{\text{diamgnetic}}{N_2} + \underset{\text{Paramagnetic}}{\frac{3}{2}O_2}$$

$M = Na, k, Rb, G$

$2NaNO_3 \xrightarrow[\Delta]{800^\circ C} Na_2O + N_2 + O_2$

$LiNO_3 \xrightarrow{\Delta} Li_2O + NO_2 + O_2$

$Mg(NO_3)_2 \xrightarrow{\Delta} MgO + NO_2 + O_2$

$Cu(NO_3)_2 \xrightarrow{\Delta} CuO + NO_2 + O_2$

$$Pb(NO_3)_2 \xrightarrow{\Delta} PbO + NO_2 + O_2$$

$$2AgNO_3 \xrightarrow{\Delta} 2Ag + 2NO_2 + \frac{1}{2}O_2$$

$$HgNO_3 \xrightarrow{\Delta} Hg + NO_2 + \frac{1}{2}O_2$$

Q.48 **Which of the s-block metal nitrate give brown colour NO_2 gas on heating**
(A) $LiNO_3$ (B) $NaNO_3$ (C) kNO_3 (D) $CsNO_3$

Sol. A

Example

(a) $\underset{\text{white vitriol}}{ZnSO_4 7H_2O} \xrightarrow{<200^\circ C} ZnSO_4H_2O \xrightarrow{>200^\circ C} ZnSO_4 \xrightarrow{>800^\circ C} ZnO + SO_2 + \frac{1}{2}O_2$

(b) $\underset{\text{Glauber's salt}}{Na_2SO_4 10H_2O} \xrightarrow{\Delta} Na_2SO_4H_2O \xrightarrow{\Delta} \underset{\text{stable}}{Na_2SO_4} \xrightarrow{\Delta} X$

(iv) HCO^- **(Bicarbonate)**

$$2MHCO_3 \xrightarrow{\Delta} M_2Co_3 + CO_2 + H_2O$$

M = Na, k, Rb, G

Example

$$\underset{\text{Solvay method}}{2NaHCO_3} \xrightarrow{\Delta} Na_2CO_3 + CO_2 \uparrow + H_2O$$

$LiHCO_3$ exist in liquid state

Metal hydroxide $(OH^-) \xrightarrow{\Delta}$ Metal oxide + H_2O

(v) **Perchlorate** (ClO_4^-)

$$\boxed{MClO_4 \longrightarrow M_2O + Cl_2 + \frac{1}{2}O_2}$$

(vi) Superoxide (O_2^-)

$$MO_2 \xrightarrow{\Delta} M_2O + \frac{3}{2}O_2$$

Where M is k, Rb, Cs

- Aqueous solution of metal hydride are basic in nature

$$M^+H^- + H^+OH^- \longrightarrow MOH + H_2 \uparrow$$

$$NaH + H^+OH \longrightarrow NaOH + H_2 \uparrow$$

$$\underset{\text{Hydrolith}}{CaH_2} + 2H^+OH^- \longrightarrow \underset{\text{Industrial method}}{Ca(OH)_2} + H_2 \uparrow$$

(vii) Nitrides (N^{3-})

$$\underset{\substack{\text{LE>EA}\\ \text{Exothermic}}}{Li_3N} > \underset{\text{unstable-endothermic reaction}}{Na_3N > k_3N}$$

$N^{-3} \rightarrow$ highly exothermic

$$NH_3 \xrightarrow{-H} \underset{\text{Amide}}{NH_2^-} \longrightarrow \underset{\text{Imide}}{NH^{2-}} \longrightarrow \underset{\text{Nitride}}{N^{-3}}$$

$$BaCN \xrightarrow{\Delta} Ba + 3N_2$$

$$2NaN_3 \xrightarrow{\Delta} 2Na(s) + 3N_2$$

* Pure nitrogen gas can be obtained by heating metal ozide because no other gas eous product is formed.

* $NaNO_3$ is used in motor car air bags

Q.49 **Anhydrous $MgCl_2$ can be obtained by**

(A) $MgCl_2 6H_2O \xrightarrow[+Cl\ gas]{\Delta} MgCl_2 + HCl \uparrow + H_2O$ **(B)** $MgCl_2 6H_2O \xrightarrow[HCl\ gas]{\Delta} MgCl_2 + HCl \uparrow + H_2O$

(C) $Mg + dilHCl \longrightarrow \underset{\text{hydrated}}{MgCl_2} + \underset{\text{(dil)}}{H_2}$ **(D)** $MgO + C + Cl_2 \longrightarrow \underset{\text{Anhydrous}}{MgCl_2} + CO \uparrow$

Sol. B

Q.50 **Which of the following contain two s-block metals?**

(A) superphosphate $CaSO_{34} + Ca(H_2PO_2)_2$

(B) Microcosmic salt $Na(NH)_4HPO_4 4H_2O$

(C) Lithophone → (ZnS + $BaSO_4$)

(D) Fusion mixture $(Na_2CO_3 + k_2CO_3)$

Sol. D

Soda lime ⇒ NaOH + CaO

Aqueous solution of $Ca(OH)_2$ = lime water

Suspension of $Ca(OH)_2$ = milk of magnesia

$$CaO + H_2O \longrightarrow \underset{\text{slaked lime}}{Ca(OH)_2}$$

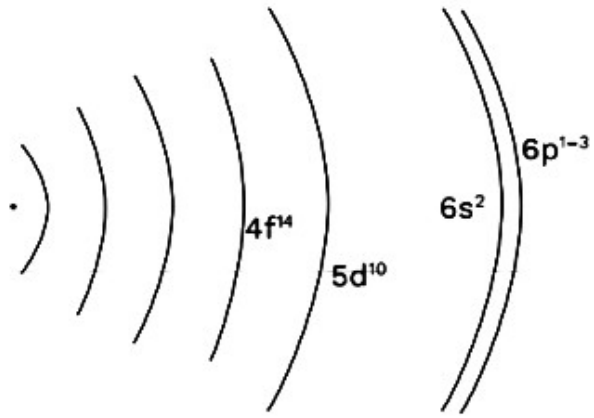
Inert pair effect

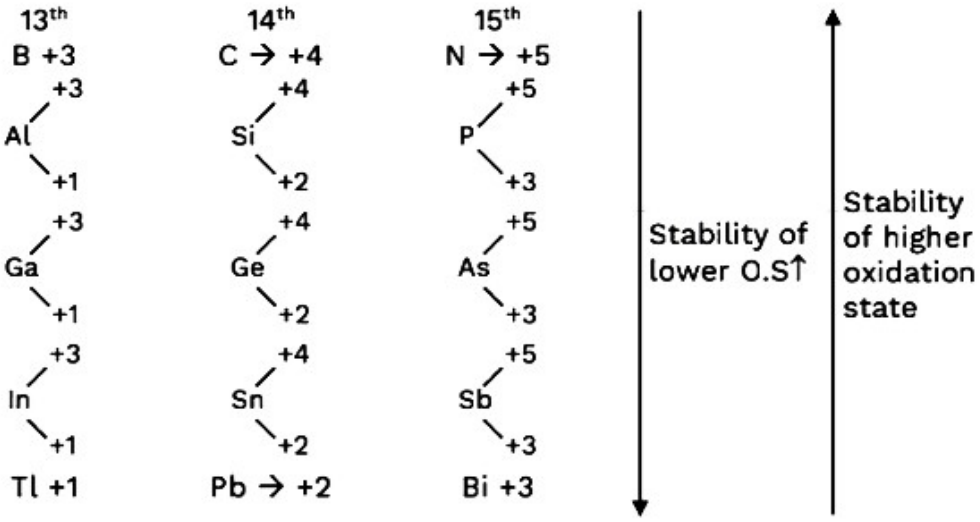
In P-block elements (group no.13 to 16), on moving down the group lower oxidation state becomes more stable as compare to higher oxidation state due to inertness of electron pair of 6s orbital is called inert pair effect. This electron pair of 6s orbital do not participate in bonding as it is strongly attracted towards nucleus due to poor shielding effect of inner d and f orbitals.

$Tl_{81}\ [Xe]4f^{14}5d^{10}6s^26p^1$

$Pb_{82}\ [Xe]4f^{14}5d^{10}6s^26p^2$

$Bi_{83}\ [Xe]4f^{14}5d^{10}6s^26p^3$





Stablity of oxidation state

Tl^{+}	>	Tl^{+3}	13th	group
Pb^{+2}	>	Pb^{+4}	14th	
Bi^{+3}	>	Bi^{+5}	15th	

Order of stability: $Tl^{+1} > In^{+1} > Ga^{+1}$ (due to inert pair effect)

Order of stability: $Pb^{+2} > Sn^{+2} > Ge^{+2}$ (due to inert pair effect)

Application of inert pair effect

1. Salt of Tl^{+3}, Pb^{+4} and Bi^{+5} are good oxidising agent.

   $Tl^{+3} + 2e^{-} \longrightarrow Tl^{+}$

   $Pb^{+4} + 2e^{-} \longrightarrow Pb^{+2}$

   $Bi^{+5} + 2e^{-} \longrightarrow Bi^{+3}$

2. PbI_4 doesnot exist because Pb^{+4} is a strong oxidising agent and I^{-} is strong reducing agent.

   $Pb^{+4} + 2e^{-} \longrightarrow Pb^{+2}$

   $\boxed{Pb^{+4} + 2I^{-} \longrightarrow Pb^{+2} + I_2}$

Reducing agent $\frac{F^- < Cl^- < Br^- < I^-}{\text{order of reducing nature}}$

$PbBr_4 \xrightarrow[\text{warming}]{\text{gentle}} PbBr_2 + Br_2$
(exist at Room temp.)

$PbCl_4 \xrightarrow{50°C} PbCl_2 + Cl_2$

PbF_4 exist at room temperature

$Pb^{+4}_{(O.A.)}(aq) + 4F^-_{(aq)} \xrightarrow{R.T.} PbF_4$
(F^-: has no reducing Character)

TlI_3 exist in Tl^{+1} and I_3^- in aqueous medium $Tl^{+3} + 2I^- \longrightarrow Tl^+ + I_2$

$I_2 + I^- \longrightarrow \underset{\text{(Tri iodide ion)}}{I_3^-}$

$\underset{L.B.}{I^-} + \underset{L.A}{I_2} \longrightarrow I_3^-$

$Sn^{+2} \xrightarrow{\text{oxidation}} Sn^{+4}$

Sn^{+2} work as reducing agent
Order of stability:

$\underset{\text{Stannous}}{Sn^{+2}} < \underset{\text{Stannic}}{Sn^{+4}}$

Example to show reducing nature of $SnCl_2$

$2\overset{+2}{HgCl_2} + \overset{+2}{SnCl_2} \longrightarrow \overset{+1}{Hg_2}Cl_2 + \overset{+4}{Sn}Cl_4$

$\underset{\text{Calomel}}{Hg_2Cl_2} + SnCl_2 \longrightarrow \underset{\text{grey ppt}}{Hg} + SnCl_4$

$2BiCl_3 + 3SnCl_2 \longrightarrow \underset{\text{Black ppt}}{2Bi\downarrow} + 3SnCl_4$

$2AuCl_3 + 3SnCl_2 \longrightarrow \underset{\text{yellow ppt}}{2Au\downarrow} + 3SnCl_4$

$Ge^{+2} < Ge^{+4}$ (order of stability)

TlF_3 —
- $Tl^{+3} + 3F^- \longrightarrow Tl^{+3} + 3F^-$
- $Tl^{+1} + F_3^- \longrightarrow Tl^{+1} + F_3^-$ (does not exist)

* CuI_2 does not exist

$$Cu^{+2}(aq) + 2I^-_{(aq)} \xrightarrow{R.T} CuI + \frac{1}{2}I_2$$

* $FeI_3 \longrightarrow$ does not exist

$$Fe^{+3}_{(aq)} + 3I^-_{(aq)} \longrightarrow FeI_2 + \frac{1}{2}I_2$$

$\rightarrow$ $Cu^{+2}_{(aq)}, Fe^{+3}_{(aq)}, Au^{+3}_{(ac)}$ exhibit oxidising character

Stability of Tl^{+3} halide

$TlF_3 > TlCl_3 > TlBr_3 > TlI_3$

Stability of Tl^{+1} halide

$TlI_3 > TlBr_3 > TlCl_3 > TlF_3$

$\downarrow$

does not exist

It is mainly applicable for 13th, 14th, 15th and 16th group

In general, elements are stable in their higher oxidation state. But in p-block elements from group no. 13 to 16 on moving down the group, lower oxidation state gradually becomes stable and higher oxidation state become less stable and last element of each group is stable in lower oxidation state rather than higher oxidation state due to inertness of ns-electrons towards bonding due to poor shielding of 4f subshell is called Inert pair effect.

Q.51 **Correct order of stability of metal cation is are:**

(A) $Al^{+3} < Sb^{+3} < Bi^{+3}$

(B) $Ge^{+4} < Sn^{+4} < Pb^{+4}$

(C) $Ga^{+} < In^{+} < Tl^{+}$

(D) $Bi^{+5} < Sb^{+5} < As^{+5}$

Sol. A, C, D

Q.52 **Correct order of stability of metal**

(A) $Sn^{+4} < Sn^{+2}$ **(B)** $Pb^{+2} < Pb^{+4}$

(C) $Pb^{+2} < Sn^{+2}$ **(D)** $Pb^{+4} < Sn^{+4}$

Sol. D

* $\overset{+3}{TlCl_3}$ $Tl^{+3} + 3Cl^- \longrightarrow$ No reaction

* $AlCl$ $Al^+ + Cl^- \longrightarrow 3AlCl$

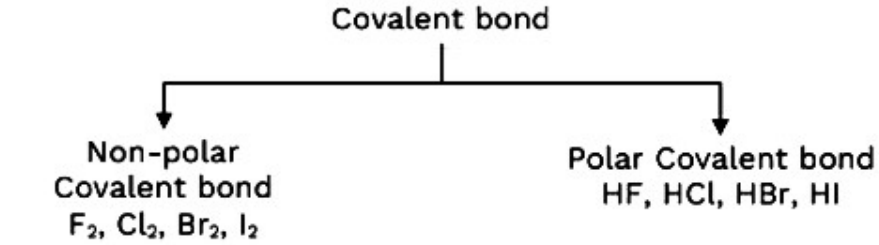
$\square$

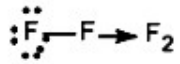
$AlCl_3 + 2\,Al$

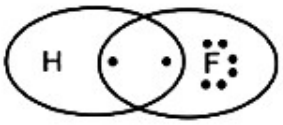
3. Covalent Bond

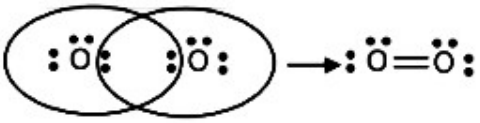
Non metal + Non metal = Covalent bond

When chemical bond is formed by equal shearing of electron between bonded atoms then it is named as covalent bond and it is directional in nature.











Theories of Covalent bond formation

Lewis Theory

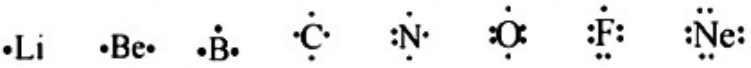
- According to earlier concept, elements form covalent bond by complete their octet (except Hydrogen) of valence shell and gets stabilize like a group 18 or noble gases. (He, Ne, Ar etc.)
- Only valence shell or outermost shell electrons will participate in the process of bond formation.

LEWIS DOT STRUCTURES OR LEWIS SYMBOLS

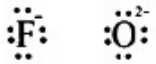
Lewis introduced simple notations which represent bond between atoms and valence election represent by dots on atom. These notations are called Lewis symbols or Lewis dot structure.

Lewis symbols for atoms and monoatomic ions :

Lewis symbol for second period elements are :



Lewis symbol for some anions are



Rules to draw Lewis dot structure (L.D.S.) of inorganic covalent compounds:

1. Selection of central atom in given compound

(i) First of all, identify the central atom in the given species. Central atom in a given molecule/ion is usually that atom which is least electronegative.

(ii) Hydrogen and Fluorine can't be central atom as its covalency is one.

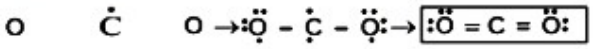
(iii) Sometimes the central atom is that atom which is less in number.

(iv) Sometimes the central atom in the given molecule/ion can't be decided on the basis of electronegativity or number of atoms (less). In such cases, that atom is central atom which appears in central position of given formula of molecule/ion. To make it more clear, central atoms in the following species are starred (*).

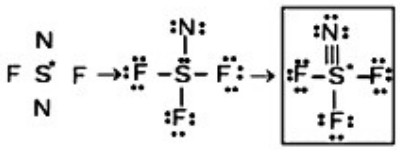
- $\dot{C}H_4$, $\dot{N}H_3$, $H_2\dot{O}$, $\dot{C}H_2Cl_2$, $\dot{C}HF_3$, $H\dot{P}O_3$
  (Note-that electronegativity of H is less than that of central atom).
- $\dot{C}O_2$, $\dot{S}O_2$, $\dot{S}O_3$, $\dot{S}OCl_2$, $\dot{P}OCl_3$
- $\dot{O}Cl_2$, $\dot{O}Br_2$ (Note that oxygen is more electronegative, but it is less in number)
- $(\dot{S}CN)^-$, (order of appearance)

2. Surrounding atoms are the atoms which are directly bonded to central atom.

3. Arrange the surrounding atoms around the central atom and first form single bond between all surrounding atom with central atom.
4. Always make sure that octet of all the surrounding atoms is complete. It is not achieved by forming single bond, then try to make the double bond or triple bond between central atom and surrounding atom, as required to complete the octet of surrounding atoms.
e.g., $\underline{CO_2}$



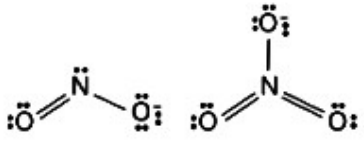
SNF_3



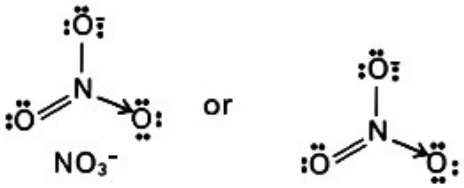
5. After the bond formation, represent the lone pair of electrons on central atom as well as on surrounding atoms.
6. Make sure that, in the structure you have made, the octet of the all surrounding atoms must be completed.
7. If the central atom belongs to second period, it can have < 8 (max.) electrons (i.e. < 4 bonds) but never greater than 8 electrons (i.e. > 4 bonds).
8. However, if the central atom belongs to third or lower period it can have > 8 electrons.
9. At last verify the covalency of central atom.

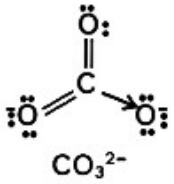
10. **Lewis Structure of ions :**

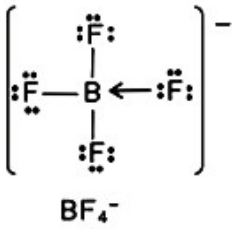
Distribute the negative charge on surrounding atom in such a way that octet of none of the surrounding atom is complete before the bond formation with central atom. If however, the octet of surrounding atom is complete by making it uni-negatively charged (particularly in case of halogen which contains seven valence electron), then such surrounding atom will attach itself with central atom through a co-ordinate covalent bond.

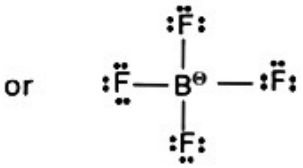


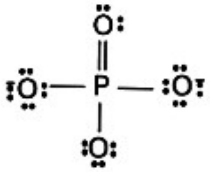
But the above structures of NO_3^- is incorrect as the central nitrogen belongs to second period, it can never form five covalent bond i.e., it can't have more than 8 electrons, so its actual structure is :

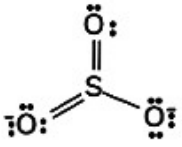


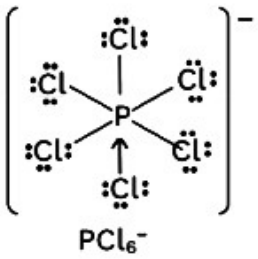


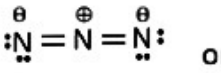
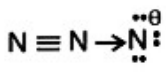
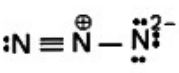


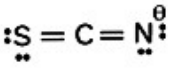
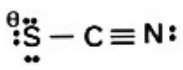








N_3^- :-  or  or 

SCN^- :-  or 

Drawbacks of Lewis Theory /Octet rule

1. Electron deficient species / hypovalent compound : The covalent compound in which central atom having less than 8 valence electron.
   e.g. BeH_2, $BeCl_2$, $AlCl_3$, B_2H_6, $ZnCl_2$, BF_3 etc.
2. Super octet / hyper valent compound : The covalent compound in which central atom having greater than 8 valence electron.
   e.g. PCl_5, SF_6, PF_7, XeF_2, SF_4 etc.
3. All odd electron species e.g. NO, NO_2, CH_3, ClO_2, ClO_3, OF etc.
   * All electron species are paramagnetic
4. Existence of compound of Xe, e.g. XeO_3, Xe_2O_4, XeO_4, XeF_4, XeF_2, XeF_6.
5. It could not tell anything about bond parameters like bond energy, bond length etc.
6. It could not give any idea about shape of the compound.

Q.53 **Which of the following statements about covalent bond is/are correct?**

(a) Covalent bond is formed between two similar atoms.

(b) Covalent bond is formed between two dissimilar atoms.

(c) Both of these.

(d) None of these.

Sol. : (c)

Covalent bond can be formed between two similar or dissimilar atoms.

Q.54 **Which of the following compounds is covalent?**

(a) H_2 **(b) CaO** **(c) KCl** **(d) Na_2S**

Sol. : (a)

H_2 is a covalent compound. Rest all are ionic compounds.

Q.55 **Which one of the following violates the octet rule?**

(a) PCl_3 **(b) CBr_4** **(c) NF_3** **(d) AsF5**

Sol. **(d)**

The electronic configuration of As : $[Ar]4s^2 3d^0 4p^3$.

It has vacant $4d$ orbital. These vacant orbitals can get involved in the hybridisation process which allows arsenic to have an expanded octet. Therefore, AsF_5 violates the octet rule.

Q.56 **The formation of compounds PF_5, SF_6, BCl_3 are not according to the rule of**

(a) octet **(b) doublet** **(c) triplet** **(d) none of above**

Sol : (a)

The octet rule states that atoms tend to form compounds in ways that give them eight valence electrons and thus, have the electronic configuration of a noble gas.

Q.57 **NaCl gives white precipitate with $AgNO_3$ solution but CCl_4 does not. This is because**

(a) NaCl is a covalent compound and gives Cl^- ions in the solution

(b) CCl_4 is an ionic compound and does not give Cl^- ions in the solution.

(c) NaCl is an ionic compound and does not give Cl^- ions in the solution.

(d) CCl_4 is a covalent compound and does not give Cl^- ions in the solution.

Sol : (d)

NaCl is an ionic compound and gives Cl^- ions in the solution. These Cl^- ions react with Ag^+ ions given by $AgNO_3$ to form while precipitate of AgCl. However, CCl_4 is a covalent compound and does not give Cl^- ions to react with Ag^+.

Q.58 **Which of the following statements is correct?**

(a) A double bond is stronger than triple bond.

(b) A triple bond is shorter than double bond.

(c) A triple bond is less reactive than double bond.

(d) A double bond is less stable than triple bond.

Sol : (b)

A triple bond is shorter than double bond.

Q.59 **In C_3O_2, there are**

(a) 2 C–C double bonds and 2 C–O double bonds.

(b) 2 C–C double bond and 3 C–O double bonds.

(c) 1 C–C triple and 3 C–O double bonds

(d) 3 C–O double bonds.

Sol : (a)

Carbon suboxide *i.e.*,

C_3O_2 has structure O=C=C=C=O.

Formal charge :

In any polyatomic ion charge is present on the complete ion and not on any individual atom e.g. NH_4^+, SO_4^{-2}, Po_4^{-3}.

To calculate charge on any individual atom, concept of formal charge introduces. This concept is helpful during draw of Lewis dot structure.

"It is defined as difference of valence electron available on atom and number of electrons available with the atom in given structure".

Formal charge = (V.E.) – [N.B.E. + (B.E./2)]

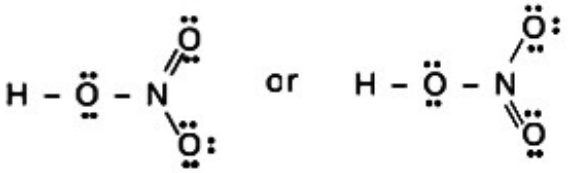
Where

V.E. = number of valence electron

N.B.E = Non-bonded electron

B.E.=Bonded electron

Examples : formal charges on every element on HNO_3



I II

The H atom : It has one valence electron, zero lone pair (or zero nonbonding electrons), and forms one bond (two bonding electrons).
Substituting in Eq. (i), we write FC = (1) – (0) -½ (2) = 0.

The O atom bonded to H: It has six valence electrons, two lone pairs (or four nonbonding electrons), and form two bonds (or four bonding electrons).
Hence, we write FC = (6) – (4) -½ (4) = 0.

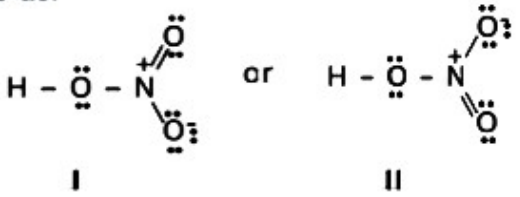
The N atom: It has five valence electrons, zero lone pair (or zero nonbonding electrons), and forms four bonds (or has eight bonding electrons).
Thus, we write FC = (5) – (0) -½ (8) = +1

The end O atom in N=O: It has six valence electrons in the free state but in the Lewis structure (I or II), it has two lone pairs (or four nonbonding electrons) and forms two bonds (or has four bonding electrons).
Thus, we can write FC = (6) – (4) -½ (4) = 0

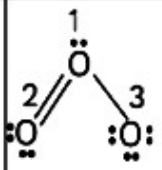
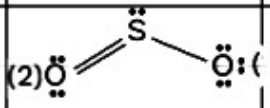
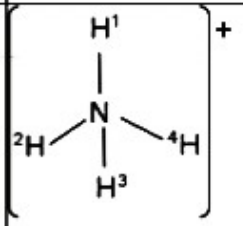
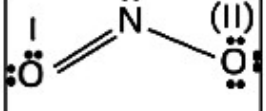
The end O atom in N–O: It has six valence electrons in the free state but in the Lewis structure (I or II), it has three lone pairs or six nonbonding electrons and forms one bond (or has two bonding electrons).

thus, we write FC = (6) – (6) -½ (2) = –1

We can now write the Lewis structures (I and II) for nitric acid molecule including the formal charges as:



I or II

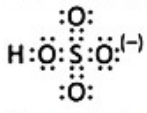
Molecule	Structure	Formal charge
O_3		O(1) = 6 – 2 -½ (6) = + 1 O(2) = 6 – 4 –½ (4) = 0 O(3) = 6 – 6 – ½ (2) = –1
SO_2		S = 6 – 2 – ½ × 6 = + 1 O(1) = 6 – 6 –½× 2 = –1 O(2) = 6 – 4 –½×4 = 0
CO		C = 4 – 2 –½× 6 = –1 O = 6 – 2 –½× 6 = + 1
CN^-		C = 4 – 2 –½× 6 = –1 N = 5 – 2 –½× 6 = 0
NH_4^+		N = 5 – 2 –½ (8) = + 1 On each H = 1 – 0 – ½ (2) = 0
NO_2^-		N = 5 – 2 – ½(6) = 0 O(I) = 6 – 4 – ½ (4) = 0 O(II) = 6 – 6 – ½(2) = –1

Q.60 **The formal charge on sulphur in HSO_4^- is**

(a) +2 (b) –2 (c) +4 (d) –4

Sol. : (a)

Lewis dot structure of HSO^- is



Formal charge $= V - L - \frac{1}{2}S$

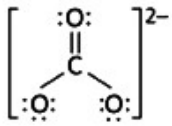
$$= 6 - 0 - \frac{1}{2} \times 8 = +2$$

Q.61 **The formal charge on double bonded O-atom in carbonate ion is**

(a) 0 (b) +1 (c) –2 (d) +2

Sol. : (a)

Lewis dot structure of CO_3^{2-} is



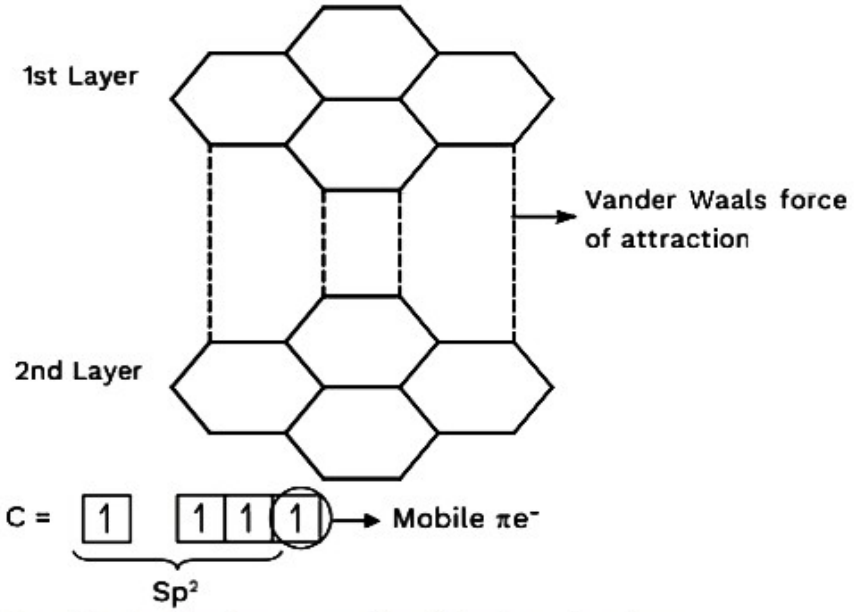
Formal charge on double bonded O - atom

$$= V - L - \frac{1}{2}S = 6 - 4 - \frac{1}{2} \times 4 = 6 - 6 = 0$$

Properties of Covalent Compound

1. Physical State: Generally, Covalent compounds exist in liquid or gases state. Some compounds having high molecular masses exist as soft solid
   Example
   S_8, P_4 and I_2
   * This is due to presence of weak force of attraction (van der waal's forces) be tween the molecule
   Exception: Diamond, SiC, SiO_2 are solid due to their polymeric nature
2. In covalent compound, atom and molecule are present but not ions.
3. Generally, they are bad conductor of electricity some polar compound such as HCl in aqueous solution conduct electricity.
4. Graphite is a covalent compound but conduct electricity due to presence of mobile

πe^- which is delocalised between hexagonal layer of graphite.



Graphite is also known as black lead or plumbago

5. Covalent compound are directional in nature, therefore it shows stereoisomerism.
6. Covalent compound are soluble in non-polar solvent (Organic solvent like C_6H_6, CCl_4, CS_2, ether etc.
(due to concept like dissolve like)
7. Covalent compound exhibit molecular reaction and these are slow than ionic reaction.

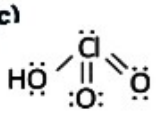
Nature of oxidising $\rightarrow F_2 > Cl_2 > Br_2 > I_2$

IF_7 ✓ ClF_7 × BrF_7 ×

Q.62 **The Lewis structure of chloric acid contains single bonds, double bonds and lone pairs of electrons are respectively**

(a) 3, 2, 5 **(b) 2, 0, 7** **(c) 2, 2, 7** **(d) 2, 3, 8**

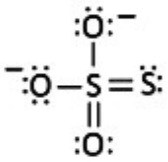
Sol. : **(c)**



Q.63 **How many electrons are involved in bonding in Lewis structure of $S_2O_3^{2-}$ ion?**

(a) 22 **(b) 20** **(c) 18** **(d) 12**

So. : (d)



No. of covalent bonds = 6

∴ Total no. of electrons involved in bonding = 6 × 2 = 12

Q.64 **Which of the following statements is correct according to Sidgwick–Powell theory?**

(a) If there are two pairs of electrons in the valence shell of the central atom, the orbitals containing them will be oriented at 120° to each other.

(b) If there are three electron pairs on the central atom, they will be at 120° to each other, giving plane triangular structure.

(c) For four electron pairs the angle is 107° and the shape is tetrahedral.

(d) All of these

Sol. : (b)

According to Sidgwick–Powell theory, if there are three electron pairs on the central atom, they will be at 120°, giving triangular structure.

Q.65 **According to Sidgwick–Powell theory prediction, the bond angle of a molecule containing 6 electron pairs in outer shell, will be**

(a) 90° **(b) 180°** **(c) 120°** **(d) 90°, 120°**

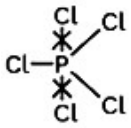
Sol. : (a)

For a molecule with 6 electron pairs, the bond angle will be 90° and the shape is octahedral.

Sudgen Concept of Singlet linkage

According to Sudgen octetrule is never violated and he gave the concept of singlet linkage and explained the thermal decomposition of PCl_5 by this concept.

$$PCl_5 \xrightarrow{\Delta} PCl_3 + Cl_3$$



Number of singlet bond = Total number of electron in bonds – 8

Example:

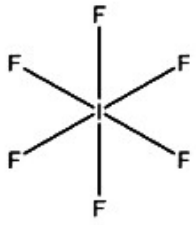
$IF_7 = 14 - 8 = 6$

* But in case of SF_6, this concept is failed. So, it has been discarded.

$$SF_6 \xrightarrow[\text{decomposition}]{\text{Thermal}}$$

Example:

IF_7



Types of Orbital

1. S-orbital

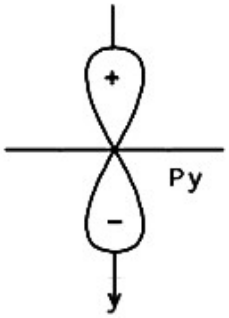
   Spherical, Non-directional gerade

2. P-orbital

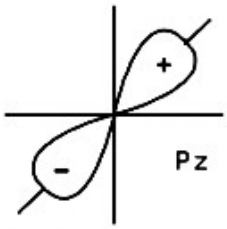
   Electron density lies along the axis.



Nodal plane - yz
ungerade



Nodal plane = xz

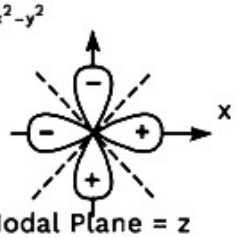


Nodal plane = xy

d-orbitals → gerade, directional

- Along the axis (Axial)
  - $d_{x^2-y^2}$
  - d_{z^2}
- Between the axis (Non-axial)
  - d_{xy}
  - d_{yz}
  - d_{zx}

$d_{x^2-y^2}$



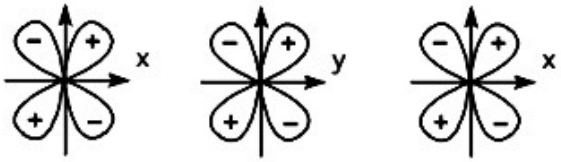
Nodal Plane = z
at 45° of x and y
axis yz, xz



Nodal Plane = 0
Nodal line = 2

Between axis in the plane

1. d_{xy} 2. d_{yz} 3. d_{zx}



Gerade	Ungerade
Centre of symmetry with respect to phase and electron density	Absent

Valency Bond Theory (VBT)

Given by Heitler and London.

It was extended by Pauling and Slater.

According to valence bond theory, atom of an element form bond only to pair up its unpaired electron present in ground state or excited state.

This pairing of unpaired electron will take place by overlapping of half filled atomic orbital each having unpaired electron with opposite spin. This combination of atomic orbitals named as overlapping.

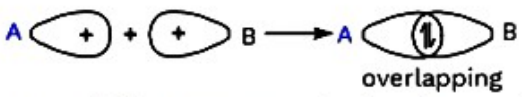
OR

According to this theory, valency shell electron participate in bond format ion. It has two parts

1. Overlapping theory
   Theory which explain formation of covalent bond.
2. Hybridisation
   Theory which explain shape of molecule and bond parameters (bond angle, bond length and bond energy.

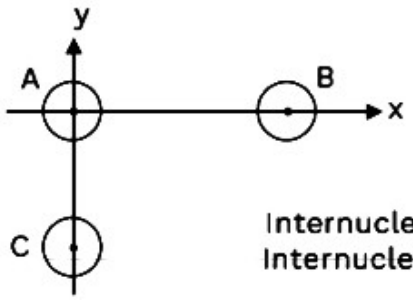
(1) Overlapping theory

Overlapping means common area sharing.



A partial interpenetration is observed when atoms at minimum energy state while forming bond. This interpenetration of orbitals is called overlapping of orbital as result pairing of electrons take place. The strength of a covalent bond is a function of extent of overlapping.

Internuclear axis: Axis on which nucleus of bond forming atom present.



Internuclear axis, for AB → x
Internuclear axis, For AC → y

Directional Properties of covalent Bonds:
With the help of overlap and hybridization of atomic orbitals, the directional nature of binds in poly atomic species is explained in VBT. For instance directional bonds in CH_4 , NH_3 , and H_2O etc

Condition for overlapping

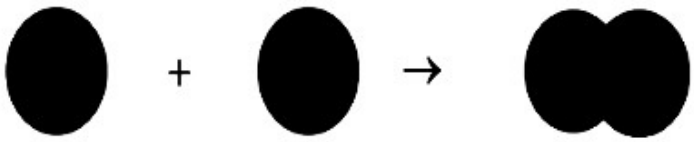
1. In homonuclear molecular species, atomic orbital of same energy level undergo overlapping while in heteronuclear molecular species, atomic orbital of comparable energy level undergo overlapping.
2. Combining atomic orbital must be in same phase and phase difference between them should be zero.

Overlapping and Nature of Covalent Bonds

Based on overlapping covalent bonds can be classified as
(i) sigma (σ) bond (ii) pi (π) bond (iii) delta (δ) bond

Sigma (σ) bond: Bonds formed due to axial overlapping are called Sigma (σ) bonds.
A sigma bond can be formed by the following types of combinations of atomic orbitals.

s-s overlapping: When two half filled s-orbitals approach along their internuclear axis.



S – orbital S – orbital S – S overlapping

Points to remember

- Phosphorus exist as P_4 but Nitrogen exist as N_2
- Oxygen exist in O_2 but S=S not possible at room temperature.

s-p overlapping: When one half filled s-orbitals app p-orbital.



s – orbital p – orbital s – p overlapping

p-p overlapping : When two half filled p-orbitals approach along their internuclear axis

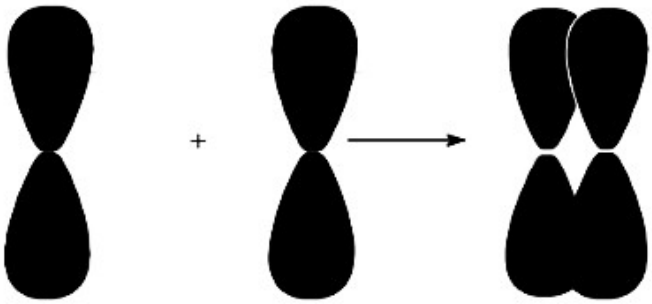


p – orbital p – orbital p – p overlapping

pi(π) bond

The bonds formed due to side wise overlapping of atomic orbitals i.e. when axes of overlapping orbitals stay parallel. The electro density gets split above and below bonding axis like two saucer type electro clouds.

(a) pπ – pπ

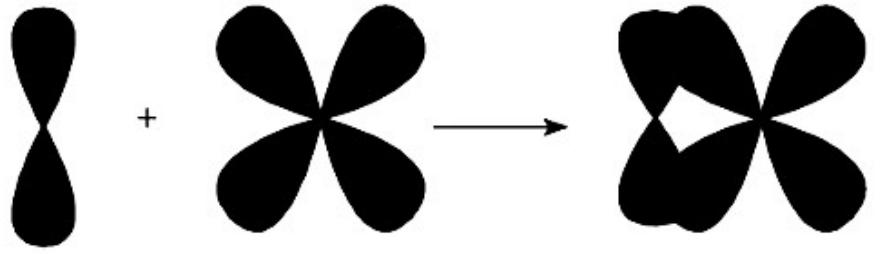


p – orbital p – orbital p-p overlapping

Overlapping	INA	Bond
$P_x + P_x$	y or z	π ($p\pi-p\pi$)
$P_y + P_y$	x or z	π
$P_z + P_z$	x or y	π

Any other p-p overlapping is not possible

(b) $\mathbf{p_\pi - d_\pi}$



p – orbital d – orbital $p_\pi - d_\pi$ orbital

(c) $\mathbf{d_\pi - d_\pi}$



d – orbital d – orbital $d_\pi - d_\pi$ overlaping

Overlapping	σ-bond	π-bond	S-bond
Py + Py	y	x Z	– –
Py + Py	-	y	–
Pz + dxz		x	–

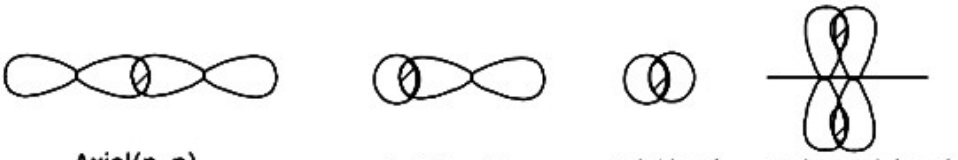
dxz + dxz		x Z	y
dyz + dyz		y Z	x
Pz + d_{z^2}	z	-	—
$d_{x^2-y^2}$ + $d_{x^2-y^2}$	x y	-	z

Comparison of s bond strength:

1. If shell number of orbitals are same

Bond strength ∝ Extent of overlapping ∝ directional property

2p–2p > 2s–2p > 2s–2s > 2p–2p ← Bond Strength
Axial axial axial Colateral



Axial(p-p) Axial(s-p) Axial(s-s) Colateral (p-p)

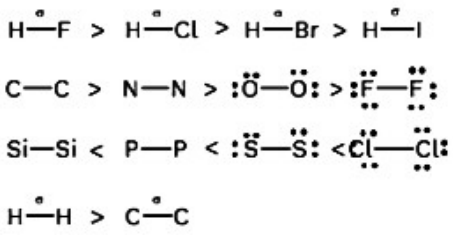
2. When value of n is different

1s–1s > 2s–2p > 2p–2p

$$\text{Bond strength} \propto \frac{1}{\text{shell number of orbitals}}$$

1s–2p > 1s–3p > 1s–4p

Example



$$C-C > N-N > :\ddot{O}-\ddot{O}: > :\ddot{F}-\ddot{F}:$$

$$Si-Si < P-P < :\ddot{S}-\ddot{S}: < :\ddot{Cl}-\ddot{Cl}:$$

Catenation property of element $\propto \sigma$ – bond strength

Cat. ↓ C > S > P **C > S > Si > P > N > O**

Comparison of p bond strength

$2p\pi - 2p\pi > 2p\pi - 3d\pi > 2p\pi - 3p\pi$
strongest stable very weak

$$\pi \text{bond strength} \propto \frac{1}{\text{Internuclear distance (Bond length)}}$$

Inter electronic repulsion must be absent between bonded atoms.

Bond Energy → **$Cl_2 > Br_2 > F_2 > I_2$**

$$\text{oxidising property of covelant oxides / oxyanion} \propto \frac{1}{T.S}$$

$$\text{Reducing property of covelant hydride} \propto \frac{1}{T.S}$$

Example

$$ClO^- > ClO_2^- > ClO_3^- > ClO_4^-$$

Oxidising property ↑ T–S↓

Example

$$CH_4 < SiH_4 < GeH_4 < SnH_4$$

Order of reducing property

Q.66 **According to VBT, which of the following statements is correct?**

(a) It is based on the knowledge of atomic orbitals and electronic configurations of elements.

(b) The inner electrons also participate in bond formation.

(c) The molecule has maximum energy at a certain distance between the atoms known as internuclear distance.

(d) None of these.

Sol. : (a)

The bond is formed due to interaction of only valence electrons, inner electrons do not participate in bonding. The molecule has minimum energy at internuclear distance.

Q.67 **The strength of covalent bond _____ extent of overlapping of orbitals.**

(a) may be or may not be related to **(b) is independent on**

(c) is dependent on **(d) is not related to.**

Sol. : (c)

As per the concept of valence bond theory, the partial merging of atomic orbitals is known as overlapping. The extent of overlapping is directly proportional to the strength of the covalent bond.

Q.68 **Which of the following overlap does not result in the formation of a bond?**

(a) Positive **(b) Negative** **(c) Zero** **(d) Both (b) and (c)**

Sol. : (d)

Zero overlap means that the orbitals do not overlap at all. When there is no overlapping the bond formation does not occur. As well the extent of overlapping is dependent on the strength of the bond. Also negative overlap does not result in the formation of a bond.

Q.69 **Which of these has lest tendency of polymerisation?**

(A) SiO_4^{4-} **(B)** PO_4^{3-} **(C)** SO_4^{2-} **(D)** ClO_4^-

Sol. $Si \overset{\pi}{-} O$ $P \overset{\pi}{-} O$ $S \overset{\pi}{-} O$ $Cl - O$

As size decreases, bond length also decreases overlapping increases, π -bond strength increases and polymer ↓

Q.70 **Give correct order of bond strength for σ bond.**
1s-1s, 2p-2p, 2s-2s, 3s-3p, 2s-3s, 3s-3s, 3p-3p, 1s-2s, 1s-2p, 2s-2p, 2s-3p

Sol. 1s-1s > 1s-2p > 1s-2s > 2p-2p > 2s-2p > 2s-2s > 2s-3p > 2s-3s > 3p-3p > 3s-3p > 3s-3s

Hint: If internuclear distance is incomparable than B.S. $\propto$ Bond distance

If internuclear distance is comparable than B.S. $\propto$ Extent of overlapping (directional of nature)

2. Hybridisation – by Pauling

It was introduced by Pauling and Slater, to explain equivalent nature of covalent bonds in a molecule.

Consider an example of $BeCl_2$ If it is formed without hybridisation then –

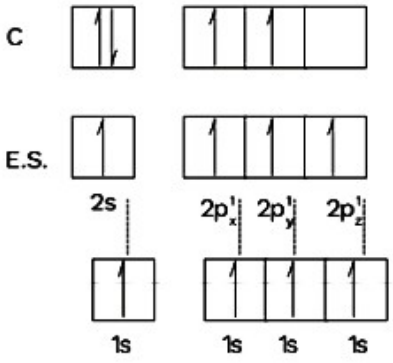
$$Cl \underset{}{\overset{p-s}{\text{———}}} Be \underset{}{\overset{p-p}{\text{———}}} Cl$$

both the Be-Cl bonds should have different parameters and p-p bond strength should be greater than s–p bond strength. But experimentally bond parameters of both the Be–Cl bonds are same. This can be explained by the concept of hybridisation.

Hybridisation : Phenomena of intermixing of atomic orbitals of same or nearly same energy and thus producing hybrid orbitals of same energy and same shape is called as hybridisation. Now after considering s-p hybridisation in $BeCl_2$ bond strength of both the bonds will be equal.

$$Cl \underset{}{\overset{p-sp}{\text{———}}} Be \underset{}{\overset{sp-p}{\text{———}}} Cl$$

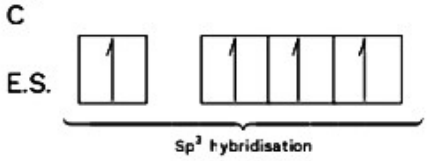
Understanding Formation of methane (CH_4) can also help us understand hybridization:-

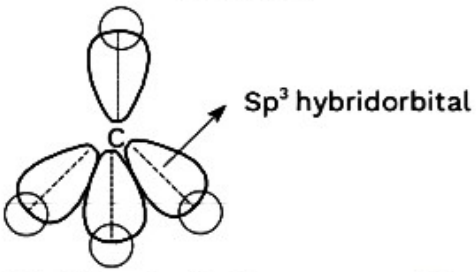


1s – 2s 1s – 2p overlapping

BE < 1s – 2s < 1s – 2p = 1s – 2p = 1s – 2p

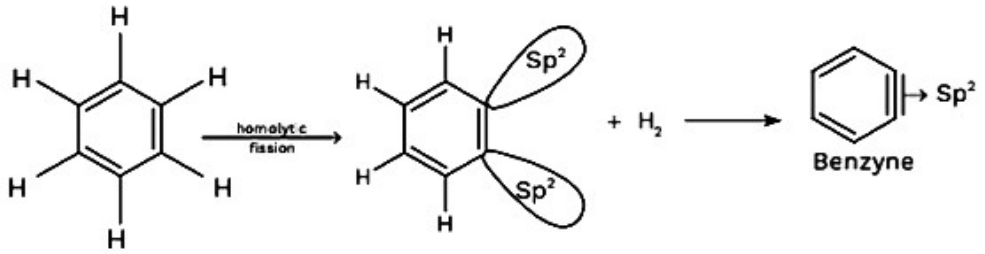
Based on above information one will expect the bond energy of three bonds to be same while fourth bond will have less energy, also bond lengths is also expected to be different in different type of overlapping. But practically all bonds are identical in shape, energy and bond angle is 109°28' which is not explained by overlapping of pure atomic orbitals. So it is explained by hypothetical phenomenon of intermixing of pure atomic orbitals named as hybridisation





Hybridisation is the process of intermixing of pure alumic orbitals having small energy difference and produce equal number of hybrid orbitals with same energy level.

→ Full filled, half filled and vacant orbital participate in hybridisation.
→ Hybrid orbitals are directional in nature and form σ bond.
→ Generally σ bond is formed by hybrid orbital while π bond is formed by pure orbitals.
→ In benzyne π bond is formed by sp^2 hybrid orbitals.



Size/Energy/directional property of hybrid orbitals $\propto \frac{1}{\text{S – character}}$

sp	sp^2	sp^3
50% s	33% s	25% s
50% p	67% p	75% p

p-character ↑, size ↑, Energy ↑, directional property ↑.

Types of Hybridisation :

Hybrid Orbital	Hybridisation	Atomic orbital involved	Shape	Bond Angle
2	sp	$sp_x / sp_y / sp_z$	Linear	180°
3	sp^2	$sp_xp_y / sp_yp_z / sp_yp_z$	Trigonal planar	120°
4	sp^3	$sp_xp_yp_z$	Tetrahedral	109°28'
5	sp^3d	$sp_xp_yp_zd_{z^2}$	Trigonal bi-pyramidal	120°,90°,180°

6	sp^3d^2	$sp_xp_yp_zd_{x^2-y^2}d_{z^2}$	Octahedral	90°
7	sp^3d^3	$sp_xp_yp_zd_{x^2-y^2}d_{z^2}d_{xy}$	Pentagonal bipyramidal	72°,90°,180°

Some examples of hybridization

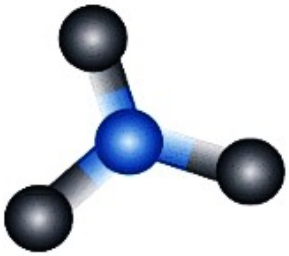
(1) sp-hybridisation → linear



Bond angle 180°

Example : $C_2H_2, BeCl_2, Co_2, C_3O_2, CdCl_2, HgCl_2$

(2) sp² hybridisation



Angle → 120°

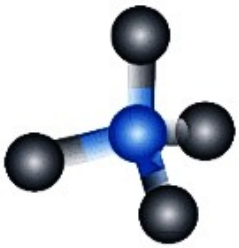
Trigonal planar

$(S + P_x + P_y)$

Example :

$BCl_3, AlCl_3, C_2H_4, BF_3, NO_3^-$, Graphite, etc

(3) Sp³ hybridisation

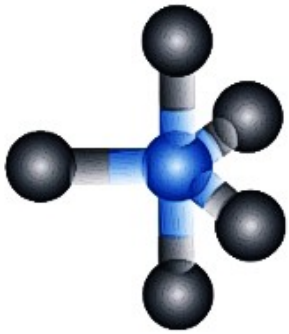


Angle 109°28'

Tetrahedral

Example : $CCl_4, NH_4^+, SO_4^{2-}, ClO_4^-, H_2O, NH_3$ diamond,

(4) Sp³d hybridisation :



Angle = 90°,120°

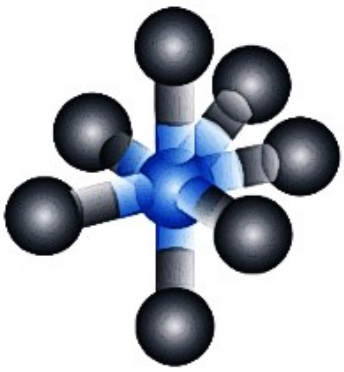
Example : $PCl_5, SF_4, ClF_3, XeF_2$

(5) $Sp^3d^2/d^2Sp^3 \rightarrow$ $S + Px + Py + Pz + d_{x^2-y^2} + d_{z^2}$



Bond angle = 90°, 180°
Example : SF_6

(6) sp^3d^3 hybridisation :



$1s + 3p + 3d\left(d_{xy} + d_{x^2-y} + d_{z^2}\right) \rightarrow$ 7 hybrid oribitals
Bond angle $\rightarrow$ 72°, 90°
Geometry $\rightarrow$ Pentagonal bipyramidal

Example : IF_7

Example	Value of bond angle	No. of B.A.
$BeCl_2$	180	1
BF_3	120	3
CH_4	109°,28	6
PCl_5	120 180	3 1
SF_6	90,90 180	6,12 3
IF_7	72 90 180	5 10 1

Calculation of Hybridisation :

Method 1
Calculate

$x = \sigma$ bond + lp

If x is

x	2	3	4	5	6	7
Hybridisation	**sp**	**sp^2**	**sp^3**	**sp^3d**	**sp^3d^2**	**sp^3d^3**

Method 2
Calculate

$$x = \frac{1}{2}\text{No. of valence } e^- \text{ oncentral atom + no. of monovalent atoms - (charge on species with sign)}$$

x	2	3	4	5	6	7
Hybridisation	**sp**	**sp^2**	**sp^3**	**sp^3d**	**sp^3d^2**	**sp^3d^3**

Example :

$$NH_3 = \frac{1}{2}[5+3-0] = 4 \rightarrow Sp^3$$

$$CO_2 = \frac{1}{2}[4+0-0] = 2 \rightarrow sp$$

$$NH_4^+ = \frac{1}{2}[5+4-(+1)] = 4 \rightarrow sp^3$$

$$ClO_4^- = \frac{1}{2}[7+0-(-1)] = 4 \rightarrow sp^3$$

$$BeCl_4^{2-} = \frac{1}{2}[2+4-(-2)] = 4 \rightarrow sp^3$$

$$SF_6 = \frac{1}{2}[6+6] = 6sp^3d^2$$

$$PCl = \frac{1}{2}[5+5-0] = 5\,sp^3d$$

$$SO_3 = \frac{1}{2}[6+0] = 3 \rightarrow sp^2$$

Method 3
In proper Lewis structure count the number of lone pairs and number of σ bonds with which that atom is directly connected to other atoms. The sum of lone pair and σ bonds

gives the number of hybrid orbitals of that atom.
For example hybridisation of following species:

(i)	CH_2Cl_2	$\rightarrow \sigma$ bonds = 4, L.P. = 0	sp^3
(ii)	PF_5	$\rightarrow \sigma$ bonds = 5, L.P. = 0	sp^3d
(iii)	$ICl_2^{\ominus}$	$\rightarrow \sigma$ bonds = 2, L.P. = 3	sp^3d
(iv)	BrF_5	$\rightarrow \sigma$ bonds = 5, L.P. = 1	sp^3d^2
(v)	$XeOF_4$	$\rightarrow \sigma$ bonds = 5, L.P. = 1	sp^3d^2
(vi)	SO_3	$\rightarrow \sigma$ bonds = 3, L.P. = 0	sp^2
(vii)	I_3^-	$\rightarrow \sigma$ bonds = 2, L.P. = 3	sp^3d

GEOMETRY OF MOLECULES

VALENCE SHELL ELECTRON PAIR REPULSION THEORY (VSEPR)

(a) Molecules having covalent bond have definite geometry as covalent bonding has directional characteristics. A simple theory was given for the molecular shape of the covalent molecules.

(b) This theory predicts the shape of the molecule by considering the most stable configuration of the bond angles in the molecule. This theory states

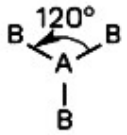
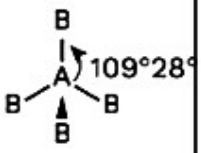
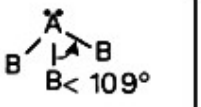
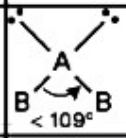
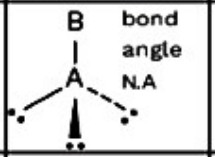
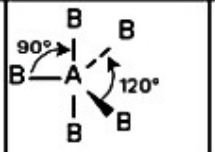
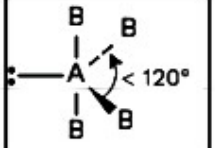
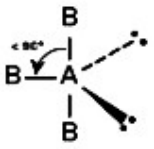
(i) Valence shell Electron pairs in the central atom of a molecule, whether bonded or unbonded are considered as occupying localized orbitals. These orbitals arrange themselves in so as to minimize the mutual electronic repulsions.

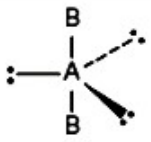
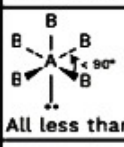
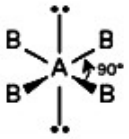
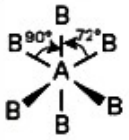
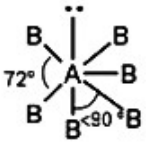
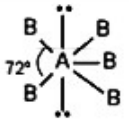
(ii) The magnitude of the different types of electronic repulsions follows the order given below:

lone pair-lone pair > lone pair - bond pair > bond pair - bond pair

(iii) The electronic repulsion between two pairs of electrons will be minimum if they are as far apart as possible.

Total no. of hybrid orbital	No. of b.p. (bond pairs)	No. of unshared pair i.e. LP	General formula	Type of hybridisation	Stereochemical formula/ str.	Shape	Example
2	2	0	AB_2	sp	$B \overset{180^\circ}{\frown} A - B$	Linear	$BeCl_2$

3	3	0	AB_3	sp^2		Trigonal planar	BF_3
3	2	1	AB_2	sp^2		V or Bent or angular	SO_2
4	4	0	AB_4	sp^3		Tetrahedron	CH_4
4	3	1	AB_3	sp^3		Trigonal pyra-midal	NH_3 CH^-_3
4	2	2	AB_2	sp^3		V or Bent or angular	H_2O SF_2
4	1	3	AB	sp^3		linear	ClO^-
5	5	0	AB_5	sp^3d		Trigonal bypy-ramidal	PCl_5
5	4	1	AB_4	sp^3d		Seesaw	SF_4
5	3	2	AB_3	$sp^3 < 90°$ d		T-shaped	ClF_3

5	2	3	AB_2	sp^3d		Linear	IF_2^-
6	6	0	AB_6	sp^3d^2	 All bond ∠= 90°	Octahedral	SF_6 IF_6^+
6	5	1	AB_5	sp^3d^2	 All less than 90°	Square pyra-midal	XeF_5^+
6	4	2	AB_4	sp^3d^2		Square planar	XeF_4
7	7	0	AB_7	sp^3d^3		Pentagonal bipyramidal	IF_7
7	6	1	AB_6	sp^3d^3		Distorted octahedral	XeF_6 IF_6^-
7	5	2	AB_5	sp^3d^3		Pentagonal	XeF_5^-

Hybridisation of species in solid state :

(1) Solid $PCl_5 = \underset{sp^3}{[PCl_4]^+}\ \underset{sp^3d^2}{[PCl_6]^-}$

(2) $PBr_5(s) = \underset{sp^3}{[PBr_4]^+}[Br^-]$

(3) $PI_5(s) = \underset{sp^3}{[PI_4]^+}[I]^-$

(4) $XeF_6(s) = \underset{sp^3d^2}{XeF_5^+} + F^-$

(5) $Cl_2O_6(s)=[\underset{sp^2}{ClO_2}]^+[\underset{sp^3}{ClO_4}]^-$

(6) $N_2O_5(s) = [\underset{Sp}{NO_2}]^+ [\underset{sp^2}{NO_3}]^-$

(7) $N_2O_4(s) = [\underset{Sp}{NO}]^+ [\underset{sp^2}{NO_3}]^-$

(8) $I_2O_4(s) = IO^+ + IO_3^-$

(9) $N_2O_3(s) = [\underset{Sp}{NO}]^+ [\underset{sp^2}{NO_2^-}]$

Participation of d-orbital in hybridisation on :

Molecular species which exist	Molecular species which donot exist
PF_5, PCl_5, PBr_5	PH_5
SF_4,SF_6,ClF_3	SH_4, SH_6
BrF_3, IF_7	ClH_3, BrH_3
XeF_2, XeF_4	IH_7, XeH_2, XeH_4
XeF_6	XeH_6
$2ICl_3$	ClF_3
PH_3, H_2S	PH_5, SH_4

Q.71 **In hybridization,**
(a) both half-filled and completely filled orbitals take part in hybridization
(b) only completely filled orbitals take part in hybridization
(c) only half-filled orbitals take part in hybridization
(d) only vacant orbitals take part in hybridization.

Sol. : (a)

Both half-filled and completely filled orbitals take part in hybridization.

Q.72 **When one s and three p orbitals hybridise,**
(a) four equivalent orbitals at 90° to each other will be formed
(b) four equivalent orbitals at 109°28' to each other will be formed
(c) four equivalent orbitals, lying in the same plane will be formed
(d) none of these.

Sol. : (b)

For sp^3 hybridisation, orbital geometry is tetrahedron with bond angle 109°28'.

Q.73 **The hybridisation of orbitals of N atom in NO_3^- and NH_4^+ are respectively**
(a) sp, sp^3 (b) sp^2, sp^3 (c) sp, sp^2 (d) sp^2, sp

Sol. : (b)

Hybridisation are : NO_2^- (sp^2) and NH_4^+ (sp^3).

Q.74 **The state of hybridisation of the central atom and the number of lone pairs over the central atom in BF_3 are**
(a) sp, 0 (b) sp^2, 0 (c) sp^3, 0 (d) dsp^2, 1

Sol. : (b)

In BF_3, the hybridization is sp^2 and number of lone pair = 0.

Q.75 **The types of hybridisation on the five carbon atoms from left to right in the molecule CH_2 CH CH_2 CH CH_2 are**
(a) sp^2, sp^2, sp^2, sp^2, sp^3 (b) sp^3, sp, sp^2, sp^2, sp^3
(c) sp^2, sp^2, sp^3, sp^2, sp^2 (d) sp^3, sp^2, sp, sp^2, sp^3

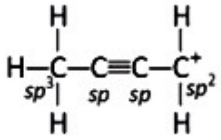
Sol. : (d) $\overset{sp^2}{CH_2} = \overset{sp^2}{CH} - \overset{sp^3}{CH_2} - \overset{sp^2}{CH} = \overset{sp^2}{CH_2}$

Q.76 **In which of the following species; all the three types of hybrid carbon atoms are present?**

(a) $CH_2 = C = CH_2$ **(b)** $CH_3 - CH = CH - CH_2^+$

(c) $CH_3 - C \equiv C - CH_2^+$ **(d)** $CH_3 - CH = CH - CH_2^-$

Sol. : (c)



Q.77 **Pick out the incorrect statement from the following.**

(a) sp^3d^2 hybrid orbitals are equivalent.

(b) sp^2 hybrid orbitals are equivalent.

(c) sp^3d hybrid orbitals are equivalent.

(d) sp^3d^3 hybrid orbitals are not equivalent.

Sol. : (c)

sp^3d orbitals are not equivalent hybrid orbitals because three lie in one plane at an angle of 120° and of the remaining two, one lies above and other lies below the plane.

BOND PARAMETERS

(I) Bond Angle (II) Bond Length (Bond distance) (III) Bond Energy

(I) Bond angle:

Angle between two adjacent bond in any species is called the Bond angle. The following factors affect bond angle.

(a) State of Hybridization :

With the change in state of hybridization relative percentage of s-character changes and with increasing percentage of s-character, the bond angle increase.

State of hybridization	% s-character	Bond angle
sp	50	180º
sp^2	33.33	120º
sp^3	25	109º 28′

If state of hybridization of central atom is same and all the surrounding atoms are identical and identically attached as well, the bond angle remains the same (as predicted by its hybridization state) no matter what the central atom and what

surrounding atoms are.

For example, bond angles in CF_4, $CC1_4$, CBr_4 is same i.e. 109º 28'

(b) Presence of lone pair :

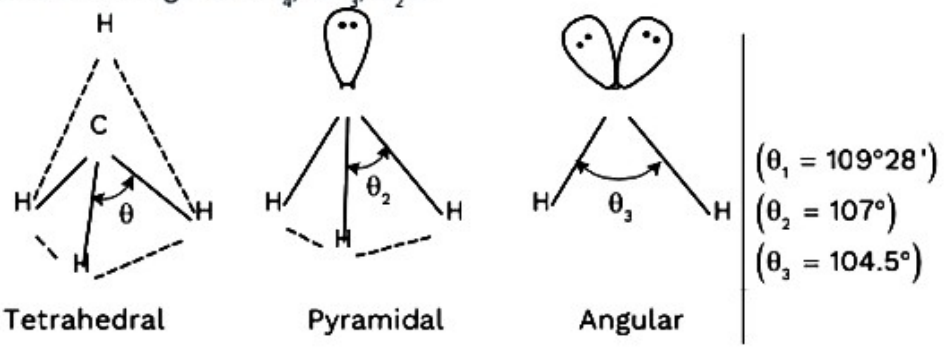
Since the lone pair occupies more space around central atom, so the space available for bond pairs is less and hence bond angle decreases.

The extent of repulsive interaction follows the order

lone pair - lone pair > lone pair – bond pair > bond pair – bond pair

Example :

The bond angle of CH_4, NH_3, H_2O.

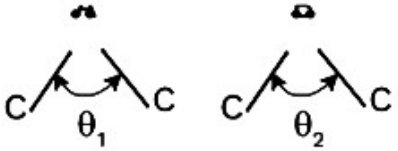


Tetrahedral Pyramidal Angular

As there is no lone pair of electron on carbon in CH_4 molecule, so all the $\hat{HCH}$ bond angles are equal (=109°28') but due to presence of one lone pair on N in NH_3 and two lone pair on O in H_2O, the lone pairs occupy more space around central atom, thus bond angle in CH_4 is maximum and in H_2O it is minimum. Hence, $q_1 > q_2 > q_3$

(c) Electronegativity of central atom :

Let us consider the molecule $\ddot{A}C_2$ and $\ddot{B}C_2$ in which electronegativity of A > B



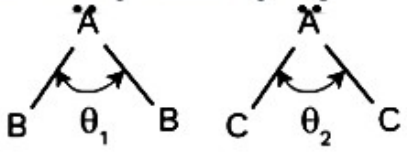
State of hybridisation of A and B = sp^2

Due to greater electronegativity of A than B, the bond pair of electron in AC_2 is more closer to A than B in BC_2. Due to decrease in distance between bond pairs in AC_2, bp-bp repulsion is more than that in BC_2.

So the bond angle order is $q_1 > q_2$

(d) Electronegativity of surrounding atom :

Let us consider two molecules $\ddot{A}B_2$ and $\ddot{A}C_2$ in both A is sp^2 hybridised and electronegativity of B > C. Due to greater electronegativity of B than that of C, the shared electron the pair is attracted towards more electronegativity B, so the repulsion between bp-bp in AB_2 is less than that in AC_2. Hence $q_1 < q_2$.



(e) Size of surrounding atom:

Let us consider the two molecules OCl_2 and OBr_2 where O-atom is sp^3 hybrid and Br atom is larger than Cl-atom, so Br will occupy more space around central atom than Cl-atom, hence bond angle in OBr_2 is more than that in OCl_2.

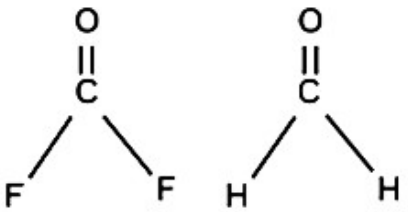
Moreover, it has been experimentally found that the bond angle in OBr_2 is 112º and in OCl_2 it is 111º, the bond angle is more than even the normal tetrahedral angle (109º28').

This may be due to steric repulsion of two Cl-atom in OCl_2 and two Br-atoms in OBr_2 and some back bonding.

(f) Multiple bond orbital repel other orbitals more strongly than single bond orbitals.

The multiple bond orbital possess extra electron density and naturally exert stronger repulsion on neighboring electron pair than a single bond does. Thus a double bond containing four electrons exerts a greater repulsive effect on other sigma bonds present in the molecule and result in a smaller bond angle.

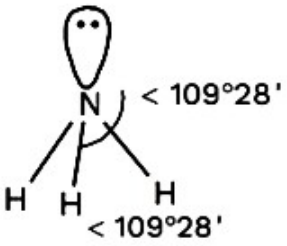
Ex.



Fluorine is more electronegative than hydrogen.

$\therefore \quad F-\hat{C}-F < H-\hat{C}-H$

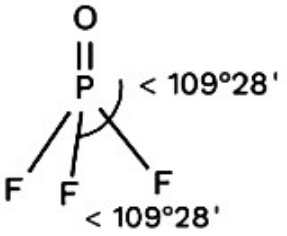
Ex.



lb-bp > bp-bp (repulsion)

shape – pyramidal

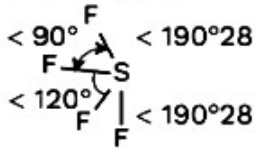
Ex.



D.B-S.B > S.B-S.B

shape – pyramidal

Ex.



F–S

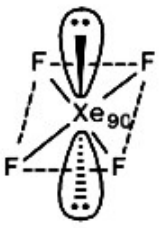
$F_{axial}-\hat{S}-F_{eq} < 90º$

$F_{axial}-\hat{S}-F_{eq} < 180º$

$F_{axial}-\hat{S}-F_{eq} < 120º$

lp–bp > bp–bp (repulsion)

Ex.

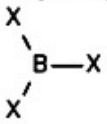


Both lone pair cancel out their effect

shape - square planar

Ex. Compare the bond angle in the following

(a) BF_3, BCl_3, BBr_3, BI_3 (X— $\hat{B}$ —X)



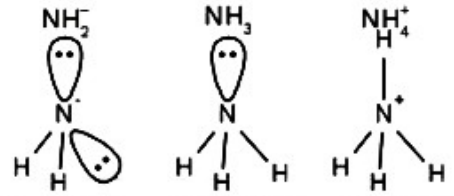
having perfect geometry therefore there is no effect of electronegativity of surrounding atom.

(b) NH_3, NF_3 (X– $\hat{N}$ –X)

solution: Fluorine is more electronegative than hydrogen therefore electron cloud is more shifted towards surrounding atom in NF_3 than NH_3.

$\therefore$ $NF_3 < NH_3$ (X– $\hat{N}$ –X)

(c)



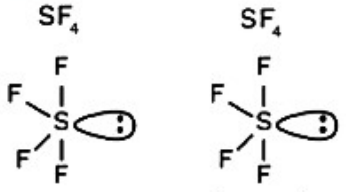
V-shape Pyramidal Tetrahedral

In all the above species N is sp^3 hybridized and all the attach surrounding atoms are identical in all species.

$\therefore$ greater the number of lone pair $NH_2^- < NH_3 < NH_4^+$ (H– $\hat{A}$ –H)

(F– $\hat{A}$ –F)

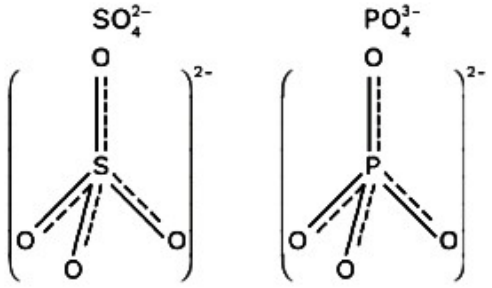
(d) In both cases central atom is sp^3d hybridized and surrounding atoms are same therefore greater the electro negativity of central atom greater will be the bond angle.



SF_4 SF_4

$SF_4 > SeF_4$ (F-A-F)

(e)



SO_4^{2-} PO_4^{3-}

Due to equivalent resonance both having perfect geometry therefore all the bond angle is same.

Q.78 **Compare the bond angle in following:**

(i) CO_2, SO_2, SO_3 **(ii) SiH_4, SiF_4, $SiCl_4$**

(iii) SO_4^{2-}, CO_3^{2-}, NO_3^- **(iv) COF_2 , $COCl_2$, $COBr_2$, COI_2**

(v) OF_2, $C1_2O$, Br_2O

Ans. **(i)** CO_2 SO_2 SO_3

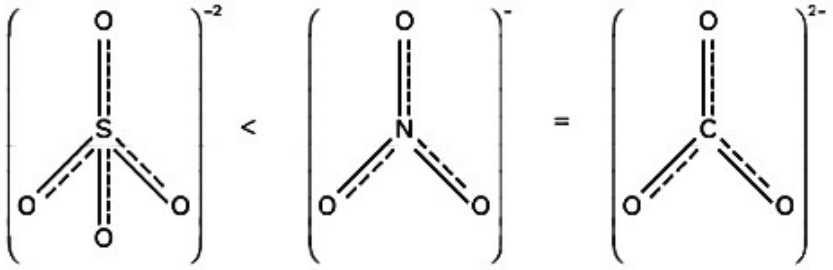
sp sp^2 sp^2

$CO_2 > SO_3 > SO_2$

In SO_2 molecule, due to presence of lone pair, bond angle decreases.

(ii) $SiH_4 = SiF_4 = SiCl_4$

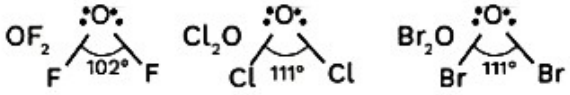
(iii)



(iv) $COF_2 < COC1_2 < COBr_2 < COI_2$

The steric crowding of the larger halogen atoms contributes in the increasing bond angles.

(v)



Due to high electronegativity of F atom, electron density gets polarized towards F in OF_2 and thus repulsion in bond pair decreases. The lone pair suppresses the bond angle to 102^0 from expected value of 109.5^0. On contrary, electron density gets polarized towards O atom in OCl_2, increasing bond pair-bond pair repulsion which results in higher bond angle of 111^0 from

expected value of 109.5^0

Points to remember

In larger halogen atoms (OBr_2) steric crowding also contributes to increasing bond angles.

Drago's rule :

When central atom of 3rd period or below elements in the periodic table do not show hybridisation while bonded with less electronegative atom like H (EN < 2.5), then lone pair is present in stereochemically inactive s-orbital and bonding will take place through pure p-orbitals.

Hence molecule like PH_3 SH_2 AsH_3 SeH_2 do not go under hybridization and bond angle of these molecules are nearly 90.

* This rule is applicable only when C.A. has lone pair

$CH_4 = SiH_4 = GeH_4 = SbH_4 = PbH_4$

Bond angle = 109°28'

$PH_3 \cong (94°)$
$AsH_3 \cong (92°)$
$SbH_3 \cong (91°)$
$BiH_3 \cong (90°)$

No. need of hybridisation

$H_2S \cong 92°$
$H_2Se \cong 91°$
$H_2Te \cong 90°$

No. hybridisation

Bent Rule :

In non-regular geometry

(i) More electronegative atom occupies that position of hybrid orbitals which has less -s – character or more p-character and less electronegative atom occupies that position of hybrid orbital which has more s-character and less p-character

(ii) Lone pair and double bond occupy that position of hybrid orbital which has more s-character.

(iii) Lone pair has slightly more percentage of s character than multiple bond..

$$\boxed{B.S \propto \frac{1}{B.L} \propto B.A. \propto \%\, s - characters}$$

$$\boxed{Cos\theta = \frac{s}{s-1} = \frac{P-1}{P}}$$ Where θ is Bond Angle

On heating PCl_5, Cl_2 is evolved due to breaking of axial bond.

$PCl_5 \xrightarrow{\Delta} PCl_3 + Cl_2$.

If all the bonds are sp^3d – 3p then why are there strength is different?

Reason: It is due to the difference in % s - character of hybrid orbital.

$$\cos\theta = \frac{s}{s-1} \quad \text{or} \quad \frac{p-1}{p}$$

Where θ is Bond Angle

Equatorial	Axial
$\theta = 120°$	$\theta = 90°$
$\cos 120° = \frac{s}{s-1}$	$\cos 90° = \frac{s}{s-1}$
% s = 33.33%	% s = 0

Key Point:

Bond Angle↑, % S ↑, attraction↑, Bond Length ↓, Bond Strength ↑

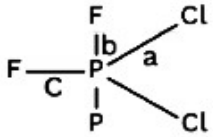
(Bond Length = Hybrid orbital length + size of Surrounding atom/central atom)

Example :

(1) $PCl_2F_3 \rightarrow sp^3d$

Bond angle $\theta = 90, \theta = 120$

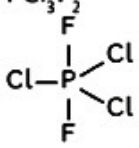
geometry Triagonal bipyramidal



a > c
b > c

$\mu \neq 0$ polar

(2) PCl_3F_2

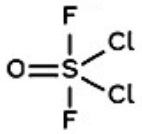


$\mu = 0$

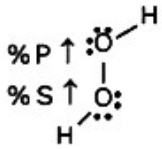
Axial B.L > equatorial B.L.

(3) $SOCl_2F_2$

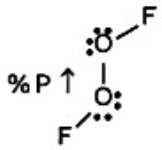
$5 + O$ sp^3d



(iv)

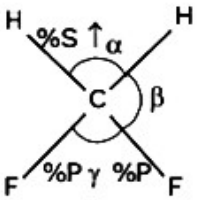


$x_1 = 1.48Å$



1.22Å

(v)



Order of bond angle $\alpha > \beta > \gamma$

Q.79 **Select the incorrect order of bond angle.**

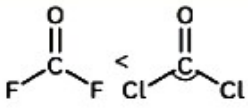
(A) $XeF_2 > XeO_3 > XeF_4$

sp^3d^2 sp^3 sp^3d^2

$180°$ $109°28'$ $90°$

(B) $H_2O < OF_2 < OCl_2$

(C) $COF_2 < COCl_2$



(D) $PH_3 < PF_3 < PCl_3$

(E) $PO_4^{3-} = SO_4^{2-} = ClO_4^-$

Sol. B

(viii) $NO_2^+ > NO_2 > NO_2^-$
180 134 120

Bent Rule :

In non-regular geometry

(i) More electronegative atom occupies that position of hybrid orbitals which has less -s – character or more p-character and less electronegative atom occupies that position of hybrid orbital which has more s-character and less p-character

(ii) Lone pair and double bond occupy that position of hybrid orbital which has more s-character.

(iii) Lone pair will always have more s-characters boiling point or melting point.

$$\boxed{B.S \propto \frac{1}{B.L} \propto B.A. \propto \%\,s - characters}$$

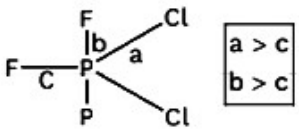
$$\boxed{Cos\theta = \frac{s}{s-1} = \frac{P-1}{P}}$$

% S and % P character

Example :

(1) $PCl_2F_3 \rightarrow sp^3d$

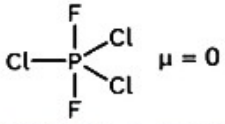
Bond angle $\theta = 90, \theta = 120$
geometry Triagonal bipyramidal



$\boxed{\begin{matrix} a > c \\ b > c \end{matrix}}$

$\mu \neq 0$ polar

(2) PCl_3F_2

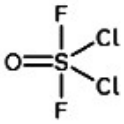


$\mu = 0$

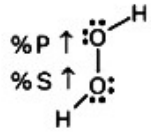
Axial B.L > equatorial B.L.

(3) $SOCl_2F_2$

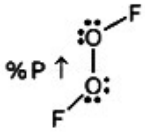
$5 + O\ sp^3d$



(iv)

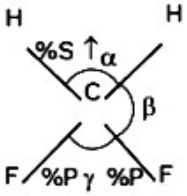


$x_1 = 1.48Å$



1.22Å

(v)

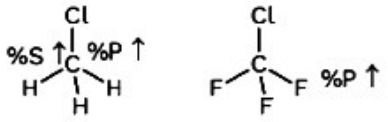


Order of bond angle $\alpha > \beta > \gamma$

$$\text{Bondangle } \alpha \%s - \text{character} \propto \frac{1}{\%P \text{ character}}$$

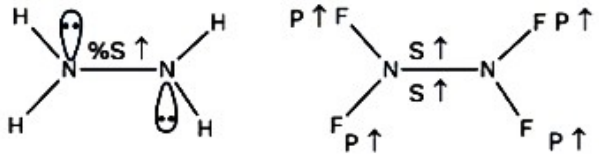
Q.80 Compare C- Cl bond length in CH_3Cl and CF_3Cl.

Sol.



$CH_3Cl > CF_3Cl$

Q.81 **Compare N – N bond length in N_2H_4 and N_2F_4.**



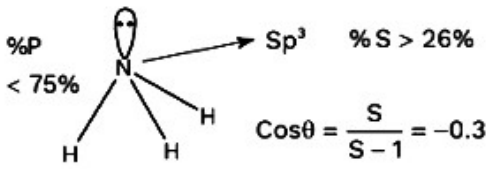
Sol. N – N bond length in N_2H_4 > N – N bond length in N_2F_4

Q.82 **Total number of molecule in which all possible bond angle are identical.**
$PF_3, CH_4, XeF_4, PF_5, IF_7, BeF_2, SF_6$

Sol. PF_3, CH_4, BeF_2, XeF_4

Q.83 **% P-character in lone pair of NH_3.**
(A) > 75% **(B) < 75%** **(C) 75%** **(D) 100%**

Sol.



$\cos\theta = \frac{S}{S-1} = -0.3$

$-0.3 = \frac{S}{S-1} = -0.3S + 0.3 = S$

$-0.3 = S + 0.3S$

$0.3 = 1.35$

$S = \frac{0.3}{1.3} = 0.23$

% s-character of lone pair $= 1 - 3(0.23)$

$= 1 - 0.64 = 0.31 > 25\%$

% p-character of lone pair $= 1 - 0.31 = 0.69$

$< 75\%$

Q.84 **In which one of the following species, the central atom has the type of hybridization which is not the same as that present in the other three?**

(a) $SbCl^{2-}$ **(b)** PCl_5 **(c)** SF_4 **(d)** I_3^-

Sol. : (a) SF_4

$SbCl^{2-}$ is sp^3d^2 while all other molecules/ions in the question involve sp^3d hybridization.

Q.85 **Predict the correct shape of BH_4^-.**

(a) Pyramidal, sp^3 **(b) Octahedral, sp^3d^2**

(c) Tetrahedral, sp^3 **(d) None of the above**

Sol. : (c)

Q.86 **BH^- has tetrahedral shape involving sp^3 hybridisation The structure and number of lone pair of IF_7 is**

(a) square pyramid, 0 **(b) trigonal bipyramid, 1**

(c) octahedral, 1 **(d) pentagonal bipyramid, 0**

Sol. : (d)

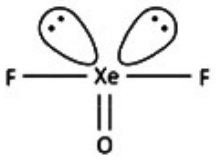
In IF_7, I atom undergoes sp^3d^3 hybridisation which results in pentagonal bipyramidal geometry. It has zero lone pair of electrons.

Q.87 **In which of the following molecules, the central atom has two lone pairs of electrons?**

(a) $XeOF_2$ **(b) BrF_5** **(c) SO2** **(d) $POCl_3$**

Sol. : (a)

In $XeOF_2$, there are two lone pairs of electrons.



Q.88 Which of the following are isoelectronic and isostructural?
NO_3^-, CO_3^{2-}, NH_4^+, CO_2
(a) CO^{2-}, CO_2 **(b) CO_2, NO^-** **(c) NH_4^+, CO_3^{2-}** **(d) NO_3^-, CO_3^{2-}**

Sol. : (d)

NO_3^- : trigonal planar, CO_3^{2-} : trigonal planar. NO_3^- and CO_3^{2-} are isoelectronic also.

Q.89 Which of the following has largest bond angle?
(a) NH_3 **(b) PH_3** **(c) H_2O** **(d) H_2S**

Sol. : (a)

The electronegativities of N (3.0) and O (3.5) are close, but NH_3 has one and H_2O has two lone pairs of electrons, therefore bond angle in NH_3 is larger (107°) than in water (104.5°). Similarly, the electronegativities of P(2.1) and S(2.5) are close but PH_3 has one while H_2S has two lone pairs of electrons, therefore, bond angle in PH_3 (93.8°) is greater than H_2S (92.1°).

Q.90 The bond angle in OF_2 out of OF_2, Cl_2O, Br_2O is minimum. It is because in case of OF_2
(a) electrons are nearer to fluorine due to high electronegativity of fluorine compare to Cl and Br

(b) lone pair-lone pair repulsion decreases bond angle
(c) both (a) and (b)
(d) none of these.

Sol. : (c)

The bond angle in OF_2 out of OF_2, Cl_2O, Br_2O is minimum. It is because in case of OF_2 electrons are nearer to fluorine due to high electronegativity of F compared to Cl and Br and because of this repulsion between electrons bond angle decreases.

Q.91 **The bond angle of Cl_2O is**
(a) greater than that of H_2O **(b) less than that of H_2O**
(c) equal to that of H_2O **(d) can't be predicted.**

Sol. : (a)

Both the compounds here have same central atom. But the equatorial atoms are different. Chlorine is more electronegative than hydrogen and thus, chlorine attracts the shared pair of electrons towards itself more strongly than oxygen does. Thus, causing more repulsion than that caused by oxygen in H_2O.

Q.92 **Which has lowest bond angle?**
(a) CH_4 **(b) NH_3** **(c) PH_3** **(d) H2O**

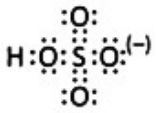
Sol. : (a)

PH_3, because it follows Drago's rule and hence, there would be no or very low hybridisation and hence, pure p-orbitals will be used. Therefore, bond angle is almost equal to 90°.

Q.93 **The formal charge on sulphur in HSO_4^- is**
(a) +2 **(b) –2** **(c) +4** **(d) –4**

Sol. : (a)

Lewis dot structure of HSO^- is



Formal charge = $V - L - \frac{1}{2}S$

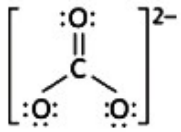
$= 6 - 0 - \frac{1}{2} \times 8 = +2$

Q.94 **The formal charge on double bonded O-atom in carbonate ion is**

(a) 0 **(b) +1** **(c) –2** **(d) +2**

Sol. : (a)

Lewis dot structure of CO_3^{2-} is



Formal charge on double bonded O - atom

$= V - L - \frac{1}{2}S = 6 - 4 - \frac{1}{2} \times 4 = 6 - 6 = 0$

4. Coordinate bond :

A coordinate bond is formed by sharing of an electron pair where shared electron pair is given by donor atom but shared by both.

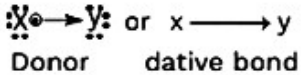
- It is formed buy intermolecular Lewis acid-base reaction.

Condition for coordinate bond :

(1) Octet of the donor atom must be completed and at least one lone pair must be present on it.

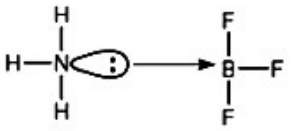
(2) There should be vacant orbital present in the acceptor.

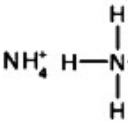
It is represented by arrowing



Donor dative bond

Example :

(i) 

(ii) NH_4^+  $H_3O^+, O_3, N_2O, N_2O_5, \left[Cu(NH_3)_4\right]^{+2}$

(iii) $K_4Fe(CN)_6, CO \quad :C \equiv O:$

Odd electron species

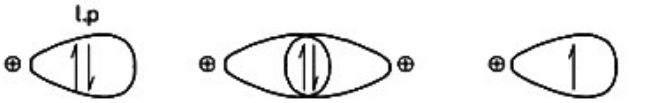
Diatomic

CN, OF, NO, O_2^-

Poly atomic

CF_3, CH_3, NO

ClO_2, ClO_3, triplet carbone



$O_2^- \rightarrow 17e^- \left(e.g\, KO_2, RbO_2, CsO_2\right)$

* Odd electron for from nucleus may or many not participate in hybridisation.

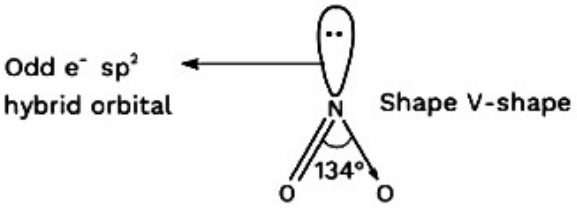
(1) Nitric Oxide (NO) : $15e^-$

$:\dot{N} \equiv O: \longrightarrow :N = O:$

Bond order = 2.5 According to MOT odd electron present in π^*MO = πABMO
At room temperature → Paramagnetic

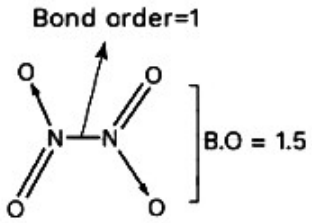
At low temperature $2NO \rightarrow N_2O_2$ (Diamagentic)

NO_2 (Nitrogen dioxide) : $23e^-$



NO_2 at room temperature – Brown colour gas and paramagnetic in nature.

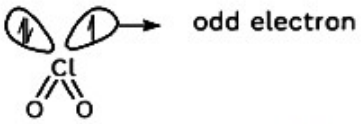
* At -11°C



$\overset{+}{N}O_2$	NO_2	NO_3^-	NO_2^-
180	134	120	115
sp	sp^2	sp^2	sp^2

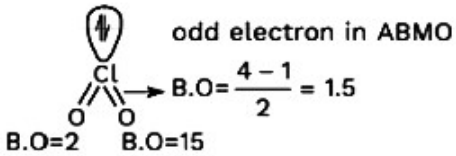
ClO_2 chlorine dioxide :

According to valence bond theory, B.O. of Cl – O = 2 but practically B.O < 2



It is explained by M.O.T

According MOT add electron present in antibonding molecular orbital.



3.0. of Cl – O = 1.75

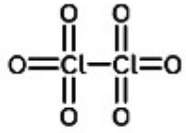
ClO_3



odd electron sp^3hybridorbital
shape pyramidal

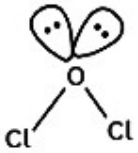
B.O=2 paramagnetic

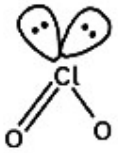
$$2ClO_3 \underset{\text{at low pressure}}{\overset{\text{at low temp}}{\rightleftharpoons}} Cl_2O_6 \xrightarrow{\text{dimer}} \text{diamagnetic}$$



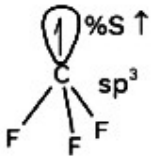
sp^3
$6d\pi - p\pi$ bonds

ClO_2	Cl_2O	ClO_2^-
118	111	$< 109°28'$





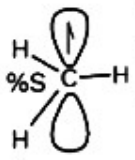
CF_3 :



%S ↑
gap ↓
due to presence of F (mixing)

Pyramidal (polar)

CH_3 :



%S ↓ gap ↑ mixing x
odd electron in pure p-orbital

Planar, Non-Polar

(1) $CF_3 > CH_3$ Hybrid>Pure orbital

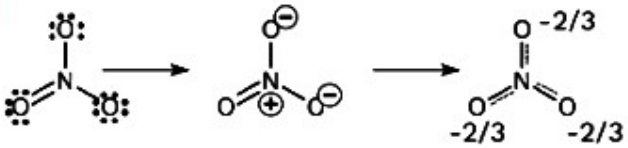
(2) $NO_2 > NO$
Localised electron Delocalised electron

(3) $\underset{\text{Localised}}{ClO_3} > \underset{\text{Delocalised}}{ClO_2}$

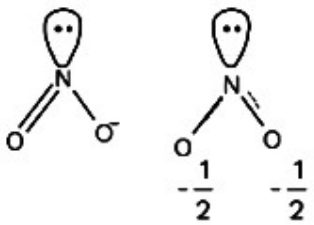
Question based on formal charge :

Q.95 **Compare formal charge on terminal oxygen atom in NO_3^- and NO_2^-ion**

Sol.



$NO_2^{\ominus}$



$\boxed{NO_3^- > NO_2^-}$

Sum of formed charge = Net charge

e.g. NO_3^-

$= 1 + 1(-1) + (-1) = -1$

Dimerization tendency :

(1) $CF_3 > CH_3$ Hybond>Pure arbital

(2) $\underset{\text{localised e-}}{NO_2} > \underset{\text{delocalised e}^-}{NO}$

(3) $\underset{\text{localised}}{ClO_3} > \underset{\text{delocalised}}{ClO_2}$

Dipole moment :

Definition: - Dipole moment is determined as the product of the magnitude of charge & distance of separation between these two charges.

$\boxed{\mu = q \times d}$ C × m (SI – unit), esu × cm. (CGS), Debye common unit

* It is a vector quantity.
Represent by arrow sign pointing towards negative end
Charge on ie = 1.6×10^{-19}C = - 4.8×10^{-10} esu.

$\boxed{1\text{debye} = 1\times10^{-18}\ \text{e.s.u.} \times \text{c.m}}$

$\boxed{1\text{debye} = 3.33\times10^{-30}\ \text{c} \times \text{m}}$

Application of dipole moment

$$\%\ \text{ionic character} = \frac{\mu_{observed}}{\mu_{theoretical}} \times 100\%$$

μobs → given

$\boxed{\mu_{theortical} = qxd}$

2. For homo diatomic molecules

$\Delta EN = 0 \quad \mu = 0$

H_2, N_2, O_2, and F_2 etc

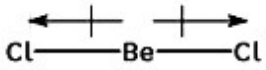
3. For poly atomic molecules :

$\boxed{\mu_R = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cos\theta}}$

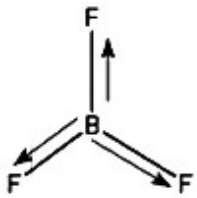
$\theta \uparrow \qquad \cos\theta \downarrow \qquad \mu \downarrow$

Direction of μ of l.p is away from central atom
Symmetrical molecular in which lone pair and bond pair are placed symmetrically has its zero dipole moment.

Example:

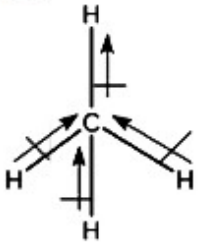


$\mu_R = 0$

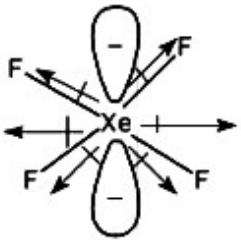


$\mu_R = 0$

Example:

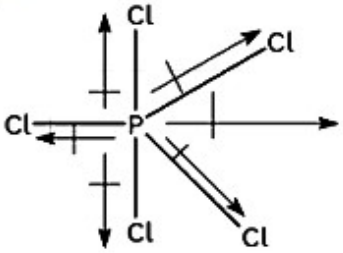


$\mu_R = 0$

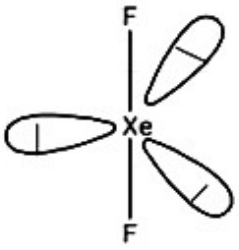


$\mu_R = 0$

Example:



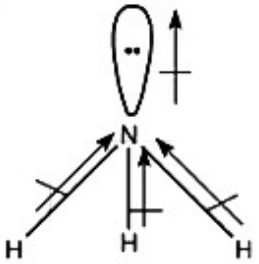
$\mu_R = 0$



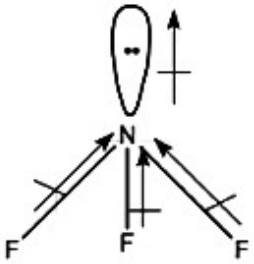
$\mu_R = 0$

Example:

.



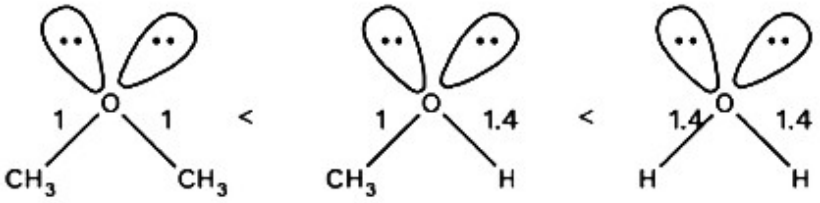
$\mu = 1.46D$



$\mu \cong 0 \quad \mu \neq 0$

$\mu = 0.23D$

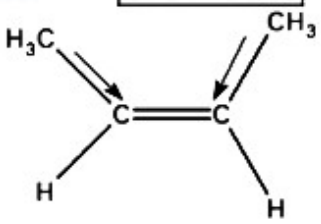
Example: $\underset{0.92}{H_2S} < \underset{1.46}{NH_3} < \underset{1.84}{H_2O} < \underset{1.82}{HF}$ all are polar $\Delta EN \uparrow$



$\cos\theta \downarrow \mu \downarrow$

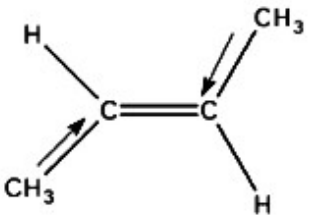
Case (1) same group

Example: CFS > trans



cis

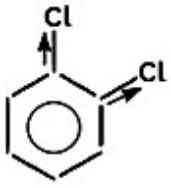
$\mu \neq 0$



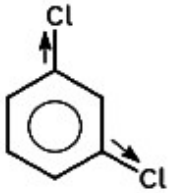
trans

$\mu \neq 0$

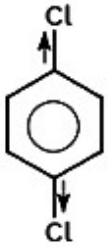
Example:



$\theta = 60^0$



$\theta = 120^0$

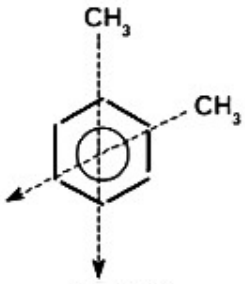


$\theta = 180^0$

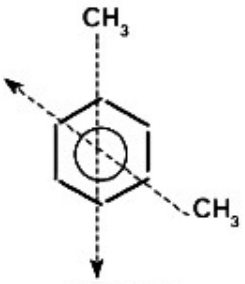
$\mu = \sqrt{3}\mu > \mu > O$

O m P

Example:



θ = 60°



θ = 120°

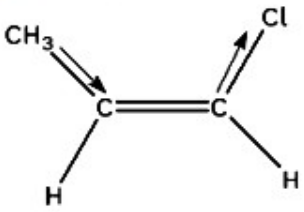


θ = 180°

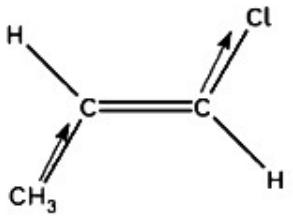
μ = O > M > P isomer

Case II diff group :

Example: trans > Cis

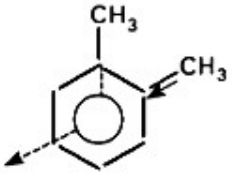


θ > 0°

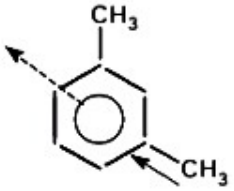


trans θ = 0°

Example:



θ = 120°



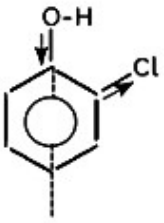
θ = 60°

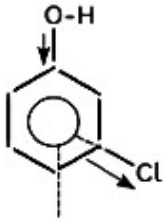


θ = 0°

μ = P > M > O ↑

Example:

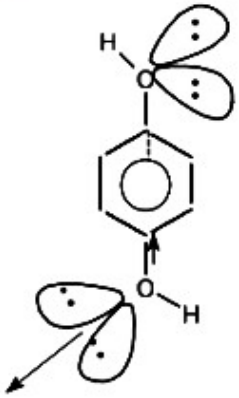




$\theta = 120$ $\theta = 60$

$\mu = \boxed{O < M < O} \uparrow$

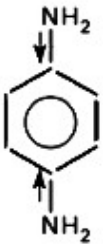
Example:



$\mu \neq O$ quinol

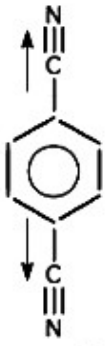
$\mu_{temp} = 4.8\,D$

Example:



$\mu \neq 0$

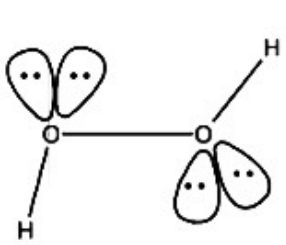
$\theta = 180°$



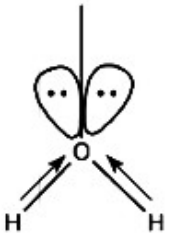
$\mu = 0$

$\boxed{\mu H_2O < \mu H_2O_2}$

Example:



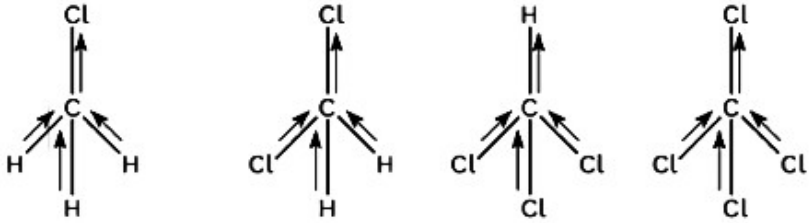
4 L.P + 2 b.P



2 L.P + 2 b.P

Q.96 **CH_3Cl, CH_2Cl_2, $CHCl_3$, CCl_4, arrange following in increasing order of dipole moment.**

Sol.



$\boxed{CH_3Cl > CH_2Cl_2 > CHCl_3 > CCl_4}$

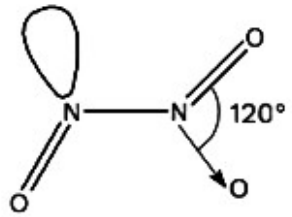
$CH_3F \quad CH_3Cl \quad CH_3Br \quad CH_3I$

Expected $CH_3F > CH_3Cl > CH_3Br > CH_3I$

Experimental $CH_3Cl > CH_3F > CH_3Br > CH_3I$

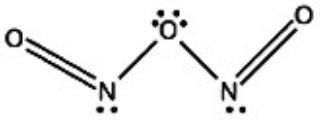
$\mu = q \times d$	$\mu = q \times d$
↓	↓
distance	charge
dominent	dominent

1. N_2O_3



Asymmetrical

l.P $\Rightarrow \mu \neq 0$



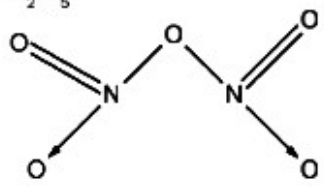
Symmetrical

V – shape $\mu \neq 0$

$$\underset{\text{brown gas}}{NO_2} + \underset{\text{Colourless}}{NO} \xrightarrow[-23^\circ C]{250K} \underset{\text{Blue solid}}{N_2O_3}$$

At – 25°C/-30°C blue liquid

2. N_2O_5



$\boxed{\mu \neq 0}$

3. $(CH_3)_3 - N \rightarrow O > (CH_3)_3 - P \rightarrow O$

4. $(CH_3)_3 - \overset{\oplus}{N} - \overset{\ominus}{O} \longrightarrow$

5. $CH_3 - C \equiv N \overset{\delta+ \; -\delta}{\longrightarrow} < CH_2 = CH - C \equiv N \overset{+\delta \; \delta-}{\longrightarrow}$

$CH_2 - CH = C = N^{\ominus}$ ⟶ charge separation

6. $C_2H_5OH < CH_3 - C(=O) - \ddot{O}H \leftrightarrow CH_3 - C(O^{\ominus}) = \overset{\oplus}{O}H$

More polar → H^+ release fast

7. $CH_3 - F < CO_3 - F$ (D is more the than H vibrational energy

Q.96 **In which of these molecule $\mu \uparrow$ to resonance.**

$C \equiv O$, $CH_2 - CH = Cl$, NH_2–⬡–NO_2, ⬡–F

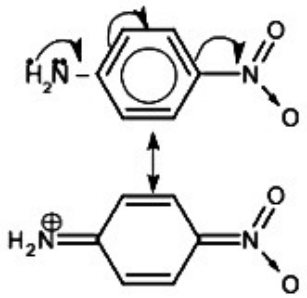
Sol.

(1) $C \equiv O$; $\overset{\ominus}{C} \equiv \overset{\oplus}{O}$; $\mu = 0.112D$

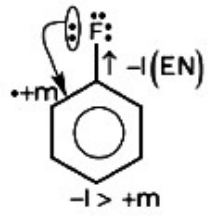
(2) $CH_2 = \overset{+\delta}{CH} - \overset{\delta-}{Cl}$ ⟵ $-I + M$ $-I + M$

$CH_2 - \overset{\ominus}{CH} = \overset{+\delta}{Cl}$

3.



4.



Q.97 **The dipole moments of CCl_4, $CHCl_3$ and CH_4 are in the order**

(a) $CH_4 = CCl_4 < CHCl_3$ **(b) $CHCl_3 < CH_4 = CCl_4$**

(c) $CH_4 < CCl_4 < CHCl_3$ **(d) $CCl_4 < CH_4 < CHCl_3$**

Sol. : (a) Dipole moment (μ) is zero for symmetrical molecules *i.e.*,

$\mu_{CCl_4} = \mu_{CH_4} = 0$ but $\mu_{CHCl_3} > 0$

Q.98 **The molecule which has zero dipole moment is**

(a) CH_2Cl_2 **(b) BF_3** **(c) NF_3** **(d) ClO2**

Sol. : (b)

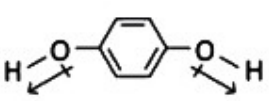
BF_3 has a symmetrical trigonal planar structure, hence it has zero dipole moment.

Q.99 **Dipole moment of cis and trans form of benzene-1, 4-diol is**

(a) μcis = 0, μtrans = 0 **(b) μcis $\neq$ 0, μtrans = 0**

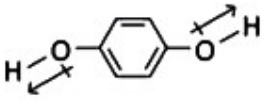
(c) μcis = 0, μtrans $\neq$ 0 **(d) μcis $\neq$ 0, μtrans $\neq$ 0**

Sol. : (b)



$\mu \neq 0$

cis-benzene-1, 4-diol



$\mu = 0$

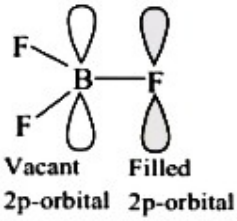
trans-benzene-1, 4-diol

Back bonding :

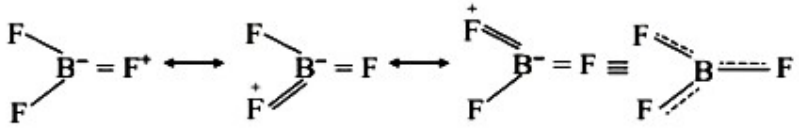
(1) BACK BONDING

It is formed due to transfer of lone pair from filled shell of an atom to unfilled shell of another bonded atom.

Consider the example of BF_3 molecule, here back bonding occurs due to transfer of 2p-electrons of fluorine into 2p empty orbital of boron atom. It completes the octet of boron.



Vacant 2p-orbital Filled 2p-orbital



Due to this delocalised p_π–p_π back bonding, B – F bond length decreases.

The extent of back bonding will be more if the orbitals involved are of same size.

For example the extent of back bonding in boron trihalides is as follows :

$BF_3 > BCl_3 > BBr_3 > BI_3$

The extent of back bonding decreases from BF_3 to BI_3 due to increasing size of p-orbitals from fluorine to iodine.

Molecules	Type of back bonding	Shape
BF_3	$2p_\pi$-$2p_\pi$	planar
$N(CH_3)_3$	no back bonding	pyramidal
$N(SiH_3)_3$	$2p_\pi$-$3d_\pi$	planar
$P(SiH_3)_3$	No back bonding	pyramidal
CH_3NCS	No back bonding	bent
SiH_3NCS	$2p_\pi$-$3d_\pi$	linear
$N(GiH_3)_3$	$2p_\pi$-$4d_\pi$	planar
GiH_3NCS	No back bonding	bent

- **Lewis Basic Order:** $N(CH_3)_3 > P(SiH_3)_3 > N(SiH_3)_3$
- **Lewis Acid Order:**
  $BF_3 < BCl_3 < BBr_3 < BI_3$
  $SiF_4 > SiCl_4 > SiBr_4 > SiI_4$
- **Carbene:**
  Triplet Carbene $:CH_2$ (Multiplicity = 3)
  Singlet Carbene $:CH_2$ (Multiplicity = 1)
  Triplet Carbene is more stable than singlet carbene.
- **Dichlorocarbene- $:CCl_2$**
  Singlet dichlorocarbene is more stable than triplet dichlorocarbene due to back bonding.

- **Effect of Back Bonding:**
  a. Bond Length must be decrease.
  b. Bond angle may be change
  c. Hybridization may be change
- When lone pair of central atom involved in back bonding, bond angle increases.
- When lone pair of surrounding atom involved in back bonding, bond angle increases due to bond pair- bond pair repulsion.
- **Comparison of Bond Length:**
  $d_{B\text{-}O}$ in $H_3BO_3 < d_{B\text{-}O}$ in $[B(OH)_4]^-$

d_{B-F} bond in BF_3 < d_{B-F} in BF_4^-

Hint:- No back bonding involved in $[B(OH)_4]^-$ and BF_4^-

Some Important Order:

- **Extent of Back Bonding:** OCl_2 < $O(SiH_3)_2$
- **Lewis Basic Nature:** NCl_3 > NF_3

  No back bonding in NCl_3 due to its explosive nature.
- **Metaboric Acid (HBO_2):** $HBO_2 \longrightarrow (BO_2^-)_n$

  $B_3O_6^-$ is aromatic, planer, O and B are sp^2 Hybridized
- $B_3N_3H_6 + 3\,HCl \longrightarrow B_3N_3H_9Cl_3$
- Inorganic Graphite $\xrightarrow{3000^\circ C}$ $(BN)_x$ Boron nitride 3 D network like structure (a- planer)

  $2p_p$-$2p_p$ back bonding involved in $(BN)_x$

Q.100 **In which of the following compound B-F bond length is found to be shortest**

(i) BF_4^- **(ii) $BF_3 \leftarrow NH_3$** **(iii) BF_3**

(iv) $BF_3 \leftarrow N(CH_3)_3$ **(v) BF_2H** **(vi) BFH_2**

Sol. : (vi) BFH_2

Q.101 **Arrange in Increasing order of B-F bond length?**

BF_3, BF_2NH_2, BF_2OH, BF_4O

Sol. $BF_3 < BF_2OH < BF_2NH_2 < BF_4O$

F < O < N

Order of e- donating tendency.

Q.102 **Order of Lewis acid strength in beryllium halide.**

Sol. $\underset{\boxed{\mu = 0}}{BeF_2} < BeCl_2 < BeBr_2 < BeI_2$

Order of L.A.S↑

Effect of back bonding > – I effect of halogen

$$\boxed{L.A.S \propto \frac{1}{B.B}}$$

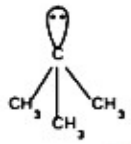
Q.103 **To predict the shape of**

(i) Trimethyl amine $(CH_3)_3N$ **(ii) Tri bolye phosphine** $(SIH_3)_3P$

(iii) Tri silyl amine $(SIH_3)_3N$

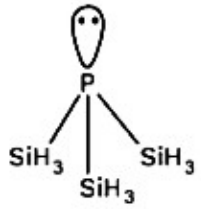
Sol.

(1)



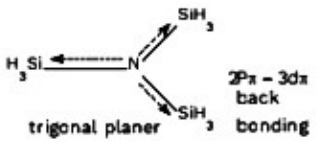
No back bonding Sp^3-pyramidal B.A. > $109^0 28'$

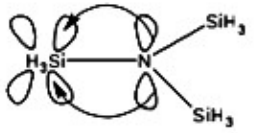
(2)



No back bonding Sp^3 pyramidal P and Si belongs to 3rd period.

(3) $N(SiH_3)_3$





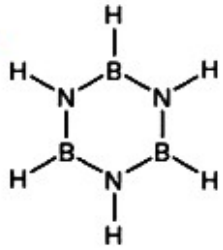
2pπ – 3dπ balk bonding

$(SiH_3)_3N < P(SiH_3)_3 < (CH_3)_3N$

Increasing order of L.B.S

Q.104 **Structure of $B_3N_3H_6$ inorganic benzene.**

Sol.



dipole moment =0

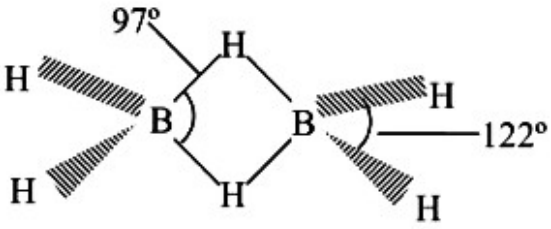
B – N Bond polar

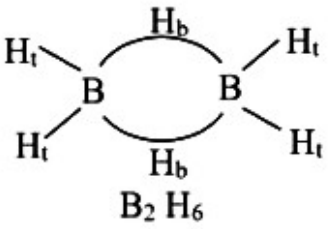
$NF_3 < NCl_3 < NBr_3 < NP_3$

L.B.S

BRIDGE BONDING

There are many compounds in which some electron deficient bonds are present apart from normal covalent bonds or coordinate bonds which are 2c-2e⁻ bonds(two centre two electron bonds). These electron deficient bonds have less number of electrons than the expected such as three centre-two electron bonds (3c-2e) which are present in diborane B_2H_6, $Al_2(CH_3)_6$, BeH_2(s) etc.]

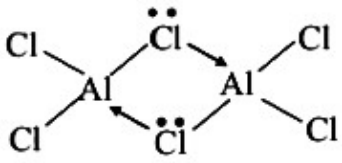




B_2H_6

The structure of diborane containing four terminal(t) and two bridging(b) hydrogen atoms. The model determined by molecular orbital theory indicates that the bonds between boron and the terminal hydrogen atoms are conventional 2c-2e⁻ covalent bonds. The bonding between the boron atoms and the bridging hydrogen atoms is, however different from that in molecules such as hydrocarbons. Having used two electrons in bonding to the terminal hydrogen atoms, each boron has one valence electron remaining for additional bonding. The bridging hydrogen atoms provide one electron each. Thus the B_2H_2 ring is held together by four electrons, an example of 3c-2e⁻ bonding. This type of bond is sometimes called as 'banana bond'. Group 13, gallium is known to form a similar compound, digallane, Ga_2H_6. But Al_2Cl_6 have covalent bond only and there is no electron deficient bonding as depicted

in the given structure.



(Al – Cl – Al is 3c – 4e bond)

Keypoints:

Two Types:

(i) 3c– 4 e⁻bond (surrounding atom has lone pair)

(ii) 3c– 2 e⁻bond (surrounding atom has no lone pair)

3c– 4 e⁻ bond remove more deficiency as compared to 3c– 2e⁻.

Vacant orbital takes part in hybridisation.

(A) Al_2Cl_6 :

Information:-

- Number of 3c–4 e⁻ bond = 2,
- Number of 2c-2 e⁻ bond = 4
- Hybridisation of central atom = sp^3. Octet: complete
- Octet complete

(B) I_2Cl_6 :

Information:-

- Hybridisation of central atom = sp^3d^2
- Maximum number of atom in a plane = 8
- Number of 3c–4 e⁻ bond = 2,
- Number of 2c-2 e⁻ bond = 4
- Octet complete

(C) B_2H_6:

Information:-

- Planarity:- Non- planar
- Maximum number of atom in a plane = 6
- Octet: incomplete, Lewis Acid
- 3c-2 e⁻ bond= 2
- 2c-2 e⁻ bond= 4
- Bridge bonds are stronger and longer than the terminal bonds.
- 2 bridging 'H' are perpendicular to the plane.

(D) $Al_2(CH_3)_6$:

Information:-

- Hybridisation of central atom = sp^3
- Planarity:- non- planar
- Number of 3c-2 e^- bond = 2
- Number of 2c-2 e^- bond = 22
- Maximum number of atoms in one plane = 10

(E) Be_2Cl_4

Information:-

- Hybridisation of central atom = sp^2
- Planarity:-Planar (all atoms are in one plane)
- e^- Deficient
- Number of 3c–4 e^- bond = 2
- Number of 2c-2 e^- bond = 2

(F) Be_2H_4 :

Information:-

- Hybridisation of central atom = sp^2
- Planarity:-Planar (all atoms are in one plane)
- Number of 3c-2 e^- bond = 2
- Number of 2c-2 e^- bond = 2

(G) $BeH_{2(s)}$(Polymer):

Information:-

- Hybridisation of central atom = sp^3
- Planarity:-Non- Planar
- Borane:

(a) NidoBorane: B_nH_{n+4} Ex:B_2H_6 (di borane), B_3H_7 (tri borane)

(b) ArachnoBorane: B_nH_{n+6} Ex:B_2H_8 (di borane –8), B_3H_9 (tri borane -9)

$$B_2H_6 + 2X \longrightarrow [X \longrightarrow BH_2 \longleftarrow X]^+ [BH_4]^-$$

Unsymmetrical cleavage is done by the molecules having H- Bond.

Q.105 **$N(SiH_3)_3$ has -**

(A) sp^3 hybridisation, pyramidal shape

(B) sp^2 hybridisation, planar shape

(C) sp^3 hybridisation, tetrahedral shape

(D) dsp^2 hybridisation, square planar shape

Q.106 **Correct statement regarding this reaction**

$BF_3 + NH_3 \rightarrow [F_3B \leftarrow NH_3]$

(A) Hybridisation of N is changed

(B) Hybridisation of B is changed

(C)NH_3 act as a lewis base

(D) (B) & (C) both

Q.107 **Shape of $N(CH_3)_3$ is**

(A) pyramidal **(B) Linear**

(C) Tetrahedral **(D) None of these**

Q.108 **Which of the following molecule is stable by back bonding**

(A) $AlCl_3$ **(B) AlH_3**

(C) BCl_3 **(D) ICl_3**

Q.109 **Which of the following represents (3C-4e$^-$) Bridge bonding**

(A) Al_2Cl_3 **(B) BH_3**

(C) B_2H_6 **(D) none of these**

Molecular orbital theory (M.O.T.)

Given by hund's & Mulliken

Drawbacks of V.B.T.

It does not explain

(1) Formation of odd e^- species

(2) Paramagnetic behaviour of O_2

(3) Existence of C_2 having only two π bond

(4) IE of $N_2 > N$ while $O_2 < O$

(5) Colour of halogen

MOT:

(1) Two atomic orbital come nearer & then overlap each other to form two molecular orbitals (MO)

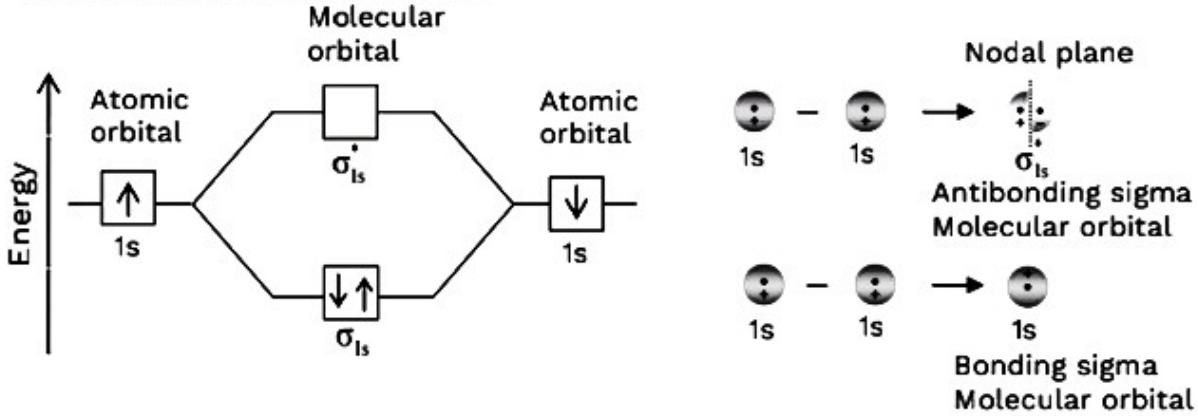
(2) Atomic orbital (AO) combines to form molecular orbital (MO).

Types of molecular orbitals

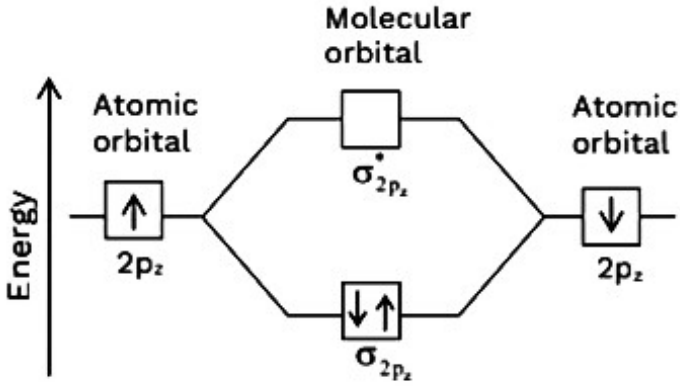
Molecular orbitals of diatomic molecules are designated as σ(sigma), π(pie), δ(delta) etc.

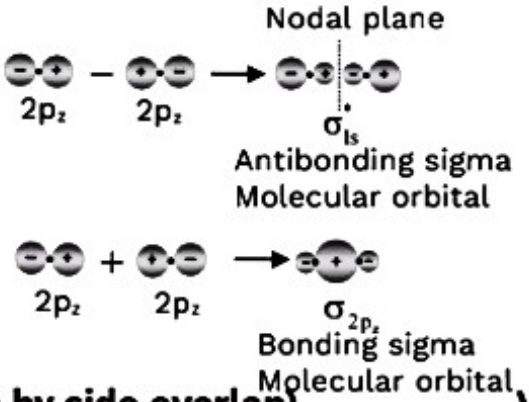
In this nomenclature, the sigma (σ) molecular orbitals are symmetrical around the inter molecular axis (assumed to be z-axis) while pi (π) molecular orbitals are not symmetrical.

(a) s-s combination of orbitals

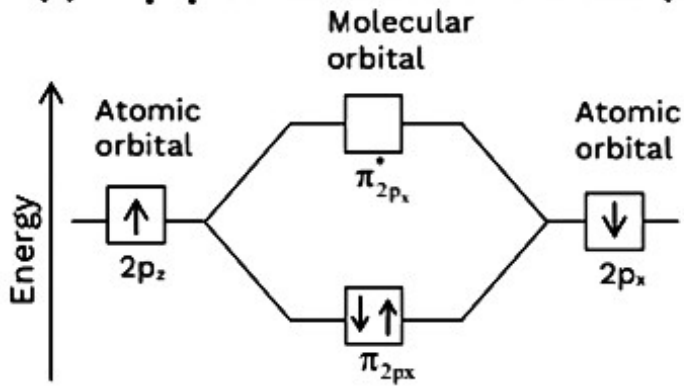


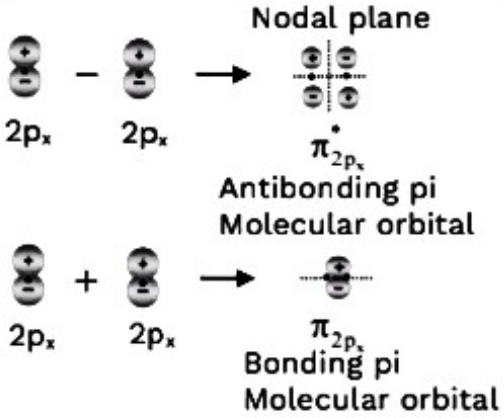
(b) p-p combination of orbital(end to end overlap)



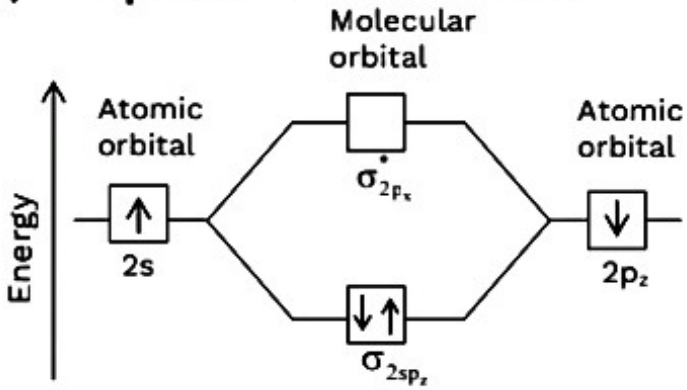


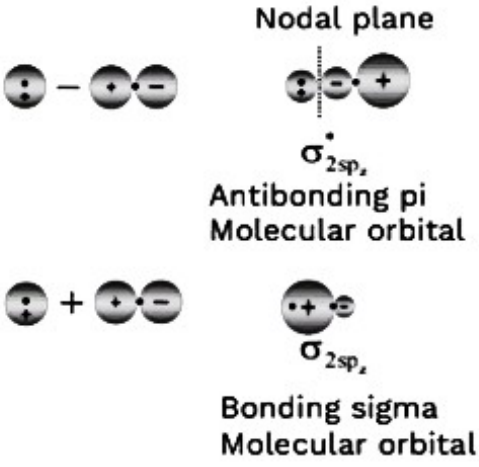
(c) p-p combination of orbitals (side by side overlap) \s





(d) s-p combination of orbitals





(3) Energy of BMO < Energy of ABMO.

(4) Molecular orbitals can be filled by electrons according to Aufbau, Hund's, Pauli's principle.

(5) Energy order of the molecular orbitals of homonuclear di-atomic molecules.

Note:1 Molecular orbital energy diagram for up to N_2 (molecule having ≤ 14 electrons)

$$\sigma_{1s} < \sigma^*_{1s} < \sigma_{2s} < \sigma^*_{2s} < \pi_{2p_x} = \pi_{2p_y} < \sigma_{2p_z} < \pi^*_{2p_x} = \pi^*_{2p_y} < \sigma^*_{2p_z}$$

Note:2 Molecular orbital energy diagram for O_2 and F_2 (molecule having > 14 electrons)

$$\sigma_{1s} < \sigma^*_{1s} < \sigma_{2s} < \sigma^*_{2s} < \sigma_{2p_z} < \pi_{2p_x} = \pi_{2p_y} < \pi^*_{2p_x} = \pi^*_{2p_y} < \sigma^*_{2p_z}$$

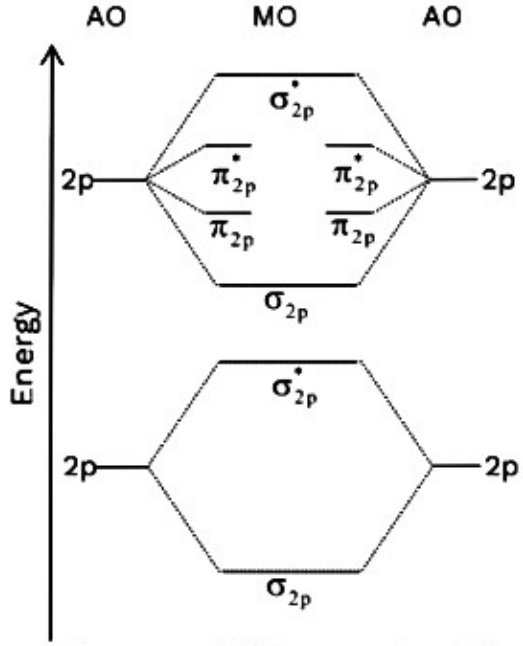
σ *, π * = antibonding molecular orbital

σ, π = bonding molecular orbital

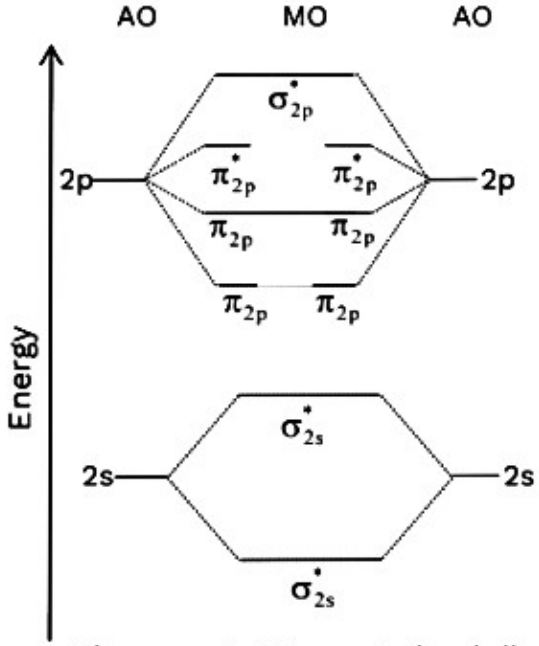
Ex. Why molecular orbitals have different order of energy in N_2 & O_2 ?

Sol. s-p mixing

Hint



The correct MO energy-lavel diagram When s-p mixing is not allowed.



The correct MO energy-lavel diagram When s-p mixing is not allowed, the energies of the σ_{sp} and π_{2p} orbitals are reversed.

B.M.O	A.B.M.O
(1) A.O. must be in same phase	A.O. must be indifferent phase
(2) e^- density max at mid pious	e^- density zero at mid point

(3) Represent attraction	Represent repulsion
(4) Over sales than A.O.	Energy > A. O
(5) Stability > A. O	Stability < A. O
(6) Represent by σ, π, S	Represent σ^*, π^*, S^*

Bond Order

Bond order can be defined as:

$$\text{Bond order} = \frac{N_b - N_a}{2}$$

N_b = No. of electron in bonding MO's

N_a = No. of electron in antibonding MO's

- If bond order = 0, it means species does not exist.
- Bond order of 1, 2 & 3 corresponds to a single bond, double & triple bond respectively.
- Bond order $\propto$ stability $\propto$ of molecule $\propto$ bond length $\downarrow$

Magnetic behaviour

- If the molecule has one or more unpaired electron, it will be paramagnetic,
- If all the electrons are paired it will be diamagnetic.
- Magnetic strength can be calculated by using spin only formula of magnetic moment (μ).
- $\mu = \sqrt{n(n+2)}$ **B.M.** (where n = number of unpaired electron)

Ex. H_2 = Configuration: $\sigma^2_{(1s)}, \sigma^{*0}_{(1s)}$

No unpaired electron
Hence H_2 is diamagnetic

$$\text{Bond order} = \frac{N_b - N_a}{2} = \frac{2-0}{2} = 1,$$

Note :

1. If bond order is fractional, then it wants to make it integer greater than previous

$$\underset{2.5}{NO} \xrightarrow{-e^-} \underset{3}{NO^+}$$

2. If bond order is greater, then molecule is more stable.

3. $O_2 < NO$ (2 < 2.5)

CO — CO^+

Bond order: 3 < 3

Bond length: 1.128A° > 1.115A°

Electron is removed from NBMO having slight anti-bonding character, so bond order slightly increase.

Q.97 **According to Molecular Orbital Theory, the shape and size of a molecular orbital depends upon_____**

(a) only shape of the combining atomic orbitals

(b) shape, size, orientation and numbers of the combining atomic orbitals

(c) only orientation of the combining atomic orbitals

(d) only size of the combining atomic orbitals.

Sol. : (b)

The shape and size of a molecular orbital depend upon the shape, size, number and orientation of the combining atomic orbitals.

Q.98 **In an anti-bonding molecular orbital, electron density is minimum**

(a) around one atom of the molecule

(b) at a point away from nuclei of the molecule

(c) between two nuclei

(d) at no place.

Sol. : (c)

Anti-bonding molecular orbitals has minimum electron density between the two nuclei of the atoms.

Q.99 **In a molecule number of electrons in BMO are more as compared to ABMO, hence**

(a) a bond will be formed **(b) no bond will be formed**

(c) information is not sufficient **(d) none of these.**

Sol. : (a)

In a molecule number of electrons in BMO are more as compared to ABMO, hence a bond will be formed.

Q.100 **The relative energies of molecular orbitals depend upon**
(a) the energies of atomic orbitals combining to form MOs.
(b) the extent of overlapping between the atomic orbitals
(c) both (a) and (b)
(d) none of these.

Sol. : (c)

The relative energies of molecular orbitals depend on both (a) and (b).

Q.101 **How many nodes are there in the antibonding molecular orbital formed by two 2s atomic orbitals?**
(a) 2 **(b) 1** **(c) 0** **(d) 3**

Sol. : (b)

Number of nodes = n – 1 where, n is principal quantum number, i.e., 2 here. Antibonding orbitals always have nodes. Normally this orbital is empty, but if it is occupied, the wave nature of electron density is out of phase which leads to destructive interference and node formation. Overall, no. of nodes present in 2s antibonding orbital is one.

Q.102 **Which of the following is incorrect statement?**
(a) sp hybrid orbital has lower energy than unhybrid p-orbitals
(b) The set of sp-orbital appears similar in shape to the original p-orbital.
(c) σ bond in HF is an example of s and p-combination.
(d) None of these

Sol. : (d)

All of the given statements are correct.

Q.103 **Which statement is false?**

A sigma molecular orbital

(a) may result from overlap of p atomic orbitals perpendicular to the molecular axis (side-on)

(b) may result from overlap of p atomic orbitals along the molecular axis (head-on)

(c) may result from overlap of one s and one p atomic orbitals

(d) may be either bonding or antibonding.

Sol. : (a)

A sigma molecular orbital may result from overlap of p atomic orbitals along to the molecular axis (head-on).

Q.104 **In which of the following pi (π) bond is weaker in comparison to other one?**

(a) p-p overlapping **(b) p-d overlapping**

(c) s-s overlapping **(d) s-p overlapping**

Sol. : (b)

Smaller orbitals have greater electron finding probability than larger orbitals. Electron dense orbital overlap strengthens bond and form strong pi bond. Therefore, strength of p-p > p-d.

Q.105 **Which of the following is correct statement?**

(a) Probability of finding the electron in bonding molecular orbitals is less than combining atomic orbitals.

(b) Bonding molecular orbitals are formed when opposite signs of orbitals overlap.

(c) d-d Combination of atomic orbitals gives δ and δ^* MO's

(d) None of these

Sol. : (c)

d-d Combination of atomic orbitals gives δ and δ^* molecular orbitals.

Q.106 **Identify the incorrect statement.**

(a) A non-bonding orbital (NBMO) is a molecular orbital for which the addition or removal of an electron does not change the energy of the molecule.

(b) The closer the energies of atomic and molecular orbitals, the more non-bonding the molecular orbitals.

(c) Electrons in a non-bonding molecular orbital are lower in energy than in a bonding molecular orbital.

(d) None of these.

Sol. : (c)

Electrons in a non-bonding molecular orbital are higher in energy than in a bonding molecular orbital, but lower in energy than electrons in an antibonding molecular orbital.

Q.107 **The bond order for carbon monoxide is**

(a) 1 **(b) 2** **(c) 3** **(d) 4**

Sol. : (c)

Carbon monoxide has 10 bonding and 4 antibonding electrons. Therefore, bond order =3

Q.108 **The bond order and the magnetic characteristic of CN– are**

(a) 3, diamagnetic **(b) 3, paramagnetic**

(c) paramagnetic **(d) diamagnetic**

Sol. : (a)

Molecular orbital configuration of CN–(14) is KK $\sigma 2s^2$, $\sigma^* 2s^2$, $\pi 2px^2 = \pi 2py^2$, $\sigma 2pz^2$

As all electrons are paired, so CN– is diamagnetic. Bond order = $\frac{1}{2}(8-2) = 3$

Q.109 **What will be the bond order of resultant species when hydrogen atom containing one electron is combined with hydrogen ion having no electron?**

(a) 1 **(b) 3** **(c)** $\frac{1}{2}$ **(d)** $\frac{3}{2}$

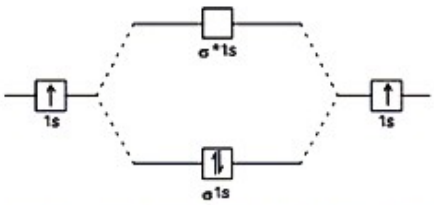
Sol. : (c)

$H + H^+ \rightarrow H^+$

The electronic configuration of H^+ is $\sigma 1s^1$

Bond order = $\frac{N_b - N_a}{2} = \frac{1-0}{2} = \frac{1}{2}$

Q.110



Calculate the bond order for given molecular orbital energy level diagram of H2.

(a)1 **(b) 2** **(c) 3** **(d) 4**

Sol. : (a)

H2 : $\sigma 1s^2$

Bond order = $\frac{N_b - N_a}{2} = \frac{2-0}{2} = 1$

Q.111 **Which of the following has bond order 0.5?**

(a) B_2 **(b) N_2** **(c) He_2^+** **(d) He_2**

Sol. : (c)

He^+ electronic configuration : $\sigma 1s^2$, $\sigma^* 1s^1$

B.O. = $\frac{2-1}{2} = 0.5$

Q.112 **In which of the following pairs of molecules/ions, both the species are not likely to exist?**

(a) H_2^-, He_2^{2+} **(b)** H_2^+, He_2^{2-} **(c)** H_2^-, He_2^{2-} **(d)** H_2^{2+}, He_2

Sol. : (d)

H_2^+ : $(\sigma 1s)^0$

Bond order = 0 He2 : $(\sigma 1s)2(\sigma^*1s)2$

Bond order = $\frac{2-2}{2}=0$

Q.113 **Total number of unpaired electrons in Li_2 molecule is**

(a) 0 **(b) 1** **(c) 2** **(d) 3**

Sol. : (a)

Electronic configuration of Li_2 = $\sigma 1s2$, σ^*1s2, $\sigma 2s2$

Number of unpaired electrons = 0

Q.114 **Which one of the following species does not exist under normal conditions?**

(a) B_2 **(b)** Li_2 **(c)** Be^+ **(d)** Be_2

Sol. : (d)

Bond order of Be_2 is zero, so it does not exist.

Q.115 **Total number of electrons present in π MOs of B_2 molecule is**

(a) 4 **(b) 2** **(c) 3** **(d) 5**

Sol. : (b)

B_2 : KK $\sigma 2s2$ σ^*2s2 $\pi 2p_x^1 = \pi 2p_y^1$

Q.116 **C_2 molecule is a diamagnetic species with bond order 2. Double bond in C_2 molecule consists of both the pi bonds because of the presence of electrons in two pi MOs.**

(a) 2 **(b) 6** **(c) 4** **(d) 8**

Sol. : (c)

Double bond in C_2 molecule consists of both π-bonds because of the presence of 4 electrons in two pi MOs.

Q.117 **Which of the following order of energies of molecular orbitals of N_2 is correct?**

(a) $(\pi 2p_y) < (\pi^* 2p_x) \approx (\pi^* 2p_y) < (\sigma 2p_z)$

(b) $(\pi 2p_y) > (\sigma 2p_z) > (\pi^* 2p_x) \approx (\pi^* 2p_y)$

(c) $(\pi 2p_y) > (\sigma 2p_z) < (\pi^* 2p_x) \approx (\pi^* 2p_y)$

(d) $(\pi 2p_y) < (\sigma 2p_z) < (\pi^* 2p_x) \approx (\pi^* 2p_y)$

Sol. : (d)

The correct order of energies of molecular orbitals in N2 molecule is

$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < \pi^* 2p_x = \pi^* 2p_y < \sigma^* 2p_z$

Q.118 **During the change : O_2 $O^+ + e^-$, bond order changes from 2 to.**

(a)1 **(b) 1.5** **(c) 2.5** **(d) 3**

Sol. : (c)

Electronic configuration of O^+ is

KK $\sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2, \pi 2p_y^2, \pi^* 2p_x^1$

B.O. $= \frac{1}{2}(8-3) = 2.5$

Q.119 **Which of the following are correctly arranged in an increasing order of their bond strengths?**

(a) $O_2^- < O_2 < O_2^+ < O_2^{2-}$ **(b)** $O_2^{2-} < O_2 - < O_2 < O_2^+$

(c) $O_2^- < O_2^{2-} < O_2 < O_2^+$ **(d)** $O_2^+ < O_2 < O_2 - < O_2^{2-}$

Sol. : (b)

Molecular orbital configuration of

$O_2 \Rightarrow (\sigma 1s)2(\sigma^*1s)2(\sigma 2s)2(\sigma^*2s)2(\sigma 2pz)2(\pi 2px)2\ \pi(2py)2(\pi^*2px)1(\pi^*2py)1;$

$$\text{B.O.} = \frac{10-6}{2} = 2$$

$O_2^+ \Rightarrow (\sigma 1s)2(\sigma^*1s)2(\sigma 2s)2(\sigma^*2s)2(\sigma 2pz)2(\pi 2px)2\ (\pi 2py)2(\pi^*2px)1$:

$$\text{B.O.} = \frac{10-5}{2} = 2.5$$

$O_2^- \Rightarrow (\sigma 1s)^2(\sigma^*1s)^2(\sigma 2s)^2(\sigma^*2s)^2(\sigma 2p_z)^2(\pi 2p_x)^2\ (\pi 2p_y)^2(\pi^*2p_x)^2(\pi^*2p_y)^1;$

$$\text{B.O.} = \frac{10-7}{2} = 1.5$$

$O_2^{2-} \Rightarrow (\sigma 1s)^2(\sigma^*1s)^2(\sigma 2s)^2(\sigma^*2s)^2(\sigma 2p_z)^2(\pi 2p_x)^2\ (\pi 2p_y)^2(\pi^*2p_x)^2(\pi^*2p_y)^2;$

$$\text{B.O.} = \frac{10-8}{2} = 1$$

Hence, increasing order of bond order or bond strength is

$$O_2^{2-} < O_2^- < O_2 < O_2^+$$

Q.120 **The number of molecules or ions having bond order 2.5 among O_2^+, CN, NO, N_2^+, NO^+, O_2^-, CN^-, N_2; is**

(a) CN, NO, CN^-, N_2 **(b)** CN, O_2^-, N_2

(c) O_2^+, NO, N_2^+, CN^- **(d)** O_2^+, CN, NO, N_2^+

Sol. : (d)

O_2^+, CN, NO and N_2^+ have bond order of 2.5.

Q.121 **If x is the number of bonding electrons and y is the number of antibonding electrons in CO molecule, then**

(a) x = 8, y = 4 (b) x = 10, y = 8 (c) x = 10, y = 4 (d) x = 8, y = 6

Sol. : (c)

CO : $\sigma 1s^2\ \sigma^*1s^2\ \sigma 2_s^2\ \sigma 2p_z^2\ \pi 2p_x^2 = \pi 2p_y^2\ \sigma^*2s^2$

For CO, x = 10 and y = 4